



IMPROVED DELAYED NEUTRON GROUP PARAMETERS FOR MOLTEN SALT REACTORS

BY

LUKE SEIFERT

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Doctoral Committee:

Associate Professor Kathryn D. Huff, Co-advisor  
Assistant Professor Madicken Munk, Co-advisor  
Professor Tomasz Kozlowski  
Assistant Professor April Novak  
Staff Scientist Samuel Walker  
Professor Daniel Katz

# Abstract

Fluid-fueled molten salt reactors (MSRs) have garnered interest since the Generation IV International Forum in the early 2000s, where the MSR was designated as one of six advanced reactors of interest. A unique consideration in MSRs is the circulation of delayed neutron precursors (DNPs). These DNPs move from neutronically-important areas in the core to areas of lower importance, such as the piping outside the core. This decreases the effective delayed neutron fraction, affecting reactor behavior. Accurate models and simulations of DNPs are important for MSR safety and design.

Many studies have examined how fuel-salt movement in MSRs alters the effective delayed neutron fraction. The movement of the DNPs is modeled using DNP groups, typically six groups, with each group  $k$  consisting of a decay constant  $\lambda_k$  and yield of delayed neutrons per fission event  $\bar{\nu}_{d,k}$ . However, these studies do not account for the effects of fuel salt movement and chemical removal rates on the DNP group parameters  $\lambda_k$  and  $\bar{\nu}_{d,k}$  used in the transient simulations. Neglecting these MSR-specific phenomena reduces accuracy in the DNP group parameters in MSR simulations.

Fuel salt movement and chemical removal rates are important to consider because these parameters directly influence safety-, security-, and safeguards-related simulations. By using DNP group parameters with greater accuracy, models can more accurately reflect physical phenomena. Having greater accuracy in predictive models is key for successful MSR deployment.

This work resolves DNP group parameters for MSRs using the microscopic method, also known as the aggregate or summation method. This method entails using individual DNP data to reconstruct the DNP groups by summing the delayed neutron count rate contribution of each DNP in simulated irradiation experiments. Pulse irradiation experiments occur at very short time scales relative to MSR physical phenomena and are used to generate the short-lived DNP group parameters. Saturation, or infinite, irradiation experiments occur at time scales several times longer than the longest DNP half-life and are used to generate the longer-lived DNP group parameters. Because the saturation irradiation lasts on the order of minutes, the thermal-hydraulic and chemical effects in MSRs affect the DNP groups. Therefore, the longer-lived DNP group parameters can be expected to vary more significantly than the shorter-lived group parameters.

A non-linear least squares solver is used to generate an improved set of DNP group parameters. The improved DNP group parameters are verified by comparing them to the data used to generate those parameters, as the total delayed neutron yields and average half-lives should be nearly identical. Additional verification of the parameters is conducted via comparison with total delayed neutron yields and average half-lives from the literature.

The proposed work includes: development and evaluation of a more sophisticated model for generating DNP group parameters; validation via comparison with the Molten Salt Reactor Experiment (MSRE) during pump start-up and coast-down; generation of delayed neutron group spectra; evaluation of various safety, security, and safeguards-related transients using improved DNP group parameters; and comparison of how different datasets affect the DNP group parameters.

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# Chapter 1

## Introduction

This thesis presents improved delayed neutron precursor (DNP) groups that capture fluid-fueled molten salt reactor (MSR) fuel salt circulation and chemical removal that improve the accuracy of MSR transient simulations. The MSR is a Generation IV reactor design [97]. MSRs are advanced reactors intended to improve safety and efficiency over traditional reactor designs. Instead of a solid fuel, MSRs use a liquid fuel composed of a molten salt in which some fissile nuclide, such as  $^{235}\text{U}$ , is dissolved. Molten salt fuels offer many benefits: high operating temperatures, low operating pressure, high thermal conductivity, and high heat capacity. Other benefits include a strong, expansion-driven negative temperature coefficient of reactivity, high resource utilization through higher burnup, reduced fabricating and transporting requirements for fuel, and improved neutron economy due to online fission product removal [173].

MSRs have several barriers to deployment which must be addressed before they are commercialized. Corrosion and material degradation pose an issue, as the high-temperature salts corrode materials. Another deployment barrier involves reactor safety, security, and safeguards by design. MSRs have distinct designs compared with traditional light water reactors, such as requiring bulk rather than discrete-item material control and accountability (MCA). To properly design MSRs, models must be able to simulate them accurately. One specific phenomenon of critical importance in MSRs is the circulating liquid fuel. When fission events occur, DNPs are formed and circulate alongside the fuel.

DNPs are nuclides that undergo  $\beta^-$  decay, producing a daughter, called an emitter, which emits a neutron [129]. Without properly modeling the DNPs, reactor simulations may predict incorrect power responses, reactivity feedback, control element worth, and reactor controllability [46, 45, 102]. Reactor physics simulations typically do not model DNPs explicitly, but instead represent them in an aggregate form using a smaller number of artificial groups, which approximates the delayed neutron emission rate at much lower computational cost.

These artificial groups are referred to as DNP groups and are organized into  $k$  groups. Each group is represented using three parameters: a decay constant  $\lambda_k$ , a neutron energy spectrum  $\chi_{d,k}$ , and an average yield of delayed neutrons per fission event  $\bar{\nu}_{d,k}$ . The neutron energy spectrum of each group is necessary only if the energy dependence is of interest for an analysis, otherwise it can be neglected. Although the number of groups is arbitrary, the use of six groups is common for capturing the general delayed neutron emission rate from all the DNPs present in the system. The DNP group parameters are intended to provide sufficient data to calculate the rate of delayed neutrons emitted by group  $k$  over a given time period.

Table 1.1 shows a brief historical timeline of DNP group parameter advancements. Early work, shown from 1946 through 1957, consisted of fissile sample irradiation in reactors to calculate DNP group parameters. From

1965 onward, investigations started to examine individual DNPs and how those DNPs could be mapped onto DNP group parameters. DNP group parameter formulation and modeling advanced rapidly thanks to increased experimental data regarding half-lives and emission probabilities of individual DNPs, which enabled more accurate group parameter formulation.

However, these formulations of DNP group parameters did not account for how MSR fuel movement and chemical removal affects the DNP group parameters. For example, if an MSR chemically removed significant quantities of an element containing important DNPs, that removal would need to be captured in the group parameters; these chemical removals are element-dependent, which the artificial group structure does not capture. Not capturing these effects biases simulations with chemical removal and fuel recirculation.

There is a gap in the literature, which can be closed by analyzing and quantifying the effects of chemical removal and recirculation on the DNP group parameters. Researchers have used the DNP groups and data discussed in Table 1.1 to analyze the flow of the traditional DNP groups in MSR kinetics [209, 214, 106, 180, 101]. However, that work has not investigated the effects of MSR phenomena on the DNP groups. Thus, the next step in group parameter development is DNP group parameters that accurately reflect recirculation and chemical removal in MSRs.

Table 1.1: Timeline of advancements in DNP groups.

| Year | Advancement                                                                                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1939 | Roberts et al. discovered delayed neutrons [153]; Bohr and Wheeler explained delayed neutrons [21].                                                                                                      |
| 1946 | Burgy et al. conducted initial spectral measurements [31].                                                                                                                                               |
| 1947 | Hughes et al. measured six DNP groups at Argonne heavy water pile [81].                                                                                                                                  |
| 1957 | Keepin et al. generated improved DNP group parameters [95, 203].                                                                                                                                         |
| 1965 | Amiel identified seven DNPs that account for 80% of delayed neutrons [4].                                                                                                                                |
| 1969 | Marmol compiled experimental data on emission probability and half-life for 34 DNPs [47].                                                                                                                |
| 1972 | Tomlinson mapped 38 DNPs onto six groups [193].                                                                                                                                                          |
| 1977 | Saphier et al. constructed delayed neutron spectra from 20 DNPs [168].                                                                                                                                   |
| 1982 | Rudstam generated DNP group parameters and spectra using 67 DNPs based on half-lives [160].                                                                                                              |
| 1983 | England et al. generated DNP group parameters and spectra using 105 DNPs based on half-lives [56].                                                                                                       |
| 1988 | Brady generated DNP group parameters and spectra using 271 DNPs with least squares [24]. Group parameters included in ENDF/B-VI.8 and remain unchanged through ENDF/B-VIII.0 [204, 29, 159, 23, 55, 25]. |
| 1999 | Parish et al. stated DNP group parameters in Evaluated Nuclear Data File (ENDF) are in need of revision [135].                                                                                           |
| 2014 | The International Atomic Energy Agency (IAEA) hosted the Coordinated Research Project (CRP) on Development of a Reference Database for $\beta$ -delayed neutron emission data [124].                     |
| 2020 | Leconte et al. used more than 400 DNPs to combine data with macroscopic experimental measurements [105].                                                                                                 |

This research aims to generate improved DNP group parameters that account for MSR recirculation and chemical removal, verify the parameters with the literature, validate the parameters by simulating transients and comparing with experimental data, and evaluate the impact of the improved parameters on safety, safeguards, and security

by design. For example, chemical removal of DNPs decreases the number of delayed neutrons emitted per fission event on average. This reduces the total delayed neutron yield, decreases the effective delayed neutron fraction, shortens the reactor period, and reduces reactor controllability. Current DNP group parameters in MSR simulations may overestimate the number of delayed neutrons in the system, resulting in non-conservative assessments of controllability.

The recirculation and chemical removal in MSRs can be incorporated into DNP group parameter calculations at varying levels of fidelity, with associated differences in cost and accuracy. The DNP group parameter calculations require simulation of fissile sample irradiation. This document describes two models, the 0D scaled and 0D flow models, that simulate irradiation of a fissile sample, from which DNP group parameters can be calculated. Two types of irradiations are considered: saturation irradiations, which irradiate a sample until the DNPs concentrations are at equilibrium, and pulse irradiations, which use a short pulse to irradiate the fissile sample. These models use different simplifications, but all of them simulate an incident neutron source that irradiates a target fissile sample consisting of a single fissile nuclide. After irradiation, the concentrations of the DNPs are extracted and used to calculate the delayed neutron count rate of the sample. Then, some selected number of DNP groups with decay constants  $\lambda_k$  and yields  $\bar{\nu}_{d,k}$  are used to fit that delayed neutron count rate.

The 0D scaled model is a 0D model of a fissile sample in an MSR where the in-core and ex-core regions are approximated by scaling the flux and chemical removal rates. The purpose of the 0D scaled model is to rapidly generate longer-lived DNP groups while approximately treating the recirculation and chemical removal rates. The 0D scaled model has been completed and is discussed in Section 3.2. This model neglects time dependence, decay chains, and cross-sections while scaling the flux and chemical removal rates based on the fraction of fuel salt in different regions of the primary loop. The 0D flow model is part of the proposed work and is also discussed in Section 3.2. This model accounts for time dependence of DNP concentrations and decay chains while applying the flux and chemical removal rates only in the relevant parts of the primary loop. It is referred to as the 0D flow model because it captures the effects of the fuel salt flow by using residence times. The purpose of the 0D flow model is to capture short and long-lived groups at higher computational cost but increased accuracy compared to the 0D scaled model. There is also another model discussed in Appendix E, the spatially resolved model, which is discussed as potential future work. This model would account for time and space dependence of DNP concentrations across the entire primary loop.

The preliminary results, in Section 4, cover impactful DNPs, DNP group parameter sensitivity studies, and 0D scaled model verification. The results show the total delayed neutron yield ( $\bar{\nu}_d$ ) and average half-life ( $\tau$ ) are within uncertainty bounds between the data used in the model and the improved DNP group parameters generated with the model. The match between the input data and the group parameters shows the effect of each DNP is captured in the DNP group parameters. Sensitivity studies reveal each DNP affects multiple DNP group parameters, not only the group with the most similar half-life. Verification with the literature shows the 0D scaled model works well for longer-lived groups but poorly for shorter-lived groups. The 0D scaled model only simulates saturation irradiations, which, in combination with the fission yield, half-life, and emission probability data used, results in short-lived groups that are not consistent with the literature.

The proposed work, in Section 5.4, starts with 0D flow model development, followed by a verification stage to ensure improved DNP group parameters are correct for a static case. The work then proposes 0D flow model validation using a transient simulation of the Molten Salt Reactor Experiment (MSRE) compared to experimental data to ensure improved DNP group parameters are physically accurate. The improved group spectra are proposed to be included in DNP group parameter formulation and to be verified for a static case. Finally, the work proposes simulation of MSRE transients for various safety, security, and safeguards scenarios using improved and pre-existing

DNP group parameters to study their impact.

Section 5.4 presents potential future work. Future work includes spatially resolved model development, uncertainty propagation of MSR physical phenomena in DNP group parameters, and investigation of the effects of varying the incident neutron energy spectrum on DNP group parameters.

## Chapter 2

# Background and Motivation

This chapter provides the background and motivation for generating more accurate DNP group parameters for MSRs and implementation of those parameters for safety, security, and safeguards purposes. An overview, including information on DNPs and MSR fuel depletion, is followed by relevant context motivating the problem, including safety, security, and safeguards. The DNP group parameters in MSRs differ from traditional group parameters in stationary systems because the fuel circulates and undergoes online chemical reprocessing, which both alter the local density and distributions of DNPs. These effects can be captured in spatially resolved MSR fuel depletion simulations. Existing methodologies for DNP group parameter generation do not account for the recirculation and chemical removal present in MSR fuel depletion [95, 195, 117, 13, 200, 24, 81, 114, 115, 143, 44, 143, 162, 204, 23, 105]. This work presents improved DNP group parameters that account for recirculation and chemical removal. These improved DNP group parameters are particularly important for safety, security, and safeguards related concerns.

### 2.1 Delayed Neutron Precursors

This section discusses what DNPs are, how they are measured, how groups are formulated, how they relate to kinetics parameters, and how they are represented in kinetics models. Before delving deeper into the DNP group parameters, it is useful to first understand DNPs conceptually because they are an important consideration in nuclear reactors [46]. DNPs are nuclides produced via fission that decay via  $\beta^-$  decay, producing a daughter, called an emitter, which emits a neutron [129]. In this work, the term DNP will be used exclusively rather than referring to emitters. Distinguishing between the emitter and the DNP is not relevant for this work because the emission time of the neutron is dominated by the  $\beta^-$  decay [21, 134].

Equation (2.1) shows a simplified equation for the reactor period  $\tau_e$ , calculated using a single effective DNP group and without the effect of prompt neutrons [40]. The delay in neutron emission significantly affects the reactor period, or the time it takes for the reactor power to increase by a factor of  $e$ , and transient behavior in general.

$$\tau_e \approx \frac{\beta_{eff} - \rho}{\rho \lambda_{eff}} \quad (2.1)$$

where

$\beta_{eff}$  = the effective delayed neutron fraction [-]

$\rho$  = the reactivity [-]

$\tau_e$  = the reactor period [s]

$\lambda_{eff}$  = the effective DNP decay constant for all DNPs [ $s^{-1}$ ].

The effective delayed neutron fraction is discussed in more detail in Section 2.1.2. The one DNP group approximation is not used in this study, so this approximation is not discussed in greater detail. However, this simplified equation illustrates that the reactor period depends greatly on the effective delayed neutron fraction, which itself is based on the DNPs.

Equation (2.2) shows the time-dependent rate of delayed neutron emission from a single DNP as:

$$\dot{n}_{d,i}(t) = P_{n,i} \lambda_i N_i(t), \quad (2.2)$$

where

$\dot{n}_{d,i}$  = the delayed neutron activity, or count rate, from DNP  $i$  [ $\# \cdot s^{-1}$ ]

$P_{n,i}$  = the average number of delayed neutrons emitted from  
decay of nuclide  $i$ , or the emission probability [#]

$\lambda_i$  = the decay constant of DNP  $i$  [ $s^{-1}$ ]

$N_i(t)$  = the number of atoms of DNP  $i$  [#].

In a solid-fueled reactor, such as a light water reactor (LWR), spatial resolution is not necessary to properly model the DNPs, as the fuel remains stationary. In MSRs, the movement of the fuel salt necessitates spatial resolution for tracking DNP distributions and concentrations. Concentrations of nuclides vary spatially as short-lived nuclides decay and longer-lived nuclides are distributed outside the core. Modeling space and time together largely increases computational cost, making problems involving DNPs in MSRs more challenging than handling DNPs in stationary fuels.

Properly modeling the DNPs in reactors is important; otherwise simulations may predict incorrect power responses, reactivity feedback, control element worth, and reactor controllability [46, 45, 102]. Section 2.1.3 shows how the power response depends on the DNPs. The reactor controllability varies based on the quantity of delayed neutrons in neutronically-important regions. If there are fewer neutronically-important delayed neutrons available due to the movement of the fuel salt, such as in an MSR, then the reactor period becomes shorter, making the reactor harder to control.

### 2.1.1 Formulation

This section discusses the methods used to formulate DNP groups. Macroscopic and microscopic approaches are used for formulating DNP groups [135, 162]. The macroscopic approach uses an experimental fissile sample irradiation to get the delayed neutron count rate. The microscopic approach uses the data from many experiments

combined to simulate a fissile sample irradiation and generate the delayed neutron count rate. Although the microscopic approach can be more challenging due to dependence on the work of many individual experiments, it offers more detail on the delayed neutron count rate.

After the delayed neutron count rate is collected, both approaches then approximate the behavior of all the DNPs with only a few artificial DNPs, known as DNP groups. These groups consist of decay constants  $\lambda_k$  and yields  $\bar{v}_{d,k}$ . These groups have historically been formed based on the DNP half-lives or the best fit to the delayed neutron count rate.

The macroscopic formulation is important in this thesis for verification and validation, as the delayed neutron count rate data are experimentally collected and can be used for comparison. The microscopic formulation is important in this thesis because it is the methodology that enables simulation of effects that target specific elements, such as chemical removal in an MSR. The microscopic approach is also used because there is no delayed neutron count-rate data for an irradiated sample in an MSR. Additionally, the microscopic approach provides useful information on how each DNP contributes and how chemical removal of certain elements affects the DNPs.

### Macroscopic Formulation of DNPs

The macroscopic approach entails irradiating a sample containing a single fissile nuclide at or near a given neutron energy. This delayed neutron count rate from this sample can then be measured and used to formulate DNP group parameters. Analysis and collection of macroscopic experiment data are common in the literature [95, 195, 117, 13, 200, 24, 81, 114, 115, 41, 170].

Small samples are used to minimize neutron multiplication within the sample to avoid spectral effects [95]. Spectral effects are caused by differing neutron energy spectra between each DNP group and the incident irradiating neutrons [95]. Another reason for using a small sample is due to physical challenges associated with the experimental sample handling (rabbit) system [30].

Two different types of irradiation are typically used: pulse and infinite. A pulse irradiation exposes the sample to neutrons with a pulse shorter than the shortest DNP half-life, and captures the short-lived DNPs more accurately than the long-lived DNPs [24]. An infinite irradiation exposes the sample to neutrons for longer than several times the longest half-life, and it captures the long-lived DNPs with more accuracy than the short-lived DNPs [24]. After irradiation, the delayed neutrons from the sample are counted as a function of time. Figure 2.1 (recreated from [95]) shows an experimental setup for collecting delayed neutrons. The delayed neutron count rate as a function of time after irradiation,  $n_d(t)$ , is given as:

$$\dot{n}_d(t) = \epsilon \sum_{i=1}^I P_{n,i} \lambda_i N_i(t), \quad (2.3)$$

where

$\dot{n}_d(t)$  = the delayed neutron count rate  $[\# \cdot s^{-1}]$

$I$  = the total number of DNPs  $[\#]$

$\epsilon$  = the detector efficiency  $[-]$ .

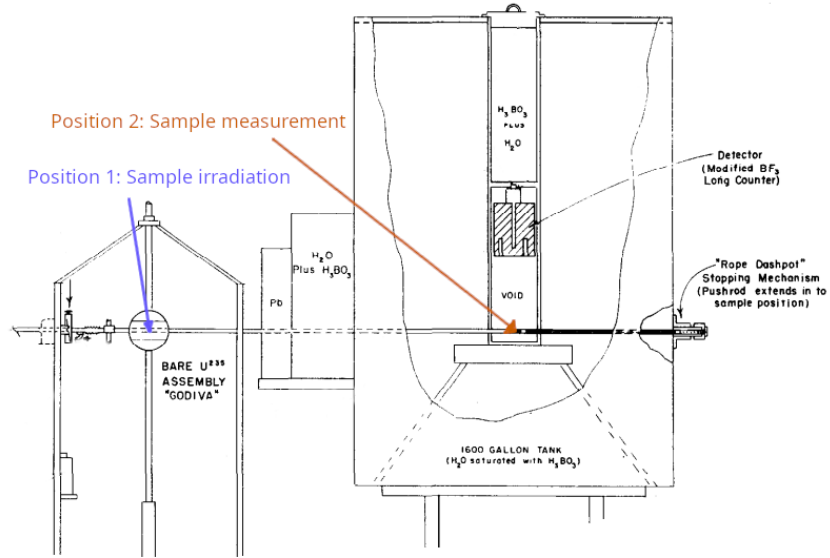


Figure 2.1: Simplified experimental setup for measuring delayed neutron emission from irradiation of a 3 gram sample in Godiva recreated from [95].

Once this delayed neutron count rate is generated, the data are used to generate DNP group parameters,  $\lambda_k$  and  $\bar{\nu}_{d,k}$ , for that fissile nuclide and neutron energy. Three different approaches have been used to generate DNP group parameters from macroscopic data: exponential stripping (or peeling) [30, 94], least squares [24, 95], and half-lives of specific DNPs [168]. Some approaches have even used a combination, such as least-squares fitting with specific half-lives [185]. The exponential stripping, or peeling, method is a more subjective form of least squares fitting, in which exponential curves are subtracted away from the raw data [147]. A semi-log form of the equation fits the tail-end of the decay curve fit with a straight line, subtracts that new exponential function off, and then repeats the process for the remaining data [185]. The use of specific half-lives to form the DNP groups is less subjective and offers a more physical interpretation of the groups. The least squares approaches are the most objective approach and generate groups which physically do not embody a single DNP. Instead, they group together the DNPs and effectively capture the overall physical effect. For this work, a least squares approach is used, as it offers an objective formulation that can be generically applied. The specifics of the least squares approach used to form the DNP parameters are discussed in Section 2.1.2.

Challenges associated with collecting experimental data include experimental noise and signal obfuscation due to moderation [129]. Experimental noise becomes significant after long periods of time relative to the half-life of the longest-lived DNP, as the delayed neutron counts decrease exponentially. The short-lived DNPs cannot be measured immediately after irradiation; they must first be transported to a delayed neutron counter. This transport time results in missing very short-lived counts. After a pulse irradiation, the long-lived groups form a background that interferes with measurement of the short-lived groups [129]. Due to these effects, the delayed neutron emission measurements are most accurate for intermediate times, while the short-lived and long-lived DNPs have greater uncertainty. Although the emission measurements have the largest uncertainty in the short-lived and long-lived DNPs, the least squares solve for the DNP group parameters leads to increased uncertainty in the intermediate-lived groups. This is because small changes in the intermediate-lived groups affect the other intermediate-lived groups

more directly, whereas the longest- and shortest-lived groups are less affected by small differences in the other group half-lives and yields.

Similarly, uncertainty arises from the measurement of the number of fissions. Although the number of fission events can be measured somewhat accurately using a fission chamber after irradiation or other techniques, the fission chamber does not account for source multiplication within the irradiated sample. Keepin et al. suggest a technique to correct for the sample self-multiplication. The sample self-multiplication requires a correction factor of up to 3% in the number of fissions.

By the year 2000, 245 distinct sets of DNP group parameters had been formulated for 20 different fissile nuclides [184]. Spriggs and Campbell went into significant detail, discussing the various analysis methods, neutron energies, fissile nuclides, counting intervals, and sample compositions for many of these DNP group parameter formulations [184]. Tuttle also provided a detailed review of many macroscopic DNP group parameters pre-1974, including details on the irradiated sample, reactor, counter, and travel time of the sample between irradiation and measurement [195]. Tables 2.1 through 2.4 show some macroscopic DNP group parameters from the literature for thermal fission of  $^{235}\text{U}$  [135].

Table 2.1: Macroscopic formulation of group yields from thermal fission of  $^{235}\text{U}$  (recreated from [135]).

| Source                      | Group [-] |         |         |         |         |         |
|-----------------------------|-----------|---------|---------|---------|---------|---------|
|                             | 1         | 2       | 3       | 4       | 5       | 6       |
| Keepin et al. [95]          | 0.00052   | 0.00346 | 0.00310 | 0.00624 | 0.00182 | 0.00066 |
| Charlton et al. [36]*       | 0.00054   | 0.00344 | 0.00319 | 0.00607 | 0.00186 | 0.00066 |
| Saleh et al. [166]          | 0.00055   | 0.00363 | 0.00297 | 0.00593 | 0.00168 | 0.00063 |
| Synetos and Williams [187]* | 0.00051   | 0.00328 | 0.00214 | 0.00583 | 0.00270 | 0.00063 |
| Tuttle [195]                | 0.00065   | 0.00362 | 0.00320 | 0.00692 | 0.00218 | 0.00044 |
| Waldo et al.[200]           | 0.00055   | 0.00366 | 0.00327 | 0.00660 | 0.00192 | 0.00070 |

\* Modified to include a sixth group by Parish et al. [135].

Table 2.2: Macroscopic formulation of group yield uncertainties from thermal fission of  $^{235}\text{U}$  (recreated from [135]).

| Source                      | Group Uncertainties [-] |               |               |               |               |               |
|-----------------------------|-------------------------|---------------|---------------|---------------|---------------|---------------|
|                             | $\sigma_1$              | $\sigma_2$    | $\sigma_3$    | $\sigma_4$    | $\sigma_5$    | $\sigma_6$    |
| Keepin et al. [95]          | $\pm 0.00006$           | $\pm 0.00025$ | $\pm 0.00041$ | $\pm 0.00047$ | $\pm 0.00019$ | $\pm 0.00013$ |
| Charlton et al. [36]*       | $\pm 0.00003$           | $\pm 0.00005$ | $\pm 0.00024$ | $\pm 0.00013$ | $\pm 0.00014$ | $\pm 0.00014$ |
| Saleh et al. [166]          | $\pm 0.00005$           | $\pm 0.00032$ | $\pm 0.00027$ | $\pm 0.00104$ | $\pm 0.00015$ | $\pm 0.00012$ |
| Synetos and Williams [187]* | $\pm 0.00004$           | $\pm 0.00020$ | $\pm 0.00036$ | $\pm 0.00037$ | $\pm 0.00029$ | $\pm 0.00012$ |
| Tuttle [195]                | $\pm 0.00007$           | $\pm 0.00013$ | $\pm 0.00041$ | $\pm 0.00019$ | $\pm 0.00021$ | $\pm 0.00007$ |
| Waldo et al.[200]           | $\pm 0.00006$           | $\pm 0.00041$ | $\pm 0.00041$ | $\pm 0.00038$ | $\pm 0.00017$ | $\pm 0.00009$ |

\* Modified to include a sixth group by Parish et al. [135].

There is reasonable agreement between the different DNP group parameters, though there is variation among them. Not only did each approach use a slightly different experimental methodology via different energy spectra, data collection methods, sample transfer time between irradiation and measurement, sample size, and sample composition, but there are also differences in how the data were processed. There are several ways that the data was processed from delayed neutron count rates into DNP group parameters among the works in these tables:

Table 2.3: Macroscopic formulation of group half-lives from thermal fission of  $^{235}\text{U}$  (recreated from [135]).

| Source                      | Group [s] |          |         |         |         |         |
|-----------------------------|-----------|----------|---------|---------|---------|---------|
|                             | 1         | 2        | 3       | 4       | 5       | 6       |
| Keepin et al. [95]          | 55.89897  | 22.72614 | 6.24457 | 2.30281 | 0.60802 | 0.23028 |
| Charlton et al. [36]*       | 55.45177  | 22.43195 | 6.02737 | 2.18658 | 0.49159 | 0.23028 |
| Saleh et al. [166]          | 55.45177  | 22.65187 | 6.24457 | 2.31049 | 0.63013 | 0.23028 |
| Synetos and Williams [187]* | 56.35343  | 22.35959 | 7.00149 | 2.72893 | 0.77016 | 0.23028 |
| Tuttle [195]                | 54.57852  | 21.86584 | 6.02737 | 2.22877 | 0.49511 | 0.17911 |
| Waldo et al.[200]           | 54.57852  | 21.86584 | 6.02737 | 2.22877 | 0.49511 | 0.17911 |

\* Modified to include a sixth group by Parish et al. [135].

Table 2.4: Macroscopic formulation of group half-life uncertainties from thermal fission of  $^{235}\text{U}$  (recreated from [135]).

| Source                      | Group Uncertainties [s] |               |               |               |               |               |
|-----------------------------|-------------------------|---------------|---------------|---------------|---------------|---------------|
|                             | $\sigma_1$              | $\sigma_2$    | $\sigma_3$    | $\sigma_4$    | $\sigma_5$    | $\sigma_6$    |
| Keepin et al. [95]          | $\pm 1.35239$           | $\pm 0.67061$ | $\pm 0.22503$ | $\pm 0.09181$ | $\pm 0.08000$ | $\pm 0.02219$ |
| Charlton et al. [36]*       | $\pm 0.44361$           | $\pm 0.50817$ | $\pm 0.10482$ | $\pm 0.06208$ | $\pm 0.03138$ | $\pm 0.02219$ |
| Saleh et al. [166]          | $\pm 3.99253$           | $\pm 0.14805$ | $\pm 0.39380$ | $\pm 0.03851$ | $\pm 0.01719$ | $\pm 0.02219$ |
| Synetos and Williams [187]* | $\pm 0.45816$           | $\pm 0.28851$ | $\pm 0.77794$ | $\pm 0.21488$ | $\pm 0.09413$ | $\pm 0.02219$ |
| Tuttle [195]                | $\pm 1.28926$           | $\pm 0.82773$ | $\pm 0.20965$ | $\pm 0.08600$ | $\pm 0.04244$ | $\pm 0.02545$ |
| Waldo et al.[200]           | $\pm 1.28926$           | $\pm 0.82773$ | $\pm 0.20965$ | $\pm 0.08600$ | $\pm 0.04244$ | $\pm 0.02545$ |

\* Modified to include a sixth group by Parish et al. [135].

simultaneous non-linear least squares fitting of saturation and pulse irradiation data, separate non-linear least squares fitting of pulse and saturation irradiation data and subsequent combination, non-linear least squares of an intermediate amount of time between a pulse and saturation, and exponential stripping.

Keepin et al. used both a simultaneous fitting of DNP group parameters from pulse and saturation irradiation data as well as a separate fitting of four groups from each irradiation. The separate fitting of four groups is normalized at the second group, and Keepin et al. reported that this method gave better results than the simultaneous fit [95].

Charlton et al. used the equation for an intermediate irradiation time, neither pulse nor saturation, as shown in Equation (2.7). They used only five groups, as the sample took half a second to move from irradiation to the detector where delayed neutron counts could be measured. The shortest-lived DNPs that would make up group six were not included, but Parish et al. added group six values for their own analysis [135]. Charlton et al. used a least squares solve to determine the DNP group parameters, according to Spriggs and Campbell [184]. Synetos and Williams used the same approach as Charlton et al. [187, 185].

Saleh et al. used the equation for an intermediate amount of time, similar to Charlton et al., where the equation is described in Equation (2.7). However, instead of using a least squares solution, they used exponential stripping [166].

Tuttle formulated suggested values based on reviewing the literature and applying empirical correlations [195].

Waldo et al. applied a transit time correction using an empirical correlation, and the DNP group parameters were fitted using a least squares solve [200]. It is unclear whether pulse or saturation irradiations were used. The conclusions indicate that the DNP group parameters are less applicable for short times or short irradiations, implying that a saturation irradiation was used.

## Microscopic Formulation of DNPs

The other approach used to formulate DNP groups is the microscopic – also referred to in the literature as the aggregate or summation – approach, which uses data from individual nuclides summed together to form a simulated delayed neutron count rate. This count rate can then be used to generate DNP group parameters, the same as in the count rate from the macroscopic method.

This approach was investigated in detail in the 1990-2000 period [143, 44, 162, 204, 23], with progress through the 2010s from more data becoming available [124, 84]. Leconte et al. used this approach as recently as 2020 [105]. This approach requires experiments on individual DNP nuclides, from which emission probabilities and half-lives can be measured. Typically, this involves rapid chemical extraction or isotope separation and magnetic tape to rapidly measure short-lived nuclides, such as in the study by Björnstad et al., in which an aluminized mylar tape is used [19]. The magnetic tape is used to transport samples or remove longer-lived decay products [177, 163]. Experimental methods include the neutron- $\beta^-$  coincidence technique, separate counting of neutrons and  $\beta$  particles, separate counting of neutrons and  $\gamma$  rays, ion counting, fission yield from a set number of fissions, gamma counting of a long-lived daughter, and normalization of the neutron counter using a known precursor [163].

Other experimental methods used to study DNPs include on-line mass spectrometry, helium-3 spectrometry, proton-recoil detectors, the time-of-flight method, and neutron scintillators [46]. On-line mass spectrometry combined with a  $^3\text{He}$  ionization chamber and a proton-recoil detector allows for measurement of the delayed neutron energy spectrum [176, 18]. The time-of-flight technique consists of measuring the flight time of a neutron over a known distance to calculate the velocity, including a neutron scintillator [18, 39, 66, 38].

These microscopic measurement data are exceptionally useful, as they allow for treatment of DNPs on a nuclide-by-nuclide basis. This is particularly important for MSRs, where the spatial DNP concentration depends on half-lives, fission yields, thermo-chemistry, and thermal-hydraulics. With data on individual nuclides, higher fidelity treatment of DNPs becomes possible.

There have been many new measurements in the past several decades investigating these nuclides. The IAEA has compiled and evaluated this data in a publicly accessible form [51, 50]. This data, when combined with a depletion simulation, can be used to model the DNPs explicitly rather than using groups. This is particularly useful for understanding which nuclides have the greatest impact, as well as for simulating experimental configurations which have yet to be thoroughly investigated for DNP group fitting.

Tables 2.5 through 2.8 show some microscopic DNP group parameters from the literature for thermal fission of  $^{235}\text{U}$  [135]. There is generally reasonable agreement between the different DNP group parameters, though it should be noted that Brady and England may be closer to ENDF/B-VI.8 because ENDF/B-VI.8 uses the Brady and England results in its formulation. Some of the half-lives differ more than others, such as the group one half-lives ranging from 51.7 to 55.3 seconds, while the macroscopic range was 54.6 to 56.4 seconds. The group three half-lives also vary noticeably, with a range of 5.7 to 7.7, while the macroscopic range was 6.0 to 7.0 seconds. The larger range can be attributed to the variety of data and methods used for forming the delayed neutron count rate as well as the fitting of the DNP group parameters, whereas the macroscopic groups have a smaller range because the delayed neutron count rate is experimentally measured and the variation is from experimental uncertainty and fitting of the group parameters.

ENDF/B-VI.8 pulls from Brady and England, as well as other works by England [159, 23, 55, 25]. The approaches in those studies collected 271 DNPs, generated a delayed neutron count rate curve, and then used the two four group aggregation approach described by Keepin et al. [95, 23]. Mills et al. used two different approaches: fitting all parameters simultaneously as described by Keepin et al. and using the Keepin et al. half-lives while fitting only the yields [125, 95]. The values presented here are from the approach where all parameters are fit simultaneously for all

Table 2.5: Microscopic formulation of group yields from thermal fission of  $^{235}\text{U}$  (recreated from [135]).

| Source                 | Group [-] |         |         |         |         |         |
|------------------------|-----------|---------|---------|---------|---------|---------|
|                        | 1         | 2       | 3       | 4       | 5       | 6       |
| ENDF/B-VI.8 [159]      | 0.00058   | 0.00302 | 0.00288 | 0.00646 | 0.00265 | 0.00111 |
| Mills et al. [125]     | 0.00056   | 0.00325 | 0.00196 | 0.00658 | 0.00287 | 0.00122 |
| Brady and England [23] | 0.00068   | 0.00341 | 0.00292 | 0.00611 | 0.00310 | 0.00158 |

\* Modified to include a sixth group by Parish et al. [135].

Table 2.6: Microscopic formulation of group yield uncertainties from thermal fission of  $^{235}\text{U}$  (recreated from [135]).

| Source                 | Group Uncertainties [-] |               |               |               |               |               |
|------------------------|-------------------------|---------------|---------------|---------------|---------------|---------------|
|                        | $\sigma_1$              | $\sigma_2$    | $\sigma_3$    | $\sigma_4$    | $\sigma_5$    | $\sigma_6$    |
| ENDF/B-VI.8 [159]      | -                       | -             | -             | -             | -             | -             |
| Mills et al. [125]     | $\pm 0.00004$           | $\pm 0.00023$ | $\pm 0.00020$ | $\pm 0.00051$ | $\pm 0.00035$ | $\pm 0.00012$ |
| Brady and England [23] | $\pm 0.00004$           | $\pm 0.00019$ | $\pm 0.00016$ | $\pm 0.00034$ | $\pm 0.00017$ | $\pm 0.00009$ |

\* Modified to include a sixth group by Parish et al. [135].

Table 2.7: Microscopic formulation of group half-lives from thermal fission of  $^{235}\text{U}$  (recreated from [135]).

| Source                 | Group [s] |          |         |         |         |         |
|------------------------|-----------|----------|---------|---------|---------|---------|
|                        | 1         | 2        | 3       | 4       | 5       | 6       |
| ENDF/B-VI.8 [159]      | 51.72740  | 21.19716 | 5.73797 | 2.28913 | 0.81595 | 0.24295 |
| Mills et al. [125]     | 55.27489  | 22.80089 | 7.67860 | 2.77148 | 1.07381 | 0.28177 |
| Brady and England [23] | 52.11633  | 21.32761 | 5.68620 | 2.18727 | 0.70114 | 0.23462 |

\* Modified to include a sixth group by Parish et al. [135].

Table 2.8: Microscopic formulation of group half-life uncertainties from thermal fission of  $^{235}\text{U}$  (recreated from [135]).

| Source                 | Group Uncertainties [s] |               |               |               |               |               |
|------------------------|-------------------------|---------------|---------------|---------------|---------------|---------------|
|                        | $\sigma_1$              | $\sigma_2$    | $\sigma_3$    | $\sigma_4$    | $\sigma_5$    | $\sigma_6$    |
| ENDF/B-VI.8 [159]      | -                       | -             | -             | -             | -             | -             |
| Mills et al. [125]     | $\pm 0.09918$           | $\pm 0.11723$ | $\pm 0.37861$ | $\pm 0.11259$ | $\pm 0.08521$ | $\pm 0.01020$ |
| Brady and England [23] | -                       | -             | -             | -             | -             | -             |

\* Modified to include a sixth group by Parish et al. [135].

DNP group parameters.

## Comparison of Macroscopic and Microscopic Approaches

Huynh and Jouanne offered a useful comparison between the macroscopic and microscopic approaches via different datasets [84]. Specifically, they compared the total delayed neutron yield from various data sources from macroscopic measurements to the calculated total delayed neutron yield from microscopic measurements. These data can also be compared to the previously shown tables, from which the total delayed neutron yield can be calculated. The total data compilation is presented in Table 2.9.

Table 2.9: Total delayed neutron yields of 100 fissions from thermal fission of  $^{235}\text{U}$  (based on [84]).

| Approach    | Data                       | Delayed Yield [-] |
|-------------|----------------------------|-------------------|
| Microscopic | JEF-2.2 [129]              | $1.708 \pm 0.12$  |
| Microscopic | JEF-3.1.1 [167]            | $1.477 \pm 0.08$  |
| Microscopic | ENDF/B-VII.1 [34]          | $1.905 \pm 0.10$  |
| Microscopic | ENDF/B-VI.8 [159]          | 1.670             |
| Microscopic | Mills et al. [125]         | $1.645 \pm 0.07$  |
| Microscopic | Brady and England [23]     | $1.780 \pm 0.05$  |
| Macroscopic | JEF-2.2 [129]              | 1.655             |
| Macroscopic | JEF-3.1.1 [167]            | 1.620             |
| Macroscopic | ENDF/B-VII.0 [33]          | 1.585             |
| Macroscopic | Keepin et al. [95]         | $1.580 \pm 0.07$  |
| Macroscopic | Charlton et al. [36]       | $1.577 \pm 0.03$  |
| Macroscopic | Saleh et al. [166]         | $1.645 \pm 0.11$  |
| Macroscopic | Synetos and Williams [187] | $1.510 \pm 0.06$  |
| Macroscopic | Tuttle [195]               | $1.700 \pm 0.05$  |
| Macroscopic | Waldo et al.[200]          | $1.670 \pm 0.07$  |

Table 2.9 shows that the total delayed neutron yield, or the number of delayed neutrons emitted per 100 fission events, for thermal fission of  $^{235}\text{U}$  ranges from 1.477 to 1.905. The microscopic values range from 1.477 to 1.905, capturing the minimum and maximum of the range of values seen in the table. The macroscopic values range from 1.510 to 1.700, presenting a tighter range of values. Another point of interest is that there does not appear to be a consistent difference between the microscopic and macroscopic values for the same dataset. For instance, JEF-2.2 has a larger microscopic yield, but JEF-3.1.1 has a larger macroscopic yield. Macroscopic yields vary less than microscopic yields, indicating larger differences in the individual DNP data that affected their concentrations, emission probabilities, or half-lives.

### 2.1.2 DNP Group and Kinetics Parameters

In Section 2.1.1, the various methods for experimentally measuring delayed neutron precursors were discussed. The next stage in analyzing these precursors, discussed in this section, specifies how they are used in kinetics models, as well as the various ways they are represented. The microscopic and macroscopic experimental approaches provide the necessary data in the form of a delayed neutron count rate to create more easily handled forms via groups. This section discusses the least squares approach to convert this data into several DNP groups.

The DNPs are commonly represented in six groups. These groups were originally developed for computational efficiency and due to a lack of data for individual precursors. Brady and England found that increasing the number of groups from six to nine caused no significant differences in a point kinetics calculation for step changes in reactivity [23]. Their study shows that increasing the number of DNP groups alone does not provide an increase in accuracy for point kinetics simulations.

Representing the DNPs in group form also simplifies the calculation of reactor kinetics, as without the groups the hundreds of DNPs would each require their own equations. Tracking all the DNPs becomes more computationally expensive when tracking all their concentrations over both space and time.

This section provides information on what the kinetics and delayed neutron group parameters physically represent and how they are calculated. The DNP group parameters directly affect many of the kinetics parameters, and both sets of data are used in kinetics and dynamics problems. Table 2.10 presents each of the relevant terms for this section.

Table 2.10: DNP group parameters and kinetics parameters along with their definitions.

| Parameter         | Description                                              | Units      |
|-------------------|----------------------------------------------------------|------------|
| $\lambda_k$       | The decay constant for DNP group $k$                     | $[s^{-1}]$ |
| $\lambda_i$       | The decay constant for DNP $i$                           | $[s^{-1}]$ |
| $\bar{\nu}_{d,k}$ | The delayed neutron yield for DNP group $k$              | [-]        |
| $\bar{\nu}_{d,i}$ | The delayed neutron yield for DNP $i$                    | [-]        |
| $\chi_k$          | The spectra of DNP group $k$                             | [-]        |
| $\chi_i$          | The spectra of DNP $i$                                   | [-]        |
| $P_{n,i}$         | The delayed neutron emission probability for nuclide $i$ | [#]        |
| $\bar{\tau}$      | The mean half-life                                       | [s]        |
| $\bar{\nu}_d$     | The total delayed neutron yield                          | [-]        |
| $\beta_k$         | The delayed neutron fraction of DNP group $k$            | [-]        |
| $\beta_i$         | The delayed neutron fraction of DNP $i$                  | [-]        |
| $\beta$           | The total delayed neutron fraction                       | [-]        |
| $\beta_{eff}$     | The effective delayed neutron fraction                   | [-]        |
| $\Lambda$         | The mean neutron generation time                         | [s]        |
| $\Lambda_{eff}$   | The effective mean neutron generation time               | [s]        |

The DNP groups come with associated parameters, such as the group decay constants,  $\lambda_k$ , the group yields,  $\bar{\nu}_{d,k}$ , and the group spectra  $\chi_k$  for DNP group  $k$ . The group yields are also known as group abundances, and can be found represented using  $a_k$ ,  $b_k$ ,  $\nu_{d,k}$ , and  $\alpha_k$  in the literature [95, 203, 134, 23]. These group yields can be converted to group delayed neutron fractions,  $\beta_k$ , by dividing the group yields by the average number of neutrons emitted per fission event, as shown in Equation (2.4) as:

$$\beta_k = \frac{\bar{\nu}_{d,k}}{\bar{\nu}}. \quad (2.4)$$

The group yields represent the number of delayed neutrons produced per fission in group  $k$ , and when summed yield the total delayed neutron yield per fission. With this data, the other terms can be calculated, such as the total delayed neutron yield,  $\bar{\nu}_d$ , and delayed neutron fraction  $\beta$ .

Although the groups are useful, there are times when individual DNP data are needed. For example, in 1986, Perry et al. demonstrated that kinetics calculations yield different results when using individual nuclides compared to the six-group parameters [141]. In 1989, Brady and England also used the individual nuclides to compare with group data [23], and updated the six-group parameters based on the explicit DNPs. The updates to the six-group parameters enabled greater consistency between experimental results and simulations.

Moving from the group parameters to the kinetics parameters, there is a convenient analogy which can be used to better understand some of the kinetics parameters. The effective multiplication factor,  $k_{eff}$ , is an important term in reactor physics. This term represents the scaling of the number of neutrons over each generation in a finite

reactor system. The infinite multiplication factor,  $k_\infty$ , represents the multiplication without considering leakage terms. This is analogous to the delayed neutron fraction,  $\beta$ , for which the effective form requires information on the specific reactor geometry.

The delayed neutron fraction,  $\beta$ , represents the ratio of delayed neutron production over all neutron production [123]. Related to the delayed neutron fraction is the effective delayed neutron fraction,  $\beta_{eff}$ . This term is more sophisticated, as rather than being a physical property of a fissile nuclide, it depends on the composition and geometry of the reactor. For MSRs, it also accounts for the movement of the DNP in different parts of the reactor.

A straightforward way to interpret the effective delayed neutron fraction is that it is the delayed neutron fraction corrected for the actual composition and geometry of the reactor. More specifically, the effective delayed neutron fraction represents the ratio between the delayed-neutron induced fission to the total number of fission events weighted by the spectrum and adjoint neutron flux [158]. Alternatively, it can be thought of as the ratio of the spectrum weighted and adjoint weighted delayed neutron production over all spectrum weighted and adjoint weighted neutron production [123]. For MSRs, the movement of the fuel salt also affects the effective delayed neutron fraction, as the integrated adjoint neutron flux is reduced due to movement of the DNPs out of the core and into less neutronically-important regions.

The adjoint flux,  $\phi^\dagger$ , is important for calculating the effective form of many kinetics parameters. The adjoint flux is not a physical quantity, but rather an importance, or weighting, function, describing the importance of neutrons in different regions of the reactor. For example, areas further from the center of the core are less important than those closer to the center, which is reflected by a decrease in the effective delayed neutron fraction in MSRs as DNPs move away from the center of the core due to fuel salt advection. The adjoint flux can be calculated by taking the adjoint of the Boltzmann transport equation, or by using the iterated fission probability (IFP) method, which provides an importance term proportional to the adjoint flux [190, 128, 194].

The mean neutron generation time,  $\Lambda$ , is the mean time between generation of a neutron and fission induced by that neutron [158]. This term also has an effective form,  $\Lambda_{eff}$ , which is the adjoint-weighted form, similar to  $\beta_{eff}$ . The effective mean neutron generation time can vary by over 200% from the mean neutron generation time [197]. The effective delayed neutron fraction generally does not differ from the delayed neutron fraction as significantly, though in an MSR the fuel circulation can cause the effective delayed neutron fraction to reduce by approximately 50% [7].

### Calculation of DNP Group Parameters

The DNP group parameters,  $\tau_k$  and  $\bar{v}_{d,k}$ , are important for simulating transients. The parameters are formulated based on the fitting of the delayed neutron count rate, as discussed in Section 2.1.1. Equation (2.3) shows the exact formulation for the delayed neutron count rate as a function of time measured by a detector for  $I$  individual DNPs for any number of fissile nuclides, recreated here as:

$$\dot{n}_d(t) = \epsilon \sum_{i=1}^I P_{n,i} \lambda_i N_i(t).$$

Equation (2.5) instead gives the delayed neutron count rate measured by the same detector for  $K$  DNP groups from a pulse irradiation as:

$$\dot{n}_d(t) \approx \epsilon F_s \sum_{k=1}^K \bar{v}_{d,k} \lambda_k e^{-\lambda_k t}. \quad (2.5)$$

For an infinite, or saturation, irradiation, Equation (2.6) is used rather than Equation (2.5), given as [24]:

$$\dot{n}_d(t) \approx \epsilon \dot{F}_s \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t}. \quad (2.6)$$

Equation (2.7) shows the intermediate case, where the irradiation time,  $T$ , is neither infinite nor a pulse [204]. This equation gives the delayed neutron count rate as:

$$\dot{n}_d(t, T) \approx \epsilon \dot{F}_s \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t} (1 - e^{-\lambda_k T}). \quad (2.7)$$

The terms in Equations (2.5), (2.6), and (2.7) are defined as follows:

- $K$  = the number of DNP groups [#]
- $\lambda_k$  = the decay constant of DNP group  $k$  [ $s^{-1}$ ]
- $F_s$  = the total number of fission events in the pulse irradiation [#]
- $\dot{F}_s$  = the fission rate for a saturation irradiation [ $s^{-1}$ ]
- $\bar{v}_{d,k}$  = the group yield for group  $k$  [-] .

These equations are specifically for a single fissile nuclide, and the derivation can be found in the work of Wilson and England [204].

The pulse irradiation approach gives better statistics on the short-lived DNPs, while the longer-lived DNPs are better represented by saturation irradiation. Brady discussed this, and used both pulse and saturation irradiation to form groups [24]. Specifically, Brady used the four long-lived groups from the saturation irradiation and the four short-lived groups from the pulse irradiation renormalized at the second DNP group, the same method discussed by Keepin et al. [95]. The half-lives for the first two groups and the ratio of the first two group yields come from the saturation irradiation, while the other values come from the pulse irradiation data.

Once the delayed neutron count rate is acquired, an over-determined non-linear least squares approach is used to calculate the group yields and decays. For example, Brady used STEPIT, an algorithm for finding local minima of a continuous function, to handle the least squares solve with both pulse and infinite irradiation experimental data [24, 35]. Waldo et al. also solved the least squares problem, though the specific approach used is not specified [200]. Using Equation (2.5), (2.6), or (2.7), the general form of the solve is given as:

$$A\vec{x} \approx \vec{b}. \quad (2.8)$$

Since the system is overdetermined, no exact solution exists; only a residual minimization, as shown in Equation (2.12), is possible.

Based on the irradiation type used, the coefficient matrix will change, as shown in Equations (2.5), (2.6), and (2.7). Equation (2.9) gives the coefficient matrix for a pulse irradiation as:

$$A = \epsilon F_s \begin{bmatrix} \lambda_1 e^{-\lambda_1 t_0} & \lambda_2 e^{-\lambda_2 t_0} & \dots & \lambda_K e^{-\lambda_K t_0} \\ \lambda_1 e^{-\lambda_1 t_1} & \lambda_2 e^{-\lambda_2 t_1} & \dots & \lambda_K e^{-\lambda_K t_1} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_1 e^{-\lambda_1 t_m} & \lambda_2 e^{-\lambda_2 t_m} & \dots & \lambda_K e^{-\lambda_K t_m} \end{bmatrix}. \quad (2.9)$$

Note that the coefficient matrix changes for a saturation irradiation, removing the  $\lambda_i$  term and changing the fission term from net fissions to the fission rate.

Within this least-squares equation, the solution vector contains only the group yields, given in Equation (2.10) as:

$$\vec{x} = \begin{bmatrix} \bar{v}_{d,1} \\ \bar{v}_{d,2} \\ \vdots \\ \bar{v}_{d,K} \end{bmatrix}. \quad (2.10)$$

The target vector contains the delayed neutron count rate data, shown in Equation (2.11) as:

$$\vec{b} = \begin{bmatrix} \dot{n}_d(t_0) \\ \dot{n}_d(t_1) \\ \vdots \\ \dot{n}_d(t_m) \end{bmatrix}. \quad (2.11)$$

The non-linear least squares solve enables simultaneous solution of both the  $\lambda_k$  and  $\bar{v}_{d,k}$  values. Using least squares with a non-relative residual would lead to early time-step count rates dominating due to their larger magnitudes. One way to resolve this is to take the natural logarithm of the target vector  $\vec{b}$ , such that the count rate evolution is linear in time. Another way to resolve this is to use minimize a relative residual, given by Equation (2.12) as:

$$r = \min_x \left\| \frac{\vec{b} - A\vec{x}}{\vec{b}} \right\|_2^2, \quad (2.12)$$

where the Euclidean norm,  $\|\cdot\|_2$ , is defined in Equation 2.13 as:

$$\|\vec{x}\|_2 = \sqrt{x_1^2 + \dots + x_n^2}. \quad (2.13)$$

Another approach that accounts for the uncertainty in the target vector  $b$ , by minimizing  $\chi^2$ , is given in Equation (2.14) [17]:

$$\chi^2 = \sum_i \left( \frac{b_i - (Ax)_i}{\sigma_i} \right)^2, \quad (2.14)$$

where

$$\begin{aligned} \chi^2 &= \text{the chi-squared statistic [-]} \\ (Ax)_i &= \text{the DNP group delayed neutron count rate [\# \cdot s^{-1}]} \\ b_i &= \text{the } i^{th} \text{ delayed neutron count rate [\# \cdot s^{-1}]} \\ \sigma_i &= \text{the standard deviation in the } i^{th} \text{ delayed neutron count rate [\# \cdot s^{-1}]} \end{aligned} \quad (2.15)$$

By accounting for the uncertainty, this gives a more informed result that accounts for uncertainties in the data.

By solving this non-linear system of equations, the group decay constants,  $\lambda_k$ , and yields,  $\bar{v}_{d,k}$ , can be generated. Once this group data is generated, and if there are multiple fissioning nuclides,  $F$ , then Equation (2.16) can be used

to determine the  $K$  group decay constants using [46] as:

$$\lambda_k = \frac{\sum_F \lambda_k^{(F)} \beta_{k,eff}^{(F)}}{\sum_F \beta_{k,eff}^{(F)}}, \quad (2.16)$$

where

$$\begin{aligned} \lambda_k^{(F)} &= \text{the decay constant of DNP group } k \text{ from fission of nuclide } F [s^{-1}] \\ \beta_{k,eff}^{(F)} &= \text{the effective delayed neutron fraction of DNP group } k \text{ from fission of nuclide } F [-]. \end{aligned}$$

This approach is effectively a fission weighting, which can be written in a more general manner. The DNP group parameters are generated for a given fuel composition, weighted by the volumetric fission rate fraction of each fissile nuclide,  $F$ , similar to Equation (2.16). The fission-weighting term is given as:

$$w_F = \frac{\int \Sigma_f^{(F)}(\vec{r}, E) \phi(\vec{r}, E) dE d\vec{r}}{\sum_{F'} \int \Sigma_f^{(F')}(\vec{r}, E) \phi(\vec{r}, E) dE d\vec{r}}, \quad (2.17)$$

the delayed neutron yield of group  $k$  is given as:

$$\bar{v}_{d,k}^{(FW)} = \sum_F \bar{v}_{d,k}^{(F)} w_F, \quad (2.18)$$

and the decay constant of group  $k$  is given as:

$$\lambda_k^{(FW)} = \sum_F \lambda_k^{(F)} w_F, \quad (2.19)$$

where

$$\begin{aligned} \bar{v}_{d,k}^{(F)} &= \text{the DNP group } k \text{ delayed neutron yield from nuclide } F [-] \\ \lambda_k^{(F)} &= \text{the DNP group } k \text{ decay constant from nuclide } F [s^{-1}] \\ \phi &= \text{the neutron flux } [cm^{-2} \cdot s^{-1}] \\ \bar{v}_{d,k}^{(FW)} &= \text{the fission-weighted DNP group } k \text{ delayed neutron yield } [-] \\ \lambda_k^{(FW)} &= \text{the fission-weighted DNP group } k \text{ decay constant } [s^{-1}] \\ \Sigma_f^{(F)} &= \text{the macroscopic fission cross section for nuclide } F [cm^{-1}]. \end{aligned}$$

The fission-weighted values can also be multiplied by an importance term to account for the difference between the delayed and prompt neutron fission spectra [149]. This is used for kinetics simulations that do not explicitly model DNPs in the transport solve, such as in point reactor kinetics (PRKs). Equation (2.18) becomes:

$$\bar{v}_{d,k}^{(AW)} = \bar{v}_{d,k}^{(FW)} \frac{\int \chi_{d,k}(E) \phi^\dagger(E) dE}{\int \chi(E) \phi^\dagger(E) dE}, \quad (2.20)$$

and Equation (2.19) becomes:

$$\lambda_k^{(AW)} = \lambda_k^{(FW)} \beta_{eff} \left( \sum_{k=1}^K \frac{\beta_k}{\lambda_k} \right)^{-1}, \quad (2.21)$$

where

$$\begin{aligned}
\beta_{eff} &= \text{the effective delayed neutron fraction [-]} \\
\beta_k &= \text{the delayed neutron fraction of group } k \text{ [-]} \\
\chi_{d,k} &= \text{the delayed neutron fission spectrum for DNP group } k \text{ [-]} \\
\bar{v}_{d,k}^{(AW)} &= \text{the adjoint-weighted DNP group } k \text{ delayed neutron yield [-]} \\
\lambda_k^{(AW)} &= \text{the adjoint-weighted DNP group } k \text{ decay constant [s}^{-1}\text{]} \\
\chi &= \text{the total fission spectrum [-]} \\
\phi^\dagger &= \text{the adjoint flux [\#} \cdot \text{cm}^{-2} \cdot \text{s}^{-1}\text{]}.
\end{aligned}$$

### Calculation of DNP Group Spectra and Concentrations

The equations thus far have not included the energy spectra, but instead have focused on the delayed neutron count rate. Including the energy in the delayed neutron count rate requires the energy spectra,  $\chi_{d,k}(E)$ , of the relevant group to be included, as shown in Equation (2.22) as:

$$\dot{n}_d(E, t) \approx \epsilon \tilde{F}_s \sum_{k=1}^K \chi_{d,k}(E) A_k e^{-\lambda_k t}, \quad (2.22)$$

where

$$\begin{aligned}
\tilde{F}_s &= \text{the generic fission term [\# or } \# \cdot \text{s}^{-1}\text{]} \\
A_k &= \text{the generic matrix term [\#} \cdot \text{s}^{-1} \text{ or } \#] \\
\chi_{d,k}(E) &= \text{the delayed neutron energy spectrum of DNP group } k \text{ [-]}.
\end{aligned}$$

This is the general form of the saturation and pulse irradiation equations, where  $A_k$  represents  $\bar{v}_{d,k}\lambda_k$  for a pulse irradiation or  $\bar{v}_{d,k}$  for saturation irradiation.  $\tilde{F}_s$  is the total number of fissions for a pulse irradiation or the fission rate for saturation irradiation.

Another consideration in calculating group parameters is calculating group concentrations from individual nuclide data. This is important for situations where reducing the problem from several hundred DNPs to a few DNP groups reduces computational cost. A specific use case for this is explicitly modeling all DNPs spatially and then using those spatially resolved concentrations with an existing six-group structure. In such a case, the DNPs must be combined in such a manner to preserve their contributions to each DNP group. The reverse is also true, where a given six-group spatially resolved concentration can be separated into individual DNPs spatially resolved concentrations.

The methods for calculating the DNP group spectra and the group concentrations directly overlap. Two different grouping methods are considered for calculation of group spectra and group concentrations: based on half-lives [56, 160] and based on fractional contribution to adjacent groups [24].

The grouping based on half-lives requires comparing each nuclide's half-life to a constant half-life bin structure formulated from analysis of individual DNPs. The groupings based on half-lives are given in the top two sets in Table 2.11. The bottom set of half-lives is from ENDF/B-VIII.0 and is used with the fractional contribution approach [29].

Table 2.11: Half-lives of DNP groups from various sources.

| Source                     | $\tau_1$ [s]   | $\tau_2$ [s]        | $\tau_3$ [s]       | $\tau_4$ [s]        | $\tau_5$ [s]          | $\tau_6$ [s]    |
|----------------------------|----------------|---------------------|--------------------|---------------------|-----------------------|-----------------|
| England et al. [56]        | $\tau > 31$    | $31 \geq \tau > 9$  | $9 \geq \tau > 3$  | $3 \geq \tau > 0.8$ | $0.8 \geq \tau > 0.3$ | $0.3 \geq \tau$ |
| Rudstam [160] <sup>*</sup> | $\tau > 30$    | $30 \geq \tau > 10$ | $10 \geq \tau > 4$ | $4 \geq \tau > 1.4$ | $1.4 \geq \tau > 0.4$ | $0.4 \geq \tau$ |
| ENDF/B-VI.8 [159]          | $\tau = 51.98$ | $\tau = 21.17$      | $\tau = 5.74$      | $\tau = 2.29$       | $\tau = 0.82$         | $\tau = 0.24$   |

<sup>\*</sup> Rudstam uses  $\tau_1 = 55.6$ , but to ensure all DNPs are included, this has been modified.

England and Wilson chose their half-life bins based on the half-lives given in ENDF/B-V, while specifying that of the reasonable choices for bounding values, “No one was completely satisfactory” [56]. Rudstam selected half-life bins for six groups, although it is clarified that “...the grouping is somewhat arbitrary” [160]. The half-life grouping method thus results in a series of comparisons; if a nuclide is within a given group’s bounds, the concentration of that nuclide is added to that group.

For the fractional grouping, the half-life selection is less arbitrary and is instead based on the DNP group parameters. The ENDF/B-VI.8 data comes from the  $^{235}\text{U}$  half-lives from the  $(n, \nu_d)$  reaction [159, 23, 55, 25]. These delayed neutron data have remained unchanged through ENDF/B-VIII.0 [29, 159]<sup>1</sup>.

The fractional grouping is more complex and assigns nuclides to groups in a more rigorous manner. The method uses two groups,  $k$  and  $k + 1$ , for each nuclide  $i$ . These groups are determined via Equation (2.23) as:

$$\lambda_k < \lambda_i < \lambda_{k+1}. \quad (2.23)$$

Nuclides which don’t fit in these bounds will have the entire fractional contribution go into the closest group. Otherwise, the nuclide will have some fraction of its concentration or spectra go to each DNP group. The fractional contribution, or weight, of nuclide  $i$  to each group is represented with  $f_{i,k}$  and  $f_{i,k+1}$ . The total fraction must be equal to 1 as shown in Equation (2.24) as:

$$f_{i,k} + f_{i,k+1} = 1. \quad (2.24)$$

To solve for the fractions, the following least-squares minimization problem is used [24]:

$$\min_{f_{i,k}, f_{i,k+1}} \int_0^\infty \left( \lambda_i e^{-\lambda_i t} - f_{i,k} \lambda_k e^{-\lambda_k t} - f_{i,k+1} \lambda_{k+1} e^{-\lambda_{k+1} t} \right)^2 dt. \quad (2.25)$$

Solving this equation for where the partial derivatives are zero for each fraction,  $f_{i,k}$  and  $f_{i,k+1}$ , combined with Equation (2.24), yields the overdetermined system of equations as:

$$A = \begin{bmatrix} \lambda_k & \frac{2\lambda_k \lambda_{k+1}}{\lambda_k + \lambda_{k+1}} \\ \frac{2\lambda_k \lambda_{k+1}}{\lambda_k + \lambda_{k+1}} & \lambda_{k+1} \\ 1 & 1 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} \frac{2\lambda_k \lambda_i}{\lambda_k + \lambda_i} \\ \frac{2\lambda_i \lambda_{k+1}}{\lambda_i + \lambda_{k+1}} \\ 1 \end{bmatrix}, \quad \vec{x} = \begin{bmatrix} f_{i,k} \\ f_{i,k+1} \end{bmatrix}. \quad (2.26)$$

Because the number of rows is greater than the number of columns in the matrix, the left inverse is used. This

<sup>1</sup>The ENDF/B-VIII.1 data is also the same as ENDF/B-VI.8, but it is currently unpublished.

system of equations is then solved as shown in Equation (2.27) as:

$$\vec{x} = (A^T A)^{-1} A^T \vec{b}. \quad (2.27)$$

With the fractional contribution terms,  $\vec{x}$ , the concentration of a given nuclide can then be multiplied by each fraction to show how much it contributes to that group. By summing the contributions of each nuclide, the group concentrations can be calculated, and conversely, the nuclide contributions can be calculated from the group concentrations. Alternatively, the spectra can be calculated by using the same approach but with individual DNP spectra instead of concentrations.

### Calculation of Kinetics Parameters

The other relevant parameters to be calculated are simpler and do not require solving least squares problems. The  $k^{th}$  DNP group delayed neutron fraction is calculated in Equation (2.4), recreated here as:

$$\beta_k = \frac{\bar{v}_{d,k}}{\bar{v}}.$$

A useful term for comparing DNP group parameters is the average half-life [144, 145, 105]. Parish et al. also used the average half-life, but with a non-standard formulation which does not include the natural log of two [135]. Equation (2.28) gives the average half-life as:

$$\bar{\tau} = \ln(2) \sum_k^K \frac{\bar{v}_{d,k}}{\lambda_k}. \quad (2.28)$$

Equation (2.29) shows how the total delayed neutron yield, or average number of delayed neutrons per fission event,  $\bar{v}_d$ , can be calculated [71]:

$$\bar{v}_d = \sum_k^K \bar{v}_{d,k}. \quad (2.29)$$

This yield can also be approximated for an individual nuclide by integrating Equation (2.2) over all time and normalizing to a single fission, where the concentration over time is assumed to be:

$$N_i(t) = N_i(0)e^{-\lambda_i t}. \quad (2.30)$$

This yields Equation (2.31), where the total number of delayed neutrons emitted from a single nuclide  $i$  is given as:

$$\int_0^\infty \dot{n}_{d,i}(t) dt = P_{n,i} N_i(0). \quad (2.31)$$

After dividing by the total number of fission events, Equation (2.32) shows that the total delayed neutron yield can be written as [84, 119]:

$$\bar{v}_d = \sum_{i=1}^I P_{n,i} CFY_i, \quad (2.32)$$

where

$CFY_i$  = the cumulative fission yield of nuclide  $i$  [-].

This also means that the delayed neutron yield of each contributing DNP can be calculated.

The delayed neutron fraction can be written as:

$$\beta = \frac{\bar{\nu}_d}{\bar{\nu}}. \quad (2.33)$$

The mean neutron generation time can be calculated as:

$$\Lambda = \frac{l}{k}, \quad (2.34)$$

where

$$\begin{aligned} l &= \text{the prompt neutron lifetime [s]} \\ k &= \text{the neutron multiplication factor [-]}. \end{aligned} \quad (2.35)$$

This can be calculated using a time-dependent Monte Carlo calculation along with PRK equations [158].

Finally, the effective parameters are introduced: the effective delayed neutron fraction,  $\beta_{eff}$ , and the effective mean neutron generation time,  $\Lambda_{eff}$ . The parameter  $\beta_{eff}$  is the spectrum- and adjoint-weighted ratio of delayed-neutron-induced fissions to the total induced fissions from all neutrons [158]. This is given in Equation (2.36) as:

$$\beta_{eff} = \frac{\int \phi^\dagger(\vec{r}, E, \vec{\Omega}) \chi_d(E) \beta \bar{\nu}(E) \Sigma_f(\vec{r}, E) \phi(\vec{r}, E, \vec{\Omega}) dE d\vec{\Omega} d\vec{r}}{\int \phi^\dagger(\vec{r}, E, \vec{\Omega}) \chi(E) \bar{\nu}(E) \Sigma_f(\vec{r}, E) \phi(\vec{r}, E, \vec{\Omega}) dE d\vec{\Omega} d\vec{r}}, \quad (2.36)$$

where

$$\begin{aligned} \phi^\dagger &= \text{the adjoint neutron flux } [\# \cdot cm^{-2} \cdot s^{-1}] \\ \chi_d &= \text{the delayed neutron energy spectrum [-]} \\ \chi &= \text{the neutron energy spectrum [-]} \\ \beta &= \text{the delayed neutron fraction [-]} \\ \bar{\nu} &= \text{the average number of neutrons released per fission [-]} \\ \Sigma_f &= \text{the macroscopic fission cross section } [cm^{-1}] \\ \phi &= \text{the neutron flux } [\# \cdot cm^{-2} \cdot s^{-1}]. \end{aligned}$$

$\Lambda_{eff}$  can also be calculated explicitly by its definition, which is the time between the birth of a neutron and the subsequent absorption that induces fission. This is shown in Equation (2.37) as [197]:

$$\Lambda_{eff} = \frac{\int \phi_0^\dagger \frac{1}{\bar{\nu}} \phi_0 dE d\vec{\Omega} d\vec{r}}{\int \phi_0^\dagger F_0 \phi_0 dE d\vec{\Omega} d\vec{r}} \quad (2.37)$$

where

$$\begin{aligned} \phi_0^\dagger &= \text{the critical adjoint neutron flux } [\# \cdot cm^{-2} \cdot s^{-1}] \\ \phi_0 &= \text{the critical neutron flux } [\# \cdot cm^{-2} \cdot s^{-1}] \\ \bar{\nu} &= \text{the average neutron speed } [cm \cdot s^{-1}] \\ F_0 &= \text{the total fission operator } [cm^{-1}]. \end{aligned}$$

The fission operator is used because only the absorptions that induce fission are of interest.

There are other methods used to calculate  $\beta_{eff}$  and  $\Lambda_{eff}$  in the literature [197]. These include the direct integration, or adjoint weighted, method [96]; the eigenvalue, or perturbation, method [77]; and the iterated fission probability method [20].  $\Lambda_{eff}$  can also be calculated using the pulsed method [158]. For  $\beta_{eff}$ , there are also the prompt, Meulekamp, and Spriggs methods which can be used [158, 123]. Aufiero et al. conducted a comparison of the Meulekamp, iterated fission probability, and prompt methods. Their results showed that the iterated fission probability and prompt methods both yield similar results while the Meulekamp method is 200 pcm higher for the Molten Salt Fast Reactor (MSFR) [7].

To account for fuel-salt movement in the effective parameters, there is typically another step required in which the static parameters, calculated with no movement, are multiplied by another term. Aufiero et al. used an analytical approach with importance-weighted spatial decay probabilities to account for the movement of the fuel salt, a deterministic approach which includes computational fluid dynamics (CFD) techniques for precursor transport, and a Monte Carlo approach which adjusts the position according to the fuel velocity field and decay constant when the emission of a delayed neutron is sampled [7]. Another approach is to account for the decay of the DNPs outside the core, as shown by Nestor [130]. Generally, an advection term is introduced to account for the circulation of the DNPs, after which the spatial distribution is incorporated into the effective parameter calculation.

### 2.1.3 Kinetics Methods

There are several different methods that have been used for analyzing transients in MSRs. These methods include multi-group neutron diffusion, PRK schemes, and quasi-statics [205]. These methods are generally coupled, either tightly or loosely, to a thermal-hydraulics solver. In general, although these methods have varying accuracy and costs, none of them include MSR specific phenomena when generating the DNP group parameters.

#### Multi-Group Diffusion

Multi-group diffusion kinetics methods are among the most accurate, and computationally expensive, approaches applied to transient problems, surpassed only by transport methods. Hurwitz and Ehrlich provided a standard formulation of multi-group diffusion written in 1953 [83].

Examples of software that used fluid-flow-velocity field coupled multi-group include GenFOAM and Griffin/Pronghorn coupling with the Multiphysics Object-Oriented Simulation Environment (MOOSE) [61, 188]. Other examples can be found in the review by Wooten et al. [100, 62, 99] and elsewhere in the literature [209, 207, 131, 137]. These studies typically used Picard iteration with multi-group neutron diffusion coupled with thermal-hydraulics to accurately capture the multi-physics effects of the reactor. However, this accuracy also comes with a higher computational cost than simpler methods. The general form of the coupling was given by Wooten and Powers for the neutron flux [205]:

$$\begin{aligned}
\frac{1}{\bar{v}_g} \frac{\partial \phi_g}{\partial t} = & \nabla \cdot D_g \nabla \phi_g + \sum_{g' \neq g}^G \Sigma_{s,g' \rightarrow g} \phi_{g'} \\
& + (1 - \beta) \chi_{p,g} \sum_{g'=1}^G (\nu \Sigma_f)_{g'} \phi_{g'} \\
& + \sum_{k=1}^K \chi_{d,g,k} \lambda_k C_k - \Sigma_{a,g} \phi_g - \sum_{g' \neq g}^G \Sigma_{s,g \rightarrow g'} \phi_g,
\end{aligned} \tag{2.38}$$

and for the group  $k$  precursor concentration,

$$\frac{\partial C_k}{\partial t} = \beta_k \sum_{g=1}^G (\bar{\nu}_g \Sigma_f)_g \phi_g - \lambda_k C_k - \nabla \cdot (C_k \vec{u}) + \nabla \cdot \frac{\nu_T}{Sc_T} \nabla C_k, \quad (2.39)$$

where

- $\bar{\nu}_g$  = the average number of neutrons produced by fission of group  $g$  [-]
- $\phi_g$  = the scalar neutron flux of group  $g$  [ $\# \cdot cm^{-2} \cdot s^{-1}$ ]
- $D_g$  = the group  $g$  diffusion term [ $cm$ ]
- $G$  = the total number of energy groups [#]
- $\Sigma_{s,g' \rightarrow g}$  = the macroscopic scattering cross section from group  $g'$  to group  $g$  [ $cm^{-1}$ ]
- $\chi_{p,g}$  = the fraction of prompt neutrons born in group  $g$  [-]
- $\chi_{d,g,k}$  = the fraction of delayed neutrons from precursor group  $k$  born in group  $g$  [-]
- $C_k$  = the concentration of DNPs in group  $k$  [ $\# \cdot cm^{-3}$ ]
- $\vec{u}$  = the velocity profile [ $cm \cdot s^{-1}$ ]
- $\Sigma_{a,g}$  = the macroscopic absorption cross section for group  $g$  [ $cm^{-1}$ ]
- $\nu_T$  = the eddy viscosity [ $cm^2 \cdot s^{-1}$ ]
- $Sc_T$  = the turbulent Schmidt number [-].

These equations solve neutron transport using multi-group neutron diffusion and solve for the DNP precursor concentrations over space and time. The precursor concentrations account for the flow field via the velocity vector and turbulent mixing using the Schmidt number and eddy viscosity. The ratio of the eddy viscosity to the turbulent Schmidt number represents diffusivity via turbulent mixing. Although the turbulent diffusion can be significant, the non-turbulent diffusion in molten salts is small relative to other spatial effects; it is not included in this equation.

### Modified Point Kinetics

PRK equations are a simpler approach to modeling a transient, with the basic form given as:

$$\frac{dn(t)}{dt} = \frac{\rho(t) - \beta_{eff}}{\Lambda} n(t) + \sum_{k=1}^K \lambda_k C_k(t), \quad (2.40)$$

and,

$$\frac{dC_k(t)}{dt} = \frac{\beta_{eff,k}}{\Lambda} n(t) - \lambda_k C_k(t), \quad (2.41)$$

where

- $n$  = the relative neutron density [#]
  - $\rho$  = the reactivity [-]
  - $\beta_{eff}$  = the effective delayed neutron fraction [-].
- (2.42)

The derivation of the PRK equations can be found in the work by Henry, with a fluid-fueled derivation in the work by Lapenta et al. [77, 103]. Modified point kinetics are used for MSRs, as standard PRK equations do not properly handle the movement of the fuel salt.

PRK generally struggle to capture spatial effects, such as flow variation, which is important for MSRs. PRK do, however, give a correct prediction for general evolution trends [205]. Wooten et al. provided four different publications which use modified point kinetics in MSR analysis [72, 122, 179, 208].

The work by Guo et al. analyzed unprotected loss of heat sink (ULOHS) and unprotected loss of flow (ULOF) transients in the Molten Salt Actinide Recycler and Transmuter reactor (MOSART). The PRK equations are modified into the delayed point reactor kinetics equations, which model precursor concentrations in-core and ex-core, with the primary set of equations given as:

$$\frac{dn(t)}{dt} = \frac{\rho - \tilde{\beta}_{eff}}{\Lambda} n(t) + \sum_{k=1}^K \lambda_k C_{k,c}(t), \quad (2.43)$$

$$\begin{aligned} \frac{dC_{k,c}(t)}{dt} = & \frac{\beta_k}{\Lambda} n(t) - \left( \lambda_k + \frac{1}{\tau_c} \right) C_{k,c}(t) \\ & + \frac{1}{\tau_l} \left( \frac{V_l}{V_c} \right) C_{k,l}(t - \tau_l), \end{aligned} \quad (2.44)$$

$$\frac{dC_{k,l}(t)}{dt} = \frac{1}{\tau_c} \left( \frac{V_c}{V_l} \right) C_{k,c}(t) - \left( \lambda_k + \frac{1}{\tau_l} \right) C_{k,l}(t), \quad (2.45)$$

where

$V_l$  = the volume of the loop [ $cm^3$ ]

$V_c$  = the volume of the core [ $cm^3$ ]

$C_{k,l}$  = the concentration of DNPs in group  $k$  in the loop [ $\# \cdot cm^{-3}$ ]

$C_{k,c}$  = the concentration of DNPs in group  $k$  in the core [ $\# \cdot cm^{-3}$ ].

These equations, shown in Equations (2.43) through (2.45) (recreated from [72, 179]), are used in standard software, such as in the Reactor Excursion and Leak Analysis Program (RELAP). Shi et al. simulated a load demand change, primary flow transient, secondary flow transient, and reactivity transient in the Molten Salt Breeder Reactor (MSBR) [179]. The modified point kinetics approach enables modeling of ex-core decay of DNPs, capturing effects during power operation and loss-of-flow conditions. However, it does not accurately capture all oscillatory effects, such as power oscillations from a flow-induced transient [205].

Intermediate terms used in Equations (2.43) through (2.45) are shown in Equations (2.46) through (2.49) as:

$$\tilde{\beta}_{eff} = \beta_{static} - \beta_{loss}, \quad (2.46)$$

$$\beta_{loss} = \frac{\Lambda(t)}{n(t)} \sum_{k=1}^K \lambda_k C_{k,l}, \quad (2.47)$$

$$\tau_c = \frac{M_c}{\dot{M}(t)}, \quad (2.48)$$

$$\tau_l = \frac{M_l}{\dot{M}(t)}, \quad (2.49)$$

where

$\beta_{static}$  = the value of  $\beta_{eff}$  without flow [-]

$M_c$  = the mass in the core [kg]

$M_l$  = the mass in the loop [kg]

$\dot{M}(t)$  = the mass flow rate [ $kg \cdot s^{-1}$ ].

A similar method is used by Suzuki and Shimazu for analyzing a reactivity initiated accident in the FUJI-12 reactor [186]. That work used a simpler form of the equations, without dividing the precursor concentrations into core and loop regions.

The work by Merle-Lucotte et al. used the IPK method<sup>2</sup>. This method used a fixed mesh for neutronics and a mobile mesh for fluid properties. Merle-Lucotte et al. found that the PRK method without any modifications was able to match the IPK method for general trends and end-states, but was unable to capture reactivity oscillations properly. The IPK method was able to capture these oscillations, but assumes uniform fission and instantaneous heat transfer [122].

The work by Zanetti et al. extended the TRACE<sup>3</sup> code, a finite volume code used for LWR transients [208]. The DNP concentration, dependent on both space and time, is discretized with an upwind scheme in space and a backward Euler method in time. Adjoint-weighted quantities discretized with the trapezoidal rule are then passed to the PRK equation. The set of DNP concentration and power, or neutron density, equations are resolved using LU factorization.

Another approach to modify the PRK is to alter the kinetics parameters based on a phase-space integration weighted with the adjoint flux [104]. Alternatively, effective flow-inclusive group yields can be calculated, as shown by Haubenreich and used in PRK equations [76].

The modified point kinetics approach provides a less computationally expensive approach to treating the DNPs while preserving general behavior. Although vortices and turbulent diffusion may not be physically captured, capturing the recirculation of the DNPs between the in-core and ex-core regions provides a higher level of fidelity than standard PRK equations.

## Quasi-Statics

Quasi-static methods seek to maintain the accuracy of multi-group diffusion approaches while also having the cost savings of PRK. The general approach of these methods is to update PRK parameters with occasional multi-group diffusion simulations to capture changes in flux and DNP spatial distributions [205]. The general formulation used for quasi-static methods is very similar among the various publications that use it [52, 188]. A complete derivation of the quasi-static method can be found in the work by Henry, who discusses the shape and amplitude flux functions, and the work by Devooght, who discussed the generalized quasi-static method [77, 49]. There are also other studies which discuss the quasi-static method in detail [11].

Wooten et al. reviewed several publications which have used quasi-statics [205]. One of these, from 2005 by Rineiski et al., documented the SIMMER code, which coupled neutronics, thermal-hydraulics, and structure codes [152]. The neutron flux was constrained in that work by a term containing the phase-space integration and the

<sup>2</sup>The "IPK method" may stand for the "Improved Point-Kinetic method," though this is not explicitly stated [122]

<sup>3</sup>TRACE stands for the "TRAC/RELAP Advanced Computational Engine", where TRAC stands for the "Transient Reactor Analysis Code" and RELAP stands for the "Reactor Excursion and Leak Analysis Program" [210, 10].

weighting function used. The neutron flux is treated as:

$$\phi(\vec{x}, t) = \phi_A(t)\psi(\vec{x}, t), \quad (2.50)$$

where

$$\begin{aligned} \phi &= \text{the neutron flux } [\# \cdot \text{cm}^{-2} \cdot \text{s}^{-1}] \\ \phi_A &= \text{the neutron flux amplitude function } [\#] \\ \psi(\vec{x}, t) &= \text{the neutron flux shape function } [\text{cm}^{-2} \cdot \text{s}^{-1}]. \end{aligned}$$

The neutron flux shape function can be approximated as time-independent, but Rineiski et al. did not use this approximation.

Rineiski et al. solved "shape" steps to handle the neutronics and DNP spatial concentrations. The equations used in the shape steps are recreated here in Equations (2.51) and (2.52). These equations were solved by time-discretizing Equation (2.51) into a set of steady-state-like equations, which are then treated using the DANTSYS module of SIMMER, which provides diffusion accelerated transport [3]. These primary equations are given as:

$$\begin{aligned} \frac{1}{\bar{v}} \frac{\partial \psi(\vec{x}, t)}{\partial t} + \frac{1}{\bar{v}\phi_A} \frac{d\phi_A}{dt} \psi(\vec{x}, t) + M(t)\psi(\vec{x}, t) = \\ (1 - \beta)\chi_p F(t)\psi(\vec{x}, t) + \frac{1}{\phi_A} \chi_d \sum_{k=1}^K \lambda_k C_k(\vec{x}, t) + \frac{1}{\phi_A} Q(\vec{x}, t), \end{aligned} \quad (2.51)$$

$$\frac{\partial C_k(\vec{x}, t)}{\partial t} = \phi_A \beta_k F(t)\psi(\vec{x}, t) - \lambda_k C_k(\vec{x}, t) - \nabla(\vec{u}(\vec{x}, t) C_k(\vec{x}, t)). \quad (2.52)$$

Within these shape steps are also sub-steps referred to as "reactivity" steps. At each of these reactivity steps, the PRK parameters and flux amplitude are calculated. The equations used in the reactivity steps are recreated here as:

$$\frac{d\phi_A}{dt} = \left( \frac{\rho - \beta_{eff}}{\Lambda} \right) \phi_A + q_{eff} + \sum_{k=1}^K \lambda_k (c_{s,k} + c_{m,k}), \quad (2.53)$$

$$\frac{dc_{s,k}}{dt} = \frac{\beta_{eff}}{\Lambda} \phi_A - \lambda_k C_{s,k}, \quad (2.54)$$

where,

$$\rho = \frac{\langle W(\vec{x}), (\chi F(t) - M(t)) \psi(\vec{x}, t) \rangle}{\langle W(\vec{x}), \chi F(t) \psi(\vec{x}, t) \rangle}, \quad (2.55)$$

$$c_{m,k} = \frac{\langle W(\vec{x}), \chi_d C_{m,k} \rangle}{\Lambda \langle W(\vec{x}), \chi F(t) \psi(\vec{x}, t) \rangle}. \quad (2.56)$$

Intermediate terms used in Equations (2.53) through (2.56) are given as:

$$\frac{\partial C_{m,k}(\vec{x}, t)}{\partial t} = -\nabla(\vec{u}(\vec{x}, t) C_{s,k}(\vec{x}, t)) - \lambda_k C_{m,k}(\vec{x}, t) - \nabla(\vec{u}(\vec{x}, t) C_{m,k}(\vec{x}, t)), \quad (2.57)$$

$$\frac{\partial C_{s,k}(\vec{x}, t)}{\partial t} = \phi_A(t) \beta_k F(t) \psi(\vec{x}, t) - \lambda_k C_{s,k}(\vec{x}, t). \quad (2.58)$$

The  $C_s(\vec{x}, t)$  term is computed at steady state by setting the left-hand side of both Equation (2.51) and (2.52) to zero as well as the fluid velocity to zero at  $t=0$ . This allows Equation (2.57) to be solved. The terms in Equations (2.50)

through (2.58) are defined as follows:

$$\begin{aligned}
M(t) &= \text{the neutron disappearance operator } [cm^{-1}] \\
F(t) &= \text{the neutron production operator } [cm^{-1}] \\
\chi_p &= \text{the prompt neutron energy spectra } [-] \\
C_k(\vec{x}, t) &= \text{the concentration of DNP group } k \text{ } [\# \cdot cm^{-3}] \\
Q(\vec{x}, t) &= \text{the external neutron source } [\# \cdot cm^{-3} \cdot s^{-1}] \\
\vec{u}(\vec{x}, t) &= \text{the DNP velocity } [cm \cdot s^{-1}] \\
C_{s,k}(\vec{x}, t) &= \text{the "standard" precursor concentration of group } k \text{ } [\# \cdot cm^{-3}] \\
C_{m,k}(\vec{x}, t) &= \text{the "movable" precursor concentration of group } k \text{ } [\# \cdot cm^{-3}] \\
\rho &= \text{the reactivity } [-] \\
c_{s,k} &= \text{the PRK "standard" precursor concentration of group } k \text{ } [\#] \\
c_{m,k} &= \text{the PRK "movable" precursor concentration of group } k \text{ } [\#] \\
q_{eff} &= \text{the PRK external neutron source } [\# \cdot s^{-1}] \\
\Lambda &= \text{the neutron generation time } [s] \\
W(\vec{x}) &= \text{the weighting function (the default in SIMMER} \\
&\quad \text{is an adjoint flux computed at steady-state conditions) } [-].
\end{aligned}$$

The quasi-statics approach provides an intermediate approach between modified point kinetics and fluid-flow-velocity field coupled multi-group diffusion. Quasi-static methods use occasional multi-group diffusion with PRK parameters, offering higher accuracy than PRK methods and less cost than full multi-group diffusion.

## 2.2 Molten Salt Reactor Fuel Depletion

Liquid-fueled MSRs are nuclear reactors which have the fissile material dissolved in a circulating molten salt. This fuel salt circulates through the in-core region, where most fissions occur, and through external piping in the ex-core region, where chemical processing occurs, after which it re-enters the in-core region. The salt is both the fuel and coolant in an MSR, where fission heat is generated in the in-core region and removed through the heat exchanger in the ex-core region.

There are many different MSR designs in the literature, as shown in Table 2.12 [37]. These designs feature a circulating salt containing fissile material. In particular, the circulating fuel salt allows for online chemical reprocessing, which causes depletion to vary spatially, leads to movement of DNPs, and requires close coupling of thermal-hydraulics to neutronics for realistic modeling.

Table 2.12: A collection of various MSR designs publicly available in the literature.

| MSR Name                                                          | Source     |
|-------------------------------------------------------------------|------------|
| FUJI series                                                       | [65]       |
| Actinides Molten Salt TransmutER (AMSTER)                         | [198]      |
| MOSART                                                            | [85]       |
| Fast Spectrum Molten Salt Reactor (FS-MSR)                        | [79]       |
| European MSFR                                                     | [121]      |
| Small Mobile Molten Salt Reactor (SM-MSR)                         | [32]       |
| MSBR                                                              | [154]      |
| MSRE                                                              | [155]      |
| Accelerator-Driven Neutron Applications (ADNA) Tier-1 reactor     | [22]       |
| Single fluid and two fluid denatured molten salt reactors (DMSRs) | [53, 2]    |
| Thorium molten salt reactor (TMSR)                                | [133, 120] |

Many studies have investigated online chemical reprocessing of MSRs, as shown in Table 2.13, which has been recreated from [169]. Table 2.13 shows which MSR depletion simulations have modeled DNPs in the "DNP" column, showing most models do not include DNP movement in depletion simulations. The table is sorted by the "Reprocessing" column, which signifies the method(s) used in the different MSR models, discussed in Section 2.2.1. It is also sorted by the publication year in the "Year" column. The table shows there are many MSR concepts for which depletion was modeled. The study by Valdez et al. investigated two MSR designs, specifically the MSBR and the Molten Salt Demonstration Reactor (MSDR). These models used a variety of spatial refinement levels, shown in the "Model" column, ranging from unit cells to full 3D models.

Table 2.13: Molten salt reactor models analyzed with methods approximating online reprocessing (recreated from [169]).

| Publication                | Reprocessing | Model     | Reactor       | DNP | Year |
|----------------------------|--------------|-----------|---------------|-----|------|
| Valdez et al. [91]         | Continuous   | Unit Cell | Several       | No  | 2020 |
| Zhuang et al. [212]        | Continuous   | Full 3D   | MSFR          | No  | 2020 |
| Xia et al. [206]           | Continuous   | Full 3D   | MSFR          | No  | 2019 |
| Zhou et al. [211]          | Continuous   | Full 3D   | MSBR          | Yes | 2018 |
| Aufiero et al. [8]         | Continuous   | Full 3D   | MSFR          | No  | 2013 |
| Rodriguez-Vieitez [156]    | Continuous   | Unit Cell | ADNA          | No  | 2002 |
| Rykhlevskii [165]          | Batchwise    | Full 3D   | MSBR          | No  | 2019 |
| Betzler et al. [16]        | Batchwise    | Unit Cell | MSBR          | No  | 2017 |
| Ridley and Chvala [151]    | Batchwise    | Full 3D   | Modified DMSR | No  | 2017 |
| Ahmad et al. [2]           | Batchwise    | Unit Cell | DMSR          | No  | 2015 |
| Park et al. [136]          | Batchwise    | Full 3D   | MSBR          | No  | 2015 |
| Sheu et al. [178]          | Batchwise    | Full 3D   | MOSART        | No  | 2013 |
| Nuttin et al. [132]        | Batchwise    | Full 3D   | MSBR          | No  | 2005 |
| Fiorina et al. [60]        | Mixed        | 2D Slice  | MSFR          | No  | 2012 |
| Nagy et al. [127]          | Mixed        | Unit Cell | Modified MSBR | Yes | 2008 |
| Merle-Lucotte et al. [120] | Mixed        | Full 3D   | TMSR          | No  | 2007 |
| Bauman et al. [73]         | Mixed        | 2D Slice  | MSBR          | Yes | 1971 |

Although these studies incorporate the effects of online chemical reprocessing on depletion, they do not explicitly model the spatial effect on depletion. Instead, a simple approximation is used to handle the in-core and ex-core regions while accounting for the entire primary salt volume. The reason why simulations do not include fully resolved spatial resolution in depletion calculations is due to the computational cost.

### 2.2.1 Depletion Simulations

There are many approaches to simulating depletion with different computational costs and levels of accuracy. Building a computational tool to handle depletion requires two components: the depletion matrix and a numerical solver.

Simulating depletion requires specific information, such as decay constants, fission yields, and fission and transmutation cross-section data. The fission and transmutation cross-section data can come from a transport calculation, while the other data can be read from a data library [107]. Simulating depletion entails numerically

solving the Bateman equations; a generalized form of which is given as [107]:

$$\begin{aligned} \frac{dN_j}{dt} = \sum_{i \neq j} [(\gamma_{i \rightarrow j} \sigma_{f,i} \phi + \lambda_{i \rightarrow j} + \sigma_{i \rightarrow j} \phi) N_i] \\ - (\lambda_j + \sigma_j \phi + \lambda_{r,j}) N_j + \dot{N}_{f,j}, \end{aligned} \quad (2.59)$$

where

- $N_j$  = the atomic density of isotope  $j$  [ $\# \cdot cm^{-3}$ ]
- $\gamma_{i \rightarrow j}$  = the fractional fission product yield of  $j$  from the fission of isotope  $i$  [-]
- $\sigma_{f,i}$  = the microscopic fission cross-section of isotope  $i$  [ $cm^2$ ]
- $\phi$  = the spectrum-averaged scalar flux in the fuel region [ $\# \cdot cm^{-2} \cdot s^{-1}$ ]
- $\lambda_{i \rightarrow j}$  = the decay constant of decay  $i \rightarrow j$  [ $s^{-1}$ ]
- $\sigma_{i \rightarrow j}$  = the microscopic transmutation cross-section of reaction  $i \rightarrow j$  [ $cm^2$ ]
- $N_i$  = the atomic density of isotope  $i$  [ $\# \cdot cm^{-3}$ ]
- $\lambda_j$  = the decay constant of isotope  $j$  [ $s^{-1}$ ]
- $\lambda_{r,j}$  = the removal constant for removal of isotope  $j$  [ $s^{-1}$ ]
- $\sigma_j$  = the microscopic total transmutation cross-section of isotope  $j$  [ $cm^2$ ]
- $\dot{N}_{f,j}$  = the feed rate for isotope  $j$  [ $\# \cdot cm^{-3} \cdot s^{-1}$ ].

Equation (2.59) includes a continuous reprocessing method, where the chemical removal rate,  $\lambda_{r,j}$ , is part of the differential equation. There are different ways of modeling this reprocessing, including batchwise methods where material is only removed at certain time intervals [169]. For this work, the continuous reprocessing method is used.

The matrix form of Equation (2.59) is given as:

$$\frac{d\tilde{N}}{dt} = A\tilde{N}, \quad (2.60)$$

where

- $\tilde{N}$  = the vector of material composition [ $\# \cdot cm^{-3}$ ]
- $A$  = the depletion matrix [ $s^{-1}$ ].

The depletion matrix contains the gain and loss terms for each nuclide in the material of interest, which vary continuously as a function of time and depend on the concentrations as well. If there are transfers between two different materials, such as the fuel salt  $N_f$  sending chemically removed nuclides to a waste tank  $N_w$ , then Equation (2.61) gives the concentrations as:

$$\begin{bmatrix} \dot{N}_f \\ \dot{N}_w \end{bmatrix} = \begin{bmatrix} A_f - T_{f \rightarrow w} & 0 \\ T_{f \rightarrow w} & A_w \end{bmatrix} \begin{bmatrix} N_f \\ N_w \end{bmatrix}, \quad (2.61)$$

where

$$\begin{aligned} A_f &= \text{the fuel salt depletion matrix [s}^{-1}\text{]} \\ A_w &= \text{the waste tank depletion matrix [s}^{-1}\text{]} \\ T_{f \rightarrow w} &= \text{the transfer matrix of nuclides from the fuel salt to waste tank [s}^{-1}\text{]}. \end{aligned}$$

There are several approaches to solving these equations, such as the Transmutation Trajectory Analysis (TTA) method and the Chebyshev Rational Approximation Method (CRAM) [108]. Other codes may use external software to compute depletion, such as REM, an in-house code used by Nuttin et al. [132] to implement 4<sup>th</sup> order Runge–Kutta. Codes may also use a power series approximation of the matrix exponential, as discussed by Aufiero et al. [8].

### 2.2.2 Stationary and Circulating Fuel Comparison

For stationary fuel, the Bateman equation as shown previously in Equation (2.59) is sufficient to simulate depletion. For circulating fuel, it is important to understand how fuel depletion can be calculated and how that relates to formulation of DNP group parameters. For instance, if spatial dependence is necessary to accurately model depletion of circulating fuel, then that will make the formation of DNPs in a circulating fuel sample more computationally expensive. For the purposes of a streamlined comparison with circulating fuel, the Bateman equation can be rewritten as:

$$\frac{dN_j}{dt} = S_j(t) - \lambda_{net,j} N_j(t), \quad (2.62)$$

where

$$\begin{aligned} S_j &= \text{the source term for nuclide } j, \\ &\quad \text{containing fission, decay, and transmutation sources [\# \cdot cm}^{-3} \cdot \text{s}^{-1}\text{]} \\ \lambda_{net,j} &= \text{the loss term for nuclide } j, \\ &\quad \text{containing decay and transmutation losses [s}^{-1}\text{]}. \end{aligned}$$

Over the course of a depletion step, the depletion matrix is assumed to be constant, so the source term varies in time only due to the effects of the changing compositions of other nuclides. The loss term,  $\lambda_{net,j}$ , does not vary with time in this equation.

This form of the Bateman equation does not have spatial or energy resolution, which makes it simpler to understand and compare with the circulating fuel version. When solving this equation, the common approach is to have different material groupings, or blocks, in different spatial regions. Each of these material blocks will have different compositions and neutronic properties, such as flux, used for that entire material. This way, the spatial resolution is represented by the material blocks, and the system of ordinary differential equations can be solved rapidly.

Circulating fuel is more complex, as instead of solving a system of ordinary differential equations, one must solve a system of partial differential equations. The simplified static Bateman equation can be modified to include advection and diffusion terms, yielding:

$$\frac{dN_j(\vec{r}, t)}{dt} = S(\vec{r}, t) - \lambda_{net,j} N_j(\vec{r}, t) + \nabla \cdot (D \nabla N_j(\vec{r}, t) - \vec{u}(\vec{r}, t) N_j(\vec{r}, t)), \quad (2.63)$$

where

$$D = \text{the mass diffusivity } [cm^2 \cdot s^{-1}].$$

The depletion matrix is assumed to have no spatial resolution in this equation. Because of this assumption, only the concentrations and material blocks provide spatial resolution to the problem. Spatial refinement can be added to this equation based on the specific approach, such as using a spatially resolved neutron flux.

Since the main challenge in solving this problem is its high computational cost, many studies have proposed methods to mitigate this issue. One common approach, the scaled flux method, is computationally inexpensive [15]. However, this method does not model the in-core and ex-core regions in a physical manner. The scaled flux method scales the flux by the fraction of fuel salt in-core compared to ex-core. This approach does not model the in-core and ex-core regions explicitly, but instead assumes the salt is perfectly mixed in the primary loop. This approximation does not introduce significant error in the spatially averaged concentrations of nuclides unless there are large chemical removal rates or near-stationary flow rates [171]. An example implementation of this approach is given in Appendix B.

Another study has shown how spatial resolution can be investigated using a system of ordinary differential equations to solve the Bateman equation. This study showed an approach that for estimating the steady-state concentration of certain nuclides across a flow loop, but the approach did not account for temporal resolution [174].

Alternatively, a more accurate treatment was recently demonstrated through MOOSE-based coupling of Griffin, a deterministic neutron transport code, with Pronghorn, a thermal fluids code, allowing for spatial and temporal resolution of certain nuclides [189]. Tano et al. presented an approach in which only insoluble, medium-lived, and high-concentration nuclides are tracked with pre-computed spatial distributions, which only need to be updated if the medium-lived isotopes are not in equilibrium after the thermal-hydraulics perturbation.

Spatially and temporally resolved approaches, such as the one used by Tano et al., are more computationally expensive than those that only require solving a system of ordinary differential equations. However, Tano et al. showed that including circulation of the fuel salt in depletion directly reduces the equilibrium concentration of  $^{135}\text{Xe}$  by 44% compared to a spatially unresolved approach without flux scaling [189]. This indicates there may be a need for spatially resolved depletion simulations to get accurate fuel salt compositions.

Tano et al. and Betzler have both described fully modeling advection in depletion as computationally intractable [189, 15]. This is because tracking all nuclides over space and time requires far more computational work than handling a traditional time dependent system of ordinary differential equations (ODEs). This spatially resolved system of partial differential equations (PDEs) requires accuracy in the short time scale flow effects. These small time steps cause multi-year depletion simulations to be exceedingly expensive. The use of small time steps is also important for numerical stability.

The numerical stability of the problem, when using an explicit time-integration scheme, uses the Courant number,  $C$ :

$$C = \frac{u\Delta t}{\Delta z}, \quad (2.64)$$

where

$\Delta t$  = the time step size [s]

$u$  = the velocity of the fuel salt [ $cm \cdot s^{-1}$ ]

$\Delta z$  = the length between spatial nodes [ $cm$ ].

The Courant number must be less than or equal to 1 for the Courant–Friedrichs–Lewy convergence condition in an explicit finite differencing scheme [42]. This equation limits the time step size based on the flow rate and spatial mesh size. Although using an implicit time solver can work around this condition, the overall computational cost remains high due to the number of nuclides, the time step size, spatial mesh resolution, and the overall amount of time simulated in depletion.

For a short depletion simulation, such as a saturation irradiation of a fuel sample, capturing these effects is computationally tractable. Because the depletion lasts only for several minutes instead of several years, the computational cost is orders of magnitude smaller. A high-fidelity, spatially resolved depletion simulation of an MSR sample irradiation experiment is feasible for generating improved DNP group parameters. However, it may be sufficient to use the scaled flux method or other approximations to reduce computational cost.

## **2.3 Safety, Security, and Safeguards**

This section discusses safety, security, and safeguards, and how each provides motivation for this work. Safety and security are considered in the context of potential transient events and their consequences. Security is examined from the perspective of potential insider sabotage, which partially overlaps with safety analysis. A saboteur could initiate an event or combination of events that threaten the general public, serving as a useful distraction for hostile agents. Finally, safeguards analyses are useful for determining if a diversion of special nuclear material (SNM) has occurred, and if so, how much and the category of the diverted material. One diversion fingerprinting technique uses a kinetics response, since diversion of fissile material alters the delayed neutron response to a transient. By improving the DNP group parameters for MSRs, previously undetectable or uncertain diversions will become more readily identifiable. Section 3.5 provides detail on what analyses will be conducted for safety, security, and safeguards.

### **2.3.1 Safety**

The current work is important for safety-related transients, as the MSR specific DNP group parameters will impact the results of these transients. Depending on the properties of the specific reactor, which affect the group parameters, the static DNP group parameter transient simulation may generate overly or insufficiently conservative results. This work will use safety-related transients to determine the effects of the improved DNP group parameters in the MSRE, and under what conditions, if any, the previous DNP groups would predict safe operation while the improved groups would predict unsafe conditions.

An important part of ensuring safe reactor operation is making sure safety-related transient events do not disrupt safe reactor operations. These transients can be caused by postulated initiating events (PIEs), which are defined in the IAEA Safety Standard Series No. NS-R-1 [86]. Of particular importance are PIEs that potentially lead to design basis accidents (DBAs). Although No. SSR-2/1 has superseded No. NS-R-1, the PIEs and DBAs for MSRs have not changed [88]. Although there are not specific events listed for MSRs, guidance is provided on categories of events that are of interest..

These events include equipment failures, such as pumps failing or pipes rupturing; human error, such as improper maintenance or faulty operator actions; external events, such as earthquakes; and combinations of events. For combinations of events, the IAEA specifies that these events should make sense in combination and in the time frame modeled. For example, if a particular event lasts a long time or has a long lasting effect, then that event could possibly overlap with another unrelated event. Alternatively, dependent events could potentially occur at the same time.

Safety-related transients for MSRs include unprotected loss of heat sink (ULOHS), unprotected loss of flow (ULOF), unprotected transient overpower (UTOP), and fuel salt overcooling (FSOC) [110, 43]. A ULOHS transient is when the heat exchanger experiences a failure, possibly caused by a pump failure in the intermediate loop, causing a loss of cooling of the reactor. This causes higher fuel salt temperatures, which can cause adverse chemical effects and increase thermal stress [12]. For the MSFR, Brovchenko et al. showed that slow ULOHS transients that take place on time scales longer than a minute minimize the increase in fuel salt temperature [27]. A ULOF transient is when the flow rate in the primary loop becomes zero, caused by pump failure in the primary loop. The ULOF transient can lead to an increase in the fuel salt temperature, similar to the ULOHS transient. A UTOP transient is a reactivity insertion event, which could be caused by various means, such as removing poisons, adding fissile material, or temperature changes. This transient also causes an increase in the fuel salt temperature, similar to the ULOHS and ULOF transients. An FSOC transient is when the fuel salt temperature decreases, which could be caused by an increased heat removal rate from the heat exchanger [12]. Overcooling the fuel salt can lead to flow blockages from the salt freezing as well as a reactivity insertion from the decrease in fuel salt temperature.

### 2.3.2 Security

This work investigates the formulation of improved DNP group parameters, which impact the reactor behavior during a transient. Sabotage-initiated transients, which are a security concern, may be improperly modeled without the improved DNP group parameters. Specifically, an actor with access to an insider may sabotage reactor operations. These improved DNP group parameters will be used to evaluate under what conditions the previous parameters would indicate a failed sabotage during security-related transients while the improved parameters indicate a successful sabotage that would compromise site security.

The goal of security, or physical protection (PP), is to prevent actors from acquiring information, stealing nuclear material, or committing sabotage [191]. Security is enforced through physical protection systems, which detect, delay, and neutralize the adversarial actor before they can complete their objective [5, 67]. There are several different categories which are important to understanding security: the actor type, their capabilities, their objectives, and their strategies. The main consideration for security in this work is protection of the public by preventing: theft of hazardous radioactive material and sabotage with the intent to damage facilities, disrupt operations, or release radioactive material. The majority of the information presented in this section comes from the Proliferation Resistance and Physical Protection Evaluation Methodology Working Group of the Generation IV International Forum [191].

The actor type involved in PP can be an individual, a group, a large organization, or a non-host state combined with any of the former. These actors may be composed of some combination of insiders and outsiders. Insiders are employees or contract staff who may be sympathetic with or suborned by the hostile actor. Outsiders can include terrorist groups, agents of proliferant states, or lone individuals. Based on the specifics of the actor type, their capabilities will vary significantly.

Actor capabilities can be classified into five categories: knowledge, skills, weapons and tools, number of actors, and dedication. The knowledge of the actor will vary depending on if they have insider knowledge or not, and how much information that insider knows. The skills of the actor will affect the strategy they use to achieve their goals, though a large organization or non-host state can be assumed to have access to a wide variety of skills. The weapons and tools of the actor will affect their strategy, but access to an insider would grant access to more more tools with less effort. The number of actors will vary with the specific strategy of that actor as well as their tactics, as a more discrete approach would require fewer actors than a forceful approach. The dedication of the actor is important to consider, as an insufficiently motivated actor may be deterred with sufficient security measures.

The objectives of the actor may include: disruption of operations, radiological release, nuclear explosives, Radiological Dispersal Devices (RDDs), and/or information theft. The objectives of the actor will influence their other decisions, particularly their strategies. If an actor's objective requires force, then the capabilities and strategies implemented by that actor will vary from one whose objective can be accomplished covertly. Assuming the actor is rational, they will select a strategy which optimizes the probability of successfully completing their objective.

The actor's strategies include their modes of attack and tactics. The modes of attack vary significantly, including cyber attacks, waterborne attacks, direct or diversionary attack using light anti-tank weapons, and more. The tactics actors can use include stealth, deceit, and overt force. The stealth and deceit tactics are more covert, including avoiding or deactivating security systems for the stealth tactic or using false identification or authorization for the deceit tactic. The overt force tactic ties more directly into many of the modes of attack, which may itself also serve as a diversion for a different tactic.

For this work, only a single type of actor and objective combination is considered – an actor with access to an insider and the objective to disrupt operations through sabotage. The specific actor capabilities and strategies are not considered, only the actor type and objective.

### **2.3.3 Safeguards**

This work creates improved DNP groups, which more accurately account for MSR recirculation and chemical removal. This will affect fingerprinting of diversion scenarios, as if fissile material is removed, then the power and reactivity response of the reactor will vary based on the treatment of the DNP groups. Without these improved DNP groups, these diversion scenarios are more challenging to detect using this method.

Preventing the spread of nuclear material for weapons is the goal of nuclear safeguards. Safeguards in this work are treated as the nuclear reactor characteristics that impede "...the diversion or undeclared production of nuclear material and the misuse of technology by the Host State seeking to acquire nuclear weapons or other nuclear explosive devices" [191]. This definition agrees with the definition made at the IAEA international workshop in 2002 [87].

In the current work, proliferation resistance (PR) is used as a way to understand and measure certain criteria used in safeguards. PR is a spectrum, so it should not be considered necessary to optimize the PR, but rather as a way to evaluate strengths and weaknesses in the design relating to safeguards.

There are six measures for PR used in this methodology: technical difficulty, cost, time, fissile material type, detection probability, and detection resource efficiency. In this work, the measures that will be analyzed include time and detection probability. The time measure is analyzed by conducting diversion scenarios over various time frames to determine the effects on diversion fingerprints. A very low PR time is 0-3 months, while a very high PR time is >30 years [191]. The detection probability measure goes hand-in-hand with time, as a protracted diversion scenario is likely to have a reduced detection probability. A very low PR detection probability ranges from 0 to 5 percent, whereas a very high PR detection probability ranges from 95 to 100 percent [191].

Based on Department of Energy (DOE) Order 474.2A Chg 1 on MCA, SNM is classified as shown in Table 2.14 [80].

Table 2.14: The SNM classification table is recreated from [80].

| Material Type          | Accountable Quantity [g] | Weight Field Used for Element | Weight Field Used for Isotope       |
|------------------------|--------------------------|-------------------------------|-------------------------------------|
| $^{235}\text{U}$       | 1.0                      | total uranium                 | $^{235}\text{U}$                    |
| $^{233}\text{U}$       | 1.0                      | total uranium                 | $^{233}\text{U}$                    |
| $^{242}\text{Pu}$      | 1.0                      | total plutonium               | $^{242}\text{Pu}$                   |
| $^{239-241}\text{Pu}$  | 1.0                      | total plutonium               | $^{239}\text{Pu} + ^{241}\text{Pu}$ |
| $^{238}\text{Pu}$      | 0.1                      | total plutonium               | $^{238}\text{Pu}$                   |
| $^{241}\text{Am}^{**}$ | 1.0                      | total americium               | $^{241}\text{Am}$                   |
| $^{243}\text{Am}^{**}$ | 1.0                      | total americium               | $^{243}\text{Am}$                   |
| $^{237}\text{Np}^{**}$ | 1.0                      | total neptunium               | -                                   |

<sup>\*\*</sup> Americium and neptunium are treated as SNM if they are separated.

Table 2.14 is used along with the graded safeguards table from the updated Order to properly classify the attractiveness level and categorization of SNM. These categories include weapons, pure products, high-grade materials, low-grade materials, and all other materials, ranging from attractiveness level "A", the most attractive, to "E", the least attractive. For this work, only materials of attractiveness level "C" and below are relevant. High-grade materials, with attractiveness level "C", include fuel elements and assemblies. Low-grade materials, with attractiveness level "D", include solutions with target concentrations of 1-25 g/L. All other materials, with attractiveness level "E", are made up of highly irradiated forms, solutions with concentrations <1 g/L, or uranium containing <20%  $^{235}\text{U}$  or <10%  $^{233}\text{U}$ .

The quantities provided in Table 2.14 show the minimum reportable, or accountable, quantities for each material type. These quantities make up the lower limit of Category IV materials for "B" through "E" attractiveness level materials. The SNM Category depends on the quantity of material as well as the attractiveness level of that material. Higher Category material has more stringent requirements due to their greater strategic importance, with Category I SNM having the most requirements. For example, if material is classified as attractiveness level "C", then any quantity of plutonium or  $^{233}\text{U} \geq 6$  kg is considered Category I SNM. Alternatively, if the same material contains  $^{235}\text{U}$ , separated  $^{237}\text{Np}$ , or  $^{241}\text{Am}$  and  $^{243}\text{Am}$ , then an amount  $\geq 20$  kg is considered Category I SNM.

If the material in the reactor is classified as attractiveness level "D", then any quantity  $\geq 16$  kg is considered Category II, as Category I quantities do not exist for this attractiveness level. Materials of attractiveness level "E" do not have Category I, II, or III quantities. Instead, reportable quantities are considered Category IV.

Significant quantities (SQs) of material are also of interest, with different classifications for the IAEA and Nuclear Regulatory Commission (NRC). The NRC classification defines an SQ in Title 10 of the Code of Federal Regulations, Part 74, Subpart D, Section 74.41 (10 CFR 74.41) as one formula kilogram of strategic SNM or at least 10 kg of 20% enriched uranium [70]. A formula kilogram is defined in 10 CFR 74.4 as one total kilogram of both  $^{235}\text{U}$  and 2.5 times the sum of  $^{233}\text{U}$  and plutonium [70].

This work will use the IAEA's classification, within which there are two categories: direct use and indirect use. Direct use nuclear material consists of  $^{233}\text{U}$ , highly enriched uranium (HEU) ( $^{235}\text{U}$  enriched over 20%), and any amount of  $^{239}\text{Pu}$  with less than 80%  $^{238}\text{Pu}$ . Indirect use nuclear material consists of thorium and low enriched uranium (LEU) ( $^{235}\text{U}$  enriched at less than 20%). SQs of the different material categories are given in Table 2.15

Table 2.15: SQs of various materials are given along with their use category.

| Material         | Use      | Significant Quantity [kg] |
|------------------|----------|---------------------------|
| Plutonium        | Direct   | 8                         |
| HEU              | Direct   | 25                        |
| $^{233}\text{U}$ | Direct   | 8                         |
| $^{235}\text{U}$ | Indirect | 75                        |
| Natural Uranium  | Indirect | 10,000                    |
| Depleted Uranium | Indirect | 20,000                    |

[191, 89, 9].

This information is important for the current work, as it provides values for quantities of diverted material to analyze. The exact classification, materials, and quantities to use will vary based on the specifics of the fuel salt and diversion scenario. However, this information provides a set of data which can be directly referenced to determine the diversion rates.

For example, a fuel salt categorized as direct use with a plutonium diversion would have 8 kilograms of plutonium as a SQ. If that fuel salt also had an attractiveness level of "D", then 16 kilograms of plutonium would be Category II. Then, the diversion time could vary over a range of months to years, based on various PR times. A method of detection for these diversion scenarios is via change in reactivity due to the change in delayed neutron precursors and fissile concentration. The detection probability is then estimated based on this measurable reactivity change. Additionally, these diversions could be detected based on measured delayed neutron count rates [172].

## 2.4 Summary

This chapter presented the background and motivation for this work. DNPs were introduced, and the methods used to form DNP groups were described. The equations used to calculate the DNP group half-lives  $\tau_k$ , yields  $\bar{\nu}_{d,k}$ , and spectra  $\chi_{d,k}$  were also provided. This information is necessary to understand the structure of the improved DNP group parameters generated in this work. Kinetic parameters and various approaches for solving kinetics problems were discussed, as these parameters and methods provide essential background information for the safety, security, and safeguards transient simulations introduced in this work. Safety, security, and safeguards were discussed because those concepts motivate the improvement to the DNP group parameters. The improved parameters enable higher accuracy kinetics modeling, which in turn improves modeling of safety, security, and safeguards-related transients. Overall, this chapter provided background information on how improved DNP group parameters can be created for MSRs, as well as motivation for why this improvement is important.

## Chapter 3

# Methodology

The previous chapter identified that existing DNP group formulations do not account for MSR circulation and chemical removal. This chapter details the tools and methods used to both generate the improved DNP group parameters that do account for these MSR phenomena and to validate them. Two models are covered in this chapter: the 0D scaled model, which is already complete; and the 0D flow model, which is proposed. Although each model varies in its fidelity and cost, the general methodology is very similar: generate concentrations of DNPs from irradiation of a sample, convert those concentrations into a delayed neutron count rate, and then use non-linear least squares to fit DNP group parameters to the count rate curve. Validation of the DNP group parameters requires solving kinetics problems that make use of the parameters. There are also transient problems of interest that affect safety, security, and safeguards by design that will be discussed in this chapter.

### 3.1 Tools

Several software tools are used to evaluate and generate the improved DNP group parameters. The Python package Molten Salt Delayed Neutron (MoSDeN) is in-progress, but it collects and consolidates data while handling calculations of delayed neutron count rates and group parameter equation solves [196]. OpenMC is used to parse nuclear data to be passed into MoSDeN, and it will also be used for the higher-fidelity 0D flow model [157]. The MOOSE application Moltres takes the DNP groups from MoSDeN and OpenMC and uses them in transient simulations [112]. Preliminary results use MoSDeN to generate the improved DNP group parameters with support from OpenMC, while the proposed work will use OpenMC more heavily in simulations and Moltres for transient simulations. Each tool is described in detail in its corresponding sub-section.

#### 3.1.1 MoSDeN

MoSDeN is a Python package designed to enable generation of DNP group parameters using the microscopic approach for any given reactor of interest. MoSDeN reads in relevant nuclear data, simulates the irradiation of a fissile sample, collects concentrations from the simulated irradiation, calculates a delayed neutron count rate of the measured sample, and uses non-linear least squares to find the best DNP group parameters that fit the delayed neutron count rate.

MoSDeN allows for varying levels of fidelity in the generation, ranging from using nuclear data and broad assumptions to a simulation with neutronics and a 0D approximation of thermal-hydraulics. The MoSDeN package relies on the SciPy, NumPy, Matplotlib, pandas, and Uncertainties packages [199, 75, 82, 201, 58]. Many features

of MoSDeN also require OpenMC to function properly [157]. MoSDeN automates data collection and processing, simulates pulse and/or saturation irradiations, converts the concentrations into a delayed neutron count rate, and generates group parameters.

### 3.1.2 OpenMC

OpenMC is an open-source Monte Carlo neutral particle transport code. The preliminary work on the 0D scaled model uses OpenMC only for accessing nuclear data, but the 0D flow model will rely more heavily on OpenMC to simulate a fixed-source simulation. OpenMC offers many useful features that are particularly helpful for the use cases presented in this work.

Its Python application programming interface (API) offers functionality that simplifies interfacing with MoSDeN. OpenMC also offers configurable options in depletion, such as a zero-power depletion step which rapidly calculates nuclide concentrations since only decay is tracked. This feature allows resolving ex-core residence time depletion steps quickly. Additionally, transport-independent depletion can be used to further speed up the irradiation simulation. OpenMC's transfer rate functionality during depletion enables simulation of chemical removal in an MSR.

For this work, MoSDeN by default uses the ENDF/B-VII.1 based depletion chains provided by OpenMC and ENDF/B-VII.1 cross section data. The thermal spectrum chain applies to neutron irradiation below 1  $eV$ , the fast spectrum chain applies to neutron irradiation above thermal and below 5  $MeV$ , and anything above that is not treated in this work [184]. The models by default assume monoenergetic neutrons at thermal energy, 0.0253  $eV$ , irradiate the sample.

### 3.1.3 Moltres

Moltres is an open-source MOOSE application that simulates MSRs by solving the multi-group neutron diffusion and Navier-Stokes equations [112]. MOOSE is an open-source, parallel, finite element framework for multiphysics applications [69]. MOOSE includes many pre-built applications that couple physical phenomena. The HEAT CONDUCTION and NAVIER-STOKES modules can couple with Moltres for general thermal-hydraulic modeling [140, 111, 142, 138]. In this work, Moltres will be used to simulate the MSRE with different DNP group parameters. This will validate the improved DNP group parameters in this work.

The MSRE reactivity data will be used for validation, and was previously modeled with Moltres [112]. The MSRE core consists of flow through channels, simplifying its modeling compared to open-pool designs due to the restricted flow path. Since the initial demonstration, Park et al. have noted general agreement in results, attributing differences to simplifications and assumptions in the geometry [138]. Additionally, Park has published a dissertation detailing 1-D, 2-D, and 3-D neutronic models of the MSRE [137].

Moltres requires various data to model various physical phenomena in a reactor. The delayed neutron modeling requires delayed neutron fractions,  $\beta_k$ , and delayed neutron decay constants,  $\lambda_k$ . Neutronics modeling requires cross-sections, prompt fission spectrum  $\chi_p$ , delayed neutron spectrum  $\chi_d$ , and other data to properly model those physical effects. Moltres provides scripts which can parse various transport solvers, such as OpenMC and Serpent, to automatically collect this information and format it for use in Moltres. In this work, the required data for Moltres will be collected from OpenMC and MoSDeN.

There are two main capabilities in Moltres which are of interest. Moltres couples three physical phenomena to model an MSR: neutron flux, DNPs, and thermal-hydraulics. This allows Moltres to model transient simulations of MSRs, which will be used to validate the improved DNP group parameters. Moltres can also simulate transients

using the MSR-specific DNP group parameters, and it can generate improved parameters with higher fidelity than the other methods. This is of interest for future work, which may use higher-fidelity multi-physics to generate improved DNP group parameters.

### Neutron Flux Modeling

The neutron flux modeling is important for this work because it determines the energy of neutrons interacting with the fissile sample and the number of interactions. It determines the fission rate and the resulting fission product concentrations, including the initial locations of DNPs. In this work, modeling the neutron flux in Moltres is not important for irradiation of the sample, but it is relevant for transient modeling to capture the changes in power. The time-dependent multi-group neutron diffusion equation used in Moltres appears in Equation (2.38), recreated here:

$$\begin{aligned} \frac{1}{v_g} \frac{\partial \phi_g}{\partial t} = & \nabla \cdot D_g \nabla \phi_g + \sum_{g' \neq g}^G \Sigma_{s,g' \rightarrow g} \phi_{g'} \\ & + (1 - \beta) \chi_{p,g} \sum_{g'=1}^G (v \Sigma_f)_{g'} \phi_{g'} \\ & + \sum_{k=1}^K \chi_{d,g,k} \lambda_k C_k - \Sigma_{a,g} \phi_g - \sum_{g' \neq g}^G \Sigma_{s,g \rightarrow g'} \phi_g \end{aligned}$$

This equation shows how the neutron flux evolves in time due to scattering  $\Sigma_s$ , capture  $\Sigma_a$ , and prompt and delayed sources. Modeling the neutron flux during a transient requires accounting for the DNPs. DNPs contribute to the neutron flux and possess their own energy spectra, necessitating their inclusion as a separate term.

### DNP Modeling

The time dependent DNP concentration is given in Equation (2.39) and repeated here:

$$\frac{\partial C_k}{\partial t} = \beta_k \sum_{g=1}^G (v \Sigma_f)_g \phi_g - \lambda_k C_k - \nabla \cdot (C_k \vec{u}) + \nabla \cdot \frac{v_T}{S_{CT}} \nabla C_k$$

This is the primary equation of interest when simulating a transient using the improved DNP group parameters. This equation shows how the concentration of precursor group  $k$  changes over time due to the source from fission, decay, movement in the flow field, and turbulent diffusion. A discussion of turbulence and how it affects the DNP group parameters and transient simulations is given in Appendix C. The turbulent effects are not considered relevant to the desired outcomes of this work for analyzing the effects of the improved DNP groups relative to the existing groups. However, it is useful to validate the results with the MSRE, as otherwise the results may not match due to the upper and lower plena mixing. Thus, turbulent modeling will be considered as a potential extension.

For the potential future work of calculating improved DNP group parameters within Moltres for the spatially resolved model, discussed in Appendix E, Equation (2.39) is less helpful. This is because it requires DNP group parameters, which Moltres would be generating. To generate the DNP group parameters using Moltres, Equation (2.63) can be used, recreated here:

$$\frac{dN_j}{dt} = S(\vec{r}, t) - \lambda_{net,j} N_j(\vec{r}, t) + \nabla \cdot (D \nabla N_j(\vec{r}, t) - \vec{u}(\vec{r}, t) N_j(\vec{r}, t))$$

This equation shows how the concentration of nuclide  $j$  evolves over time due to various source  $S$ , loss  $\lambda_{net}$ ,

diffusion  $D$ , and advection  $\vec{u}$  terms. The spatially resolved model could use spatially resolved depletion over the irradiation simulation. Equation (2.39) models DNP groups, while Equation (2.63) can be used to model individual DNPs.

### Thermal-Hydraulic Modeling

Thermal hydraulic modeling is important for this work because transient simulations of MSRs require thermal-hydraulics to properly account for the movement of the DNPs. Moltres offers time-dependent, incompressible, buoyancy-driven flow [137]:

$$\rho \frac{\partial \vec{u}}{\partial t} = -\rho(\vec{u} \cdot \nabla) \vec{u} - \nabla p + \mu \nabla^2 \vec{u} + \rho \alpha \vec{g} (T - T_{\text{ref}}), \quad (3.1)$$

$$\nabla \cdot \vec{u} = 0, \quad (3.2)$$

where

$\rho$  = fluid density [ $kg \cdot m^{-3}$ ]

$p$  = pressure [ $Pa$ ]

$\vec{u}$  = salt velocity [ $m \cdot s^{-1}$ ]

$\mu$  = dynamic viscosity [ $Pa \cdot s$ ]

$\alpha$  = coefficient of thermal expansion [ $K^{-1}$ ]

$\vec{g}$  = gravitational force vector [ $N$ ]

$T$  = fluid temperature [ $K$ ]

$T_{\text{ref}}$  = reference temperature at which the nominal density is provided [ $K$ ].

The temperature distribution is given by [137] as:

$$\rho c_p \frac{\partial T}{\partial t} = -\rho c_p \vec{u} \cdot \nabla T + \nabla \cdot (k \nabla T) + Q_f - Q_s, \quad (3.3)$$

with the fission heat source term defined as:

$$Q_f = \sum_{g=1}^G \epsilon_g \Sigma_g^f \phi_g, \quad (3.4)$$

where

$c_p$  = specific heat capacity of molten salt [ $J \cdot kg^{-1} \cdot K^{-1}$ ]

$k$  = effective thermal conductivity of molten salt [ $W \cdot m^{-1} \cdot K^{-1}$ ]

$Q_f$  = fission heat source [ $W \cdot m^{-3}$ ]

$\epsilon_g$  = average fission energy released per fission in group  $g$  [ $J$ ]

$Q_s$  = heat sink/removal [ $W \cdot m^{-3}$ ].

### Ex-core and In-core Coupling

While Moltres treats the in-core region as described above with 2D or 3D resolution, Moltres treats the ex-core region as a tightly-coupled 1D pipe geometry with coupling using Picard iterations and the inlet/outlet boundary conditions. The specific equations to treat the variables of interest in this geometry can be derived from their full forms. For example, the flux and eddy viscosity are zero in the ex-core region, leading to the precursor concentration in that region:

$$\frac{\partial C_i}{\partial t} = -\lambda_i C_i - u_x \frac{\partial C_i}{\partial x}. \quad (3.5)$$

## 3.2 DNP Group Generation

Current DNP group parameters are tabulated by fissile nuclide and incident neutron energy. For the improved DNP group parameters, it is challenging to continue using tabulated parameters due to the number of variables involved in spatially resolved fuel salt recirculation and element-specific chemical removal rates. In this work, the flow is simplified to two parameters: an in-core average residence time,  $\tau_{in}$ , and an ex-core residence time,  $\tau_{ex}$ .

This approach captures fuel salt advection around the primary loop and approximates turbulent and diffusive mixing. The terms derive from the averaged behavior of a fuel salt control volume, treating thermal-hydraulic effects, such as vortices, as changes in residence times. This method allows for approximate fitting of any circulating fuel reactor, requiring only the generation of relevant data for determining the DNP group parameters.

Chemical removal rates are challenging to represent with only a small number of parameters while containing a useful amount of information. For any given set of removal rates for a reprocessing scheme, DNP group parameters can be generated. This does not provide easily tabulated data for every MSR parameter, but offers greater accuracy for specific reprocessing schemes.

In this work, two different models are considered using MoSDeN: the 0D scaled and the 0D flow model. The 0D scaled model has been completed and offers a rapid, but less accurate, approach to calculate the improved DNP group parameters that account for fuel salt recirculation and chemical removal. The 0D flow model will be completed, providing a slower but more accurate approach to calculating the improved DNP group parameters.

An irradiation time of several minutes suffices to reach equilibrium DNP concentrations, as the saturation time frame is on the order of multiple half-lives of the longest-lived DNP. Elements with cycle times on the order of months to years contribute negligibly, as Section 4.2.1 shows. Elements with cycle times on the order of seconds have greater effects and must be included for accurate data generation. The cycle time refers to a linear approximation of the time it takes for an element to be completely removed from a volume, and it is equivalent to the inverse of the removal rate.

The MSBR reprocessing scheme serves as the basis for analyzing the largest cycle times in this work. This is because the MSBR presents very large chemical removal rates, providing useful conservative estimates as to the impact these large removal rates could have on DNP group parameters. Because the MSRE is the primary reactor of interest in this work, the MSBR reprocessing scheme is only used for isolated analysis on the effects of large chemical removal rates in Section 4.2.1. These chemical removal rates are given in Table 3.1. Rare earths, semi-noble metals, volatile fluorides, and higher nuclides have small removal rates relative to the time scales of interest for saturation irradiations. Section 4.2.1 examines these longer cycle times to confirm their negligible impact.

Table 3.1: MSBR Online Reprocessing Removal Rates from [154].

| Reprocessing Group | Element(s)                             | Removal Rate [ $s^{-1}$ ] |
|--------------------|----------------------------------------|---------------------------|
| Rare Earths        | Y, La, Ce, Pr, Nd, Pm, Sm, Gd          | $2.32 \times 10^{-7}$     |
| Rare Earths        | Eu                                     | $2.32 \times 10^{-8}$     |
| Noble Metals       | Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te | $5.00 \times 10^{-2}$     |
| Seminoble Metals   | Zr, Cd, In, Sn                         | $5.79 \times 10^{-8}$     |
| Gases              | Kr, Xe                                 | $5.00 \times 10^{-2}$     |
| Volatile Fluorides | Br, I                                  | $1.93 \times 10^{-7}$     |
| Discard            | Rb, Sr, Cs, Ba                         | $3.37 \times 10^{-9}$     |
| Salt Discard       | Th, Li, Be, F                          | $3.37 \times 10^{-9}$     |
| Protactinium       | Pa                                     | $3.86 \times 10^{-6}$     |
| Higher Nuclides    | Np, Pu                                 | $1.98 \times 10^{-9}$     |

With the chemical reprocessing scheme in place, the next step to generate the improved DNP group parameters is to decide on the level of fidelity. The two levels of fidelity are the 0D scaled model, which is complete, and the 0D flow model, which is proposed. There is also another level of fidelity discussed in Appendix E, the spatially resolved model, which could be developed as future work.

Although online reprocessing is of interest for this work, only the removal of non-fissile nuclides is considered. Online refueling is not examined, as it only affects the fission-weighting of the DNP group parameters, as discussed in Section 2.1.2. For example, if the irradiated sample is  $^{235}\text{U}$  and the refuel material is  $^{239}\text{Pu}$ , then the addition of  $^{239}\text{Pu}$  would only alter the fission-weighting and not the actual DNP concentrations from each fissile nuclide. Even if  $^{235}\text{U}$  is added as a refuel material, this could simply be treated as multiple identical individual fissile control volumes circulating in the system or as a change in the sample's mass. Similarly, the chemical removal of some fissile nuclide does not impact this analysis, since the DNP group parameters are normalized per fission. Overall, the addition or removal of fissile nuclides only affects the fission-weighting, while the addition or removal of non-fissile nuclides will affect the DNP group parameters.

A Monte Carlo sampling approach is used to propagate uncertainties in the various inputs, such as in emission probabilities, half-lives, fission yields, reprocessing terms, or residence times. The sampling approach simply repeats the process for generating the DNP group parameters, but with randomly sampled input data. The data itself must be perturbed, even though uncertainties propagated to the delayed neutron count rate could in theory be perturbed instead. This is because the uncertainties in the delayed neutron count rate are not independent from time step to time step, but instead they all depend on the underlying data which changes the shape of the delayed neutron count rate at all times.

For generation of the improved DNP group parameters, the main goal is to formulate them for individual fissile nuclides. For some of the analyses that will be conducted in this work, the fissions are induced in several fissile nuclides. For these analyses, a fission-weighted approach is used, where the DNP group parameters are scaled based on the fraction of fissions that lead to those group parameters. Equations (2.18) and (2.19) show how the

fission-weighted DNP group parameters can be calculated.

### 3.2.1 0D Scaled Model

This is a simple model that primarily serves as a way to verify the microscopic approach is working as expected by comparing with existing literature. This approach can only model saturation concentrations, as the decay chain is not modeled. It is called the 0D scaled model because it uses the scaled flux method, discussed in Section 2.2.2. The cumulative fission yield (CFY) is used to calculate the equilibrium DNP concentrations, with a scaling term based on the residence time of fuel salt in the core relative to the net residence time. The CFY gives the amount of each nuclide that is formed from both fission and decay. An example solution with no reprocessing is shown in Equation (3.6) [84]:

$$N_i = \frac{f_{in} CFY_i}{\lambda_i}, \quad (3.6)$$

where

$f_{in}$  = the fraction of time fuel-salt spends in the core [-]

$CFY_i$  = the cumulative fission yield of nuclide  $i$  [-].

The derivation for this equation is given in Appendix B. It is important to note that the flux is scaled when calculating the fission rate as well, by the same  $f_{in}$  term that scales the cumulative fission yield. The  $f_{in}$  term scaling the CFY and fission rate cancels out when calculating the number of delayed neutrons per fission, which leads to no deviation in the results based on the residence time.

No cross-sections are included in this model, nor is time-dependent decay of nuclides into DNPs included. There is a possible approach to include the effects of DNP absorption cross-sections in this model, but this approach is not used [174, 14]. The 0D flow model will account for absorption and transmutation cross-sections.

Figure 3.1 shows how the different data is loaded into MoSDeN, including external data and user defined input data. The green circles represent input data required for generating the DNP group parameters. The red diamond represents MoSDeN. MoSDeN takes in the data to simulate irradiation of the target fissile sample to generate the irradiated concentrations, the delayed neutron count rate, and finally the DNP group parameters that fit that count rate. The blue squares represent the outputs, in this case the DNP group parameters: the DNP group yields and the DNP group half-lives.

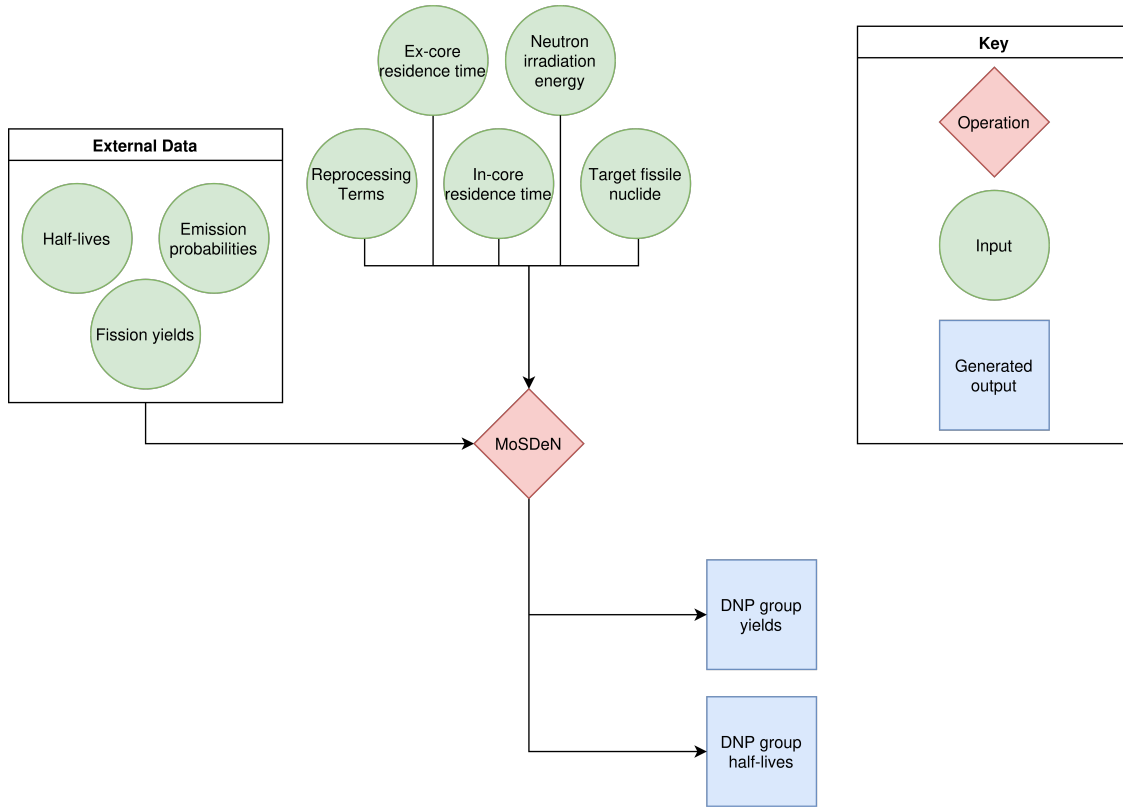


Figure 3.1: Flow diagram of 0D scaled model used to generate improved DNP group parameters.

After all this data is loaded into MoSDeN, it generates the concentration of each DNP. This concentration is then combined with emission probabilities and decay constants to form a delayed neutron count rate. A non-linear least squares solve then fits the improved DNP group parameters to the delayed neutron count rate, generating the group yields and half-lives, given as blue squares in Figure 3.1.

This model is only valid for saturation irradiation simulations, as the concentration calculated in Equation (3.6) is the equilibrium concentration. For a pulse irradiation, this approach is not valid. This prevents the 0D scaled model from simulating pulse irradiations. The proposed 0D flow model can simulate pulse irradiations, and it will be used to evaluate whether pulse irradiations provide more accurate yields and half-lives for the shorter-lived DNPs.

### 3.2.2 0D Flow Model

The 0D flow model enables improved analysis of the general trends in nuclide evolutions. It treats the flow in a zero-dimensional manner using in-core and ex-core residence times, and uses OpenMC to track the full depletion matrix. Figure 3.2 shows the 1D in-core and ex-core flow paths, which this 0D model emulates by irradiating the sample for the in-core residence time and letting it decay for the ex-core residence time. The fission history function is approximated in this work as a square wave, with a constant fission rate during the in-core residence time and no fission during the ex-core residence time. This approximation simplifies the analytical form of the fission history when forming the non-linear least squares problem, as shown in Appendix A. Appendix A derives the delayed

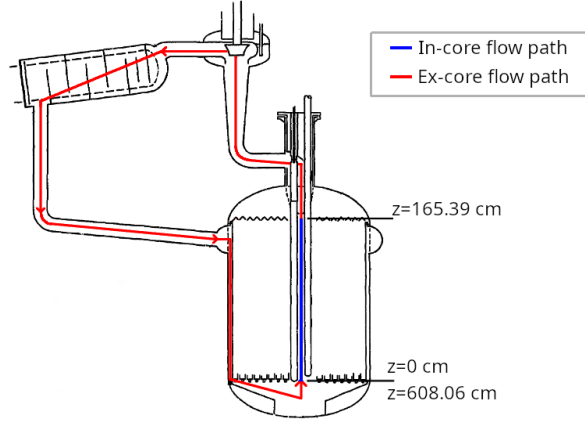


Figure 3.2: The flow path and bounds for the in-core region of the MSRE, overlaid on the MSRE loop diagram from [54].

neutron count rate after  $T$  seconds of irradiation in Equation (3.7):

$$\dot{n}_d(T, t) = F_0 \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t} \left( (J_{in} - J_{ex}) - \frac{e^{-\lambda_k T}}{1-d} \left( e^{\lambda_k \tau_{in}} d^{J_{ex}+1} - e^{\lambda_k \tau_{in}} + 1 - d^{J_{in}+1} \right) \right), \quad (3.7)$$

where,

$$\begin{aligned} T &= \text{the total fission history, or net irradiation, time [s]} \\ \tau_{in} &= \text{the in-core residence time [s]} \\ \tau_{ex} &= \text{the ex-core residence time [s]} \\ \tau_{net} &= \tau_{in} + \tau_{ex} \\ &= \text{the net residence time [s]} \\ J_{in} &= \left\lfloor \frac{T}{\tau_{net}} \right\rfloor \\ &= \text{the number of times the sample enters the core minus one [#]} \\ J_{ex} &= \left\lfloor \frac{T - \tau_{in}}{\tau_{net}} \right\rfloor \\ &= \text{the number of times the sample exits the core minus one [#]} \\ d &= e^{\lambda_k \tau_{net}}. \end{aligned} \quad (3.8)$$

The values of  $J_{in}$  and  $J_{ex}$  use the floor function to ensure the integer number of times is accurately counted. This closed-form solution is numerically unstable for very short ex-core irradiation times due to the positive exponential terms. Although computationally slower to solve, Equation (AA.9) offers a more stable form, recreated here as:

$$\begin{aligned} \frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{v}_{d,k}} &= e^{-\lambda_k t} \sum_{j=0}^{J_{ex}} \left( e^{-\lambda_k (T - \tau_{in} - j \cdot \tau_{net})} - e^{-\lambda_k (T - j \cdot \tau_{net})} \right) \\ &\quad + e^{-\lambda_k t} \sum_{j=J_{ex}+1}^{J_{in}} \left( 1 - e^{-\lambda_k (T - j \cdot \tau_{net})} \right). \end{aligned} \quad (3.9)$$

Figure 3.3 shows the flow diagram for the 0D flow model. The model provides eight parameters for OpenMC [157]: in-core residence time, ex-core residence time, neutron irradiation energy, target fissile nuclide, target density, total irradiation time, temperature, and reprocessing terms. The depletion matrix is passed into OpenMC, while the emission probabilities are passed into MoSDeN. The depletion matrix contains information on half-lives, fission yields, cross-sections, branching ratios, and other useful data needed for OpenMC depletion. Section 3.3 provides further details on this process.

After OpenMC completes the fixed-source simulation, the final nuclide concentrations in the irradiated sample are saved for analysis in MoSDeN. MoSDeN combines the concentrations with data from the IAEA beta-delayed neutron emission database to generate a delayed neutron count rate [51]. A non-linear least squares fitting function from SciPy then calculates the group half-lives and yields using the Trust Region Reflective algorithm [199, 26]. These group parameters correspond to a specific nuclide, neutron energy, in-core residence time, ex-core residence time, reprocessing scheme, and set of nuclear data. Section 3.4 provides further details on this block. Some additional factors to consider include self-shielding due to sample density (while avoiding self-multiplication), temperature-induced cross-section variations, and under- or over-irradiation, all of which can affect the results [95].

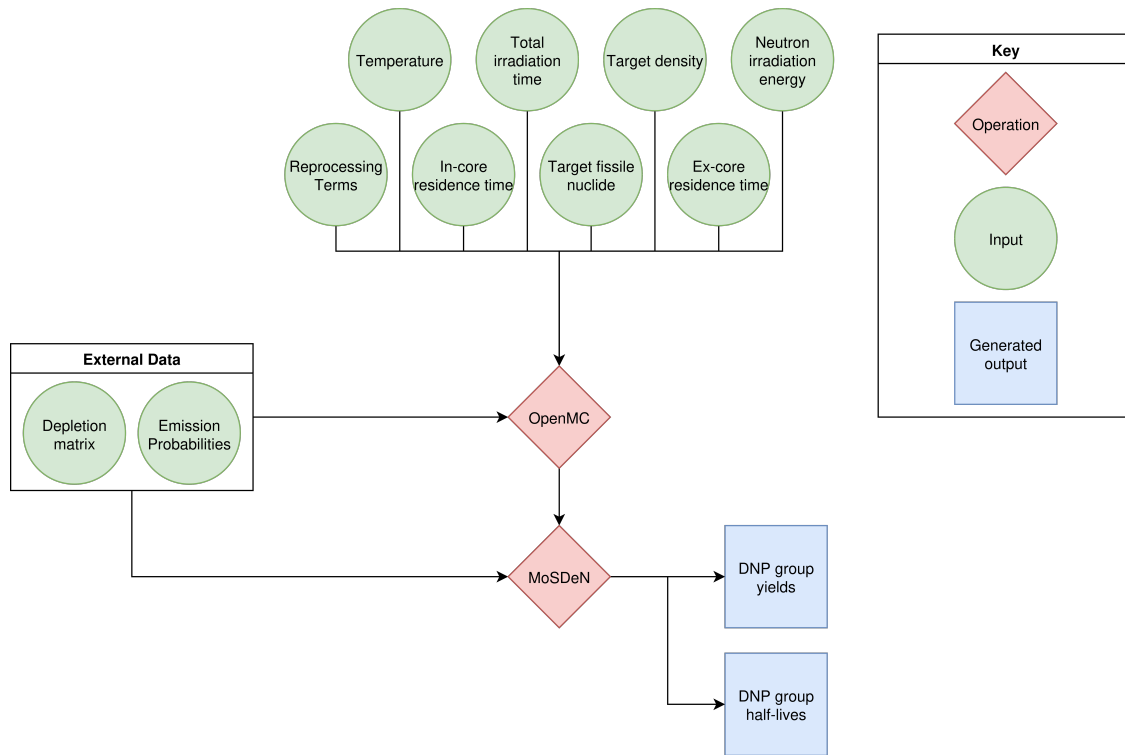


Figure 3.3: Flow diagram of 0D flow model used to generate improved DNP precursor group parameters.

The 0D flow model will use a three-gram spherical sample with vacuum boundary conditions. It will treat the problem as a fixed-source simulation, with a uniform volumetric neutron source. This approach is not physical, as a real neutron source would have a spectrum and originate externally. However, the neutron spectrum for irradiating samples to generate DNP group parameters is not provided in the literature, and is generally provided as "thermal", "fast", or an average energy rather than a fully-defined spectrum. The spatial distribution of the source has a negligible impact, as the sample is small and exhibits minimal self-multiplication. Thus, this approximation does not limit the effectiveness of the 0D flow model.

### 3.3 Delayed Neutron Count Rate Generation

Now that the various models have been discussed, execution details will be explored in the following sub-sections to provide specific information on the methodology behind the generation of the improved DNP group parameters. This includes a detailed discussion on the concentration generation and the data used with those concentrations to produce a delayed neutron count rate.

#### 3.3.1 DNP Concentrations

The concentrations from the 0D scaled model are calculated using Equation 3.6. The 0D flow model will use OpenMC to handle the calculation of the concentrations over time.

The 0D flow model will use a fixed-source irradiation of a three-gram spherical fissile sample. A uniformly distributed source will be placed inside the target with a vacuum boundary condition at the edge of the sphere. The sample must have a sufficiently large volume to ensure a high number of fissions, but excessive volume increases self-shielding and self-multiplication. Adjusting the density mitigates this issue: higher density increases self-shielding and self-multiplication, while lower density reduces fission probability and worsens statistical accuracy. By using a uniformly distributed neutron source within the sample, self-shielding is lessened while retaining a small sample. Additionally, since the fission counts can be taken directly from OpenMC tallies, self-multiplication and self-shielding can be accounted for directly without introducing error.

The fixed source in this model will consist entirely of neutrons at a single energy, rather than a spectrum. For the proposed analyses, only thermal fission of  $^{235}\text{U}$  is studied. This energy influences fission cross sections, transmutation cross sections, and fission yields. Traditionally, tabulated group parameter data is formatted for a single energy or energy range, such as fast or thermal. However, macroscopic experiments, discussed in Section 2.1.1, use reactors with varying neutron energy spectra, not monoenergetic neutrons. This could be addressed by modeling a reactor in OpenMC and using a k-eigenvalue calculation to generate a neutron energy spectrum. This analysis is considered outside the current scope, but is suggested as future work in Section 5.4. Photons can also produce neutrons through photofission, though this is also considered outside the scope of this work [98]. There are also neutrons emitted after a delay from photons through non-fissile material, known as photoneutrons [92]. Photoneutrons are also considered outside the scope of this work.

The neutron source distribution does not replicate the experimental process, where neutrons irradiate the sample externally rather than internally. However, an evenly distributed source reduces self-shielding concerns. The exact spatial locations of fission events within the sample should have a negligible effect on the nuclide concentrations, which are the outputs of the OpenMC simulation. With a density of  $10 \text{ g} \cdot \text{cm}^{-3}$ , the spherical sample has a radius of approximately  $0.4 \text{ cm}$ . After running the OpenMC simulation, the concentrations are collected from the depletion results for the final irradiation time step. These final concentrations are then converted into a delayed neutron count rate.

#### 3.3.2 Concentrations to Count Rate

After receiving the concentrations from OpenMC, MoSDeN determines the concentration of each nuclide as a function of time during the measurement period. The concentrations are calculated based on half-life, as shown in Equation (3.10), where the concentration as a function of time is:

$$N(t) = N_0 e^{-\lambda t}. \quad (3.10)$$

Equation (3.10) shows the approach used in the 0D scaled model, and does not account for decay chains between nuclides. However, the general form of the equation can still be used, but with inclusion of the various decay chains and branching ratios. A simple way of implementing the decay chain, while also accounting for transmutations and absorptions, is using zero-power depletion simulations in OpenMC, which will be used in the 0D flow model. These concentrations will be available at coarse time steps and linearly interpolated using SciPy, which will allow MoSDeN to evaluate concentrations at any time after irradiation [199]. The specific time steps at which concentrations are written will be investigated after the development of the 0D flow model.

The concentrations are combined with Equation (2.3), recreated here:

$$\dot{n}_d(t) = \epsilon \sum_{k=1}^K P_{n,k} \lambda_k N_k(t).$$

The efficiency term,  $\epsilon$ , is set to 1 in this work, as the simulated model captures every produced delayed neutron in the count rate. The delayed neutron emission rates of each DNP over time are summed to generate the delayed neutron count rate.

### 3.4 DNP Group Reconstruction

Once the delayed neutron count rate is generated, it is used to generate the improved DNP group parameters,  $\bar{v}_{d,k}$  and  $\lambda_k$ . Fitting the DNP group parameters to the delayed neutron count rate curve employs a non-linear least squares method, as both the group half-lives and yields must be determined. The equation describing the problem appears in Equation (2.5), recreated here:

$$\dot{n}_d(t) \approx \epsilon F_s \sum_{k=1}^K \bar{v}_{d,k} \lambda_k e^{-\lambda_k t}.$$

An important consideration for this process is the selection of time values for generating the DNP group parameters. Increasing the number of time nodes raises computational costs, but too few nodes result in a poorer fit [135]. Section 4.2.2 shows the sensitivity study for both the node spacing and the maximum time to measure delayed neutrons.

Another consideration for the non-linear least squares solution is the covariance, which is larger for the pulse irradiation than for the saturation irradiation. This results from multiplying the group yield by the group decay constant. Replacing this multiplication with an independent term produces worse fits, as the multiplication is not independent and is necessary for accurate modeling.

The residual to minimize with in the solution process is given in Equation (2.12), recreated here as:

$$r = \min_x \left\| \frac{\vec{b} - A\vec{x}}{\vec{b}} \right\|_2^2, \quad (3.11)$$

with bounds of:

$$\tau_k \in [1 \text{ ms}, 1000 \text{ s}]$$

$$\bar{v}_{d,k} \in [0, 1].$$

Absolute convergence tolerances for the cost function, change in independent variables, and gradient norm are all

set to  $1 \times 10^{-12}$  to ensure tight convergence. The Trust Region Reflective method, which allows for bounds to be included for the parameters, is used to solve this least squares problem [26]. Another least squares fitting technique is the Levenberg-Marquardt method, though the bounds cannot be set with this method [109, 118, 124].

The final consideration in DNP group reconstruction is uncertainty. This work tracks uncertainties using the uncertainties package [58]. These uncertainties are in the cumulative fission yields  $CFY$ , half-lives  $\tau$ , and emission probabilities  $P_n$  for the 0D scaled model. For the 0D flow model, only uncertainties in the emission probabilities and half-lives for the delayed neutron count rate calculation will be propagated. OpenMC does not propagate concentration uncertainties through depletion, and developing this extension is out of scope for this work.

The uncertainties are propagated through the count rate calculation, but for the non-linear least squares solve, Monte Carlo sampling is used. Each count rate is generated using randomly sampled fission yields, half-lives, and emission probabilities within their uncertainty bounds per DNP. This enables formation of a normal distribution of group parameters, from which the mean and standard deviation of the mean can be extracted.

Once the DNP group parameters and their uncertainties are generated, the final step to obtain the finalized group parameters is to combine the saturation and pulse irradiation results. In macroscopic measurements, saturation irradiation provides better statistics for longer-lived groups, while pulse irradiation is better for shorter-lived groups. Brady combines the results to create a six-DNP group structure by using four long-period groups from saturation irradiation and four short-period groups from pulse irradiation. The groups are normalized at the second group, where the two longest-lived group half-lives and yield ratios use the saturation results [24]. The remaining groups use the pulse irradiation results.

The 0D scaled model uses only saturation irradiation data for the least squares solve. However, the 0D flow model will enable pulse irradiations to be simulated, so the proposed work includes analysis of how inclusion of pulse irradiation data changes the DNP group parameters.

### 3.5 Impact of DNP Groups on Transients

The impact of the DNP groups on transients will be evaluated using Moltres with fluid-flow-velocity field coupled multi-group diffusion kinetics, as discussed in Section 2.1.3. This approach provides high-fidelity results by coupling thermal-hydraulics and neutronics, a necessity for MSR modeling. Modeling every DNP in space and time would be prohibitively expensive, so they are instead modeled using DNP group parameters. Moltres allows direct comparison between static DNP group parameters, which do not account for MSR phenomena, and parameters that do. This comparison will be performed by varying the DNP group parameters in Moltres and simulating the same transient, followed by comparison of temperature and power responses.

The results of interest from this analysis include the reactivity and power responses to various initiating events. The initiating events and transients vary, as different reactor responses are studied. The events of interest depend on the analytical perspective. Specifically, there are four different perspectives of interest in this work: validation, safety, security, and safeguards. Each perspective involves a unique combination of motivation, proposed initiating event, and investigated response. These are discussed in detail in their respective sections.

The effective delayed neutron fraction  $\beta_{eff}$  is also of interest and will be calculated. It can be calculated from Moltres using the prompt method, discussed in Section 2.1.2. Moltres uses fission-weighted DNP group parameters, and the effective multiplication factor will capture the spatial distribution of each DNP group concentration. Adjoint-weighting is not used for these parameters because Moltres explicitly captures the delayed neutrons in the diffusion solve. The effective delayed neutron fraction will be compared between the improved DNP groups and existing groups to offer more detail for comparison between the groups.

Moltres will provide an effective delayed neutron fraction that captures the effects of fuel circulation. The fission weighting values will be generated using fission rate tallies in OpenMC.

### 3.5.1 Validation

The motivation for validation is to confirm that the proposed methodology and results align with experimental data. This ensures that the demonstrated improved DNP group parameters are applicable to actual MSRs. Discrepancies between experimental results and the model highlight potential weaknesses in the approach. If no discrepancies exist, confidence in the results increases for problems similar to those validated.

In this work, the validation of the DNP group parameters will follow a multi-stage approach. The first stage will compare the irradiated sample's DNP group parameters from thermal fission of  $^{235}\text{U}$  with no ex-core time to the values in the literature [95, 36, 159, 125, 23, 166, 187, 195, 200, 135]. This comparison will demonstrate that the model accurately aligns with experimentally measured values for a stationary case. If the parameters do not match, the issue lies in the data or the aggregation of individual DNPs into the count rate. If the parameters are accurately predicted, the formulation is considered valid.

The second stage of the validation will use the MSRE modeled in Moltres to compare the effective delayed neutron fraction. The literature reports an effective delayed neutron fraction in the MSRE of 0.0045, which will be compared to values calculated using the previous and improved DNP group parameters. To ensure the energy spectra of the MSRE is properly captured in the DNP group parameters, OpenMC will first be used to run a Monte Carlo k-eigenvalue solve. This will enable generation of four sets of DNP groups: a stationary group using thermal neutrons, an improved group using thermal neutrons, and then the same two groups but with the MSRE neutron spectrum. These results will then be compared with DNP groups in the literature for the MSRE to determine which set of parameters most closely fits the measured effective delayed neutron fraction [106].

The third stage of the validation that will be conducted in this work applies the improved DNP group parameters to the MSRE pump start-up and coast-down transient simulation to compare against experimental data on reactivity changes from DNPs [148]. The purpose of this comparison is to demonstrate that the non-stationary model generates valid DNP group parameters. This experiment was previously studied by Reynolds and Palmer, as well as by Park using Moltres and Quasimolto<sup>1</sup>, allowing comparison with their results [150, 137]. However, previous models excluded the inlet and outlet plena in the MSRE, leading to discrepancies with experimental data. This work will develop a model including the plena, to be combined with updated DNP group parameters, to generate delayed neutron reactivity responses closer to the experimental data.

### 3.5.2 Safety

The motivation for studying safety is to protect the public and meet regulatory requirements. Safety is discussed in detail in Section 2.3.1, which showed that many safety-related transients exist. The DNP group parameters are critical in these transients, as they influence the reactor period, power, and temperature.

The transients that will be investigated in the MSRE are the ULOHS, ULOF, UTOP, and FSOC events [110, 43]. A ULOHS transient can occur when the heat exchanger experiences a failure, possibly triggered by a pump failure in the intermediate loop, leading to a loss of cooling of the reactor. This will be modeled in Moltres by changing the inlet temperature to match the outlet temperature, rather than using a constant inlet temperature. This will simulate a loss of heat sink, as rather than the inlet temperature being constant and lower than the outlet temperature, it will simulate no cooling. A ULOF transient is when the flow rate in the primary loop becomes zero, caused by pump

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<sup>1</sup>Quasimolto is an open-source multiphysics code that models multi-group neutron transport, DNP balance, and energy balance.

failure in the primary loop. This will be modeled in Moltres by slowing the velocity field, which will alter both the temperature and DNP distribution. A UTOP transient is a reactivity insertion event, which could be caused by various means, such as removing poisons, adding fissile material, or temperature changes. This will be modeled in Moltres by having a step decrease in the removal cross section, which will represent a control rod withdrawal. An FSOC transient occurs when the fuel salt temperature decreases, which could be caused by an increased heat removal rate from the heat exchanger. This will be modeled in Moltres by decreasing the inlet fuel salt temperature.

Each transient will be simulated using Moltres in the MSRE using existing DNP groups and the improved DNP groups. For each transient, the power response, maximum temperature, minimum temperature, and reactivity response will be recorded. The power response shows how changes in the DNP group parameters affect other responses, such as temperature. It is important to evaluate if MSRs may have unexpected unsafe temperatures. This could occur if traditional DNP group parameters predict safe operation while the improved parameters predict unsafe operation.

### 3.5.3 Security

Security is discussed in detail in Section 2.3.2, which details that, for this work, scenarios of interest are when actors use an insider to disrupt the reactor through sabotage. This ties directly into the safety design study, as sabotage aims to create unsafe reactor conditions. Ensuring the MSR operates safely after sabotage is critical. If static DNP group parameters predict safe operation while improved parameters predict unsafe operation, correcting this discrepancy in the reactor design is essential.

Since the safety study covers most transients relevant to sabotage, additional considerations can be incorporated into these transients. As sabotage may initiate multiple safety-related transients simultaneously, it is reasonable to investigate such scenarios. Specifically, the attack scenario assumes that multiple simultaneous events are instantiated by an insider in combination with a hostile actor [1]. This aligns with specifications provided in 10 CFR 73.1 on a design basis threat, which includes an internal threat for radiological sabotage as a diversionary action [70]. The radiological sabotage design basis threat includes well-trained, armed, and informed individuals with insider assistance, an internal threat, and a cyber attack.

The four unique transients that will be considered are the ULOHS, ULOF, UTOP, and FSOC. Combining them to form all unique combinations results in six double-transients, three triple-transients, and one quadruple-transient, totaling ten combined security-related sabotage-initiated transients. However, some of these combinations do not result in a change, such as the ULOF and ULOHS, as the loss of heat sink would not be important for a stationary fuel salt that does not circulate out of the core to the heat exchanger. The combinations that will be considered are each transient on its own and the UTOP combined with each other transient. This results in four single transients and three double transients, totaling seven transient simulations that will be analyzed. It is feasible that the attack scenario for the radiological sabotage design basis threat could cause these seven transients in an attempt to disrupt operations and divert attention. Each of these transients will be simulated using Moltres in the MSRE using existing DNP groups and the improved DNP groups.

These security analyses will demonstrate if certain combinations of transients predicted to be safe and non-disruptive with previous DNP groups may actually be disruptive when using the improved DNP groups. The transient will be considered disruptive if safety limits on temperature are exceeded. If there are sabotage events which are predicted to be non-disruptive by the previous groups but not by improved groups, then this would indicate an area of improvement for modeling and reactor design. If the improved DNP groups do not demonstrate disruptive effects, then this analysis will still provide useful insight into how much variation in temperature could be expected due to different transient combinations.

### 3.5.4 Safeguards

Safeguards are discussed in Section 2.3.3, with a focus on the importance of preventing SNM diversion in MSRs. Detection methods, such as neutron detectors and surveillance systems, play a key role in safeguarding the reactor [213]. The improved DNP group parameters in this work affect the effective delayed neutron fraction and reactivity response. Using static DNP group parameters to detect SNM diversion predicts less accurate reactivity responses than the improved parameters. If the difference is significant, relying on static parameters may reduce the detection probability of a diversion scenario.

To investigate this, diversion scenarios will be studied for a single significant quantity of  $^{235}\text{U}$  and another scenario of a single significant quantity of plutonium removed over the course of one year, simulated using OpenMC. The diversion will be from the  $^{235}\text{U}$ -fueled MSRE from a fresh fuel salt over the course of one year at nominal power. The change in the fuel salt composition will affect the DNP group parameters, as discussed in Section 2.1.2. The fission-weighted DNP group parameters can be used in Moltres, as discussed in Section 3.1.3, for simple step reactivity insertion transients. The following four sets of fission-weighted DNP group parameters will be compared: existing parameters without diversion, existing parameters with diversion, improved parameters without diversion, and improved parameters with diversion. The power responses will be compared using a simple negative step reactivity insertion will be simulated for each set of DNP group parameters.

The power responses will demonstrate if power response fingerprinting could be obfuscated for an MSR model that does not use improved DNP groups. This obfuscation will be investigated by comparing power responses, which are dependent on the delayed neutron fraction. The delayed neutron yield, and therefore the delayed neutron fraction, will be smaller when accounting for chemical removal and recirculation, which will also be true for measured responses in the MSR. This smaller delayed neutron fraction will cause a larger power response, which will more accurately reflect what would be measured in an MSR. A model using existing DNP groups would have a larger delayed neutron fraction than it should, which would lead to a smaller than expected power response.

If there is a diversion of uranium, then there will be more plutonium fission, which will decrease the effective delayed neutron fraction. This will lead to a larger power response. The opposite is true for a  $^{235}\text{U}$  diversion, which will lead to a smaller power response.

Based on these effects, a case with no diversion while using existing DNP group parameters would predict a smaller power response than the reactor will measure. This will increase the false-positive detection probability for  $^{235}\text{U}$  diversion. Additionally, a plutonium diversion would lead to the measured reactor power response aligning with the response using existing DNP groups. By using the improved DNP group parameters, this difference between the modeled and actual delayed neutron fraction can be reduced, allowing the power response to more accurately fingerprint diversion of nuclear material.

## Chapter 4

# Preliminary Results

The preliminary results use the 0D scaled model, discussed in Section 3.2.1, to generate improved DNP group parameters. The results verify the DNP group parameters using stationary thermal fission of  $^{235}\text{U}$  for which extensive experimental data exist. The verification results indicate potential issues with the short-lived DNPs, which is likely from a combination of the DNP nuclear data used and the assumptions made in the 0D scaled model. These preliminary results investigate individual DNPs, DNP group parameter sensitivities, and DNP group parameter verification.

The following results use the 0D scaled model parameters listed in Table 4.1, unless otherwise specified. The row labeled "sampled simulations" shows the number of non-linear least squares evaluations used in the Monte Carlo sampling. Each simulation uses emission probabilities, half-lives, and fission yields independently sampled within uncertainties from a normal distribution based on the nuclear data.

Table 4.1: Default parameters for 0D scaled model simulations.

| Parameter               | Value             | Units |
|-------------------------|-------------------|-------|
| Fissile Nuclide         | $^{235}\text{U}$  | [-]   |
| Incident Neutron Energy | 0.0253            | [eV]  |
| Fission Yields          | ENDF/B-VII.1 [34] | [-]   |
| Half-lives              | IAEA [51]         | [-]   |
| Emission Probability    | IAEA [51]         | [-]   |
| Final Decay Time        | 1200              | [s]   |
| Decay Time Spacing      | Logarithmic       | [-]   |
| Decay Time Nodes        | 300               | [-]   |
| Sampled Simulations     | 1000              | [-]   |
| $\tau_{in}$             | 9 [74]            | [s]   |
| $\tau_{ex}$             | 16 [74]           | [s]   |

## 4.1 Important DNPs

As discussed in Section 2.1.1, the microscopic approach uses individual DNPs to generate DNP group parameters. Because of this, the contribution of each DNP to the total delayed neutron yield can be evaluated. This is given in Equation (2.32):

$$\bar{\nu}_d = \sum_{i=1}^I P_{n,i} C F Y_i.$$

The total delayed neutron yield is a quantity of interest, as it is used for calculation of the delayed neutron fraction.

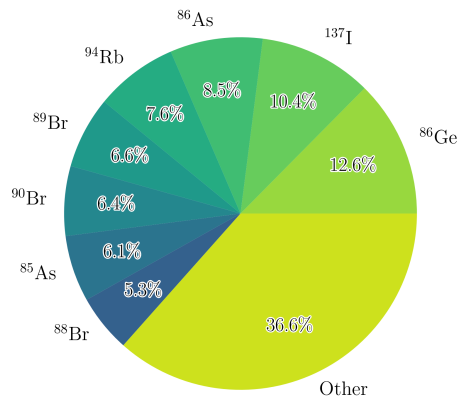
Another quantity of interest is the number of delayed neutrons from the existing concentration of nuclides, rather than the yield per fission. This quantity is particularly useful for saturation irradiations, where concentrations of different DNPs will accumulate over time when compared to the quantities present for a single fission event. This can be calculated by integrating the delayed neutron count rate, as shown in Equation (2.31), recreated here as:

$$\int_0^\infty \dot{n}_{d,i}(t) dt = P_{n,i} N_i(0).$$

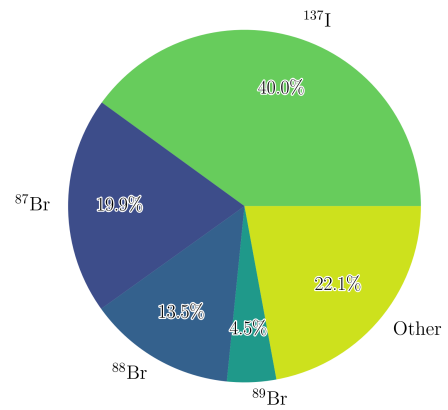
Figure 4.1 shows both approaches and how they differ. Figure 4.1(a) shows the delayed neutron yield contributions from each DNP, which indicate the DNPs which emit the most delayed neutrons after a single fission event. Figure 4.1(b) shows each DNPs' contribution to the delayed neutron count rate for a saturation irradiation. The count rate contribution is weighted more heavily towards longer-lived DNPs because they build up during the saturation irradiation. Figure 4.1(c) shows the concentrations of each DNP, correlated with Figure 4.1(b) as shown in Equation (2.31). DNPs with a large concentration may have a small emission probability and contribute few delayed neutrons to the count rate, such as  $^{141}\text{Cs}$ .

For the delayed neutron yield contributions in Figure 4.1(a),  $^{86}\text{Ge}$  contributes the most, meaning that for each thermal fission event in  $^{235}\text{U}$ , approximately 12.5% of delayed neutrons come from  $^{86}\text{Ge}$ .  $^{86}\text{Ge}$  is exceedingly short-lived, with a half-life of approximately 200 milliseconds. In Figure 4.1(b), it contributes less than 4% of the total delayed neutrons after the sample has been irradiated. Longer-lived DNPs, such as  $^{137}\text{I}$ , accumulate during irradiation, dominating the delayed neutron count rate. This helps explain why pulse irradiation data are important in experimental measurements, as saturation irradiations are dominated by long-lived DNPs. Figure 4.1(c) shows the relative concentration of each DNP. The concentrations are generally consistent with the literature, though  $^{141}\text{Cs}$  appears to be an outlier [105]. This may result from the 0D scaled model simplifications, such as excluding parasitic absorption, or because other works ignore  $^{141}\text{Cs}$  due to its extremely small emission probability.

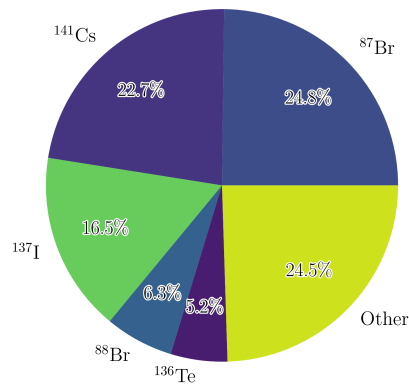
Figure 4.2 shows the delayed neutron count rate in Figure 4.2(a), along with time-integrated forms in Figures 4.2(b) and 4.2(c), representing the total delayed neutron counts at each time. The DNPs presented in this figure are based on the delayed neutron count rate; at each time step, the two DNPs with the largest delayed neutron count rate are added to the list of DNPs to be included in the figure. Figure 4.2(a) shows the  $^{86}\text{Ge}$  count rate dominates at short time scales due to its short half-life, but it becomes less important after a tenth of a second. This again demonstrates how saturation irradiation enables better analysis of the longer-lived DNPs, but reduces the relative importance of shorter-lived DNPs. Figure 4.2(b) shows total counts varying over time on a log scale.  $^{86}\text{Ge}$  initially dominates but quickly becomes unimportant compared to other DNPs. Figure 4.2(c) mirrors Figure 4.1(b), as both show  $^{86}\text{Ge}$  contribution is minimal after saturation irradiation. The total delayed neutron count contribution is dominated by longer lived DNPs like  $^{137}\text{I}$  and  $^{87}\text{Br}$ .



(a) Delayed neutron yield contributions.

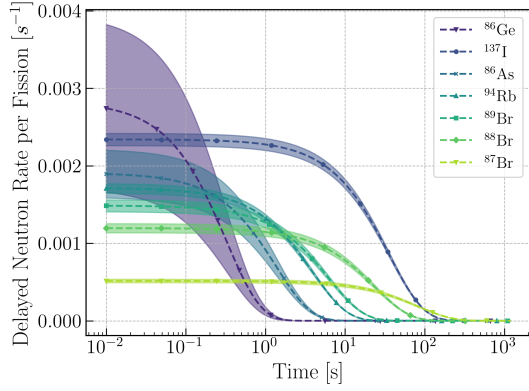


(b) Delayed neutron count rate contributions.

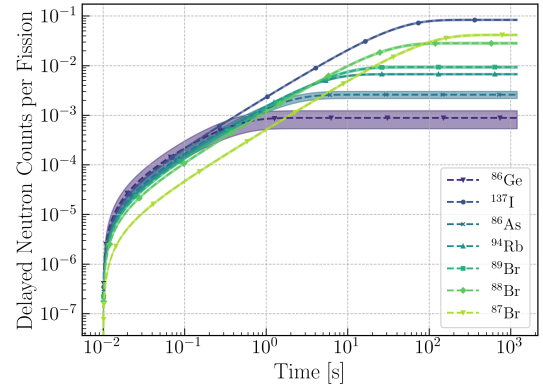


(c) DNP relative concentrations.

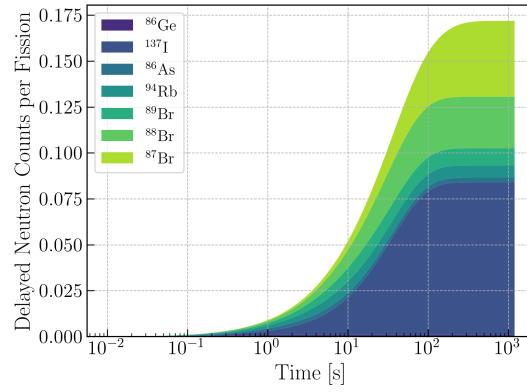
Figure 4.1: Delayed neutron contributions from each DNP after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.



(a) The delayed neutron count rate.



(b) The delayed neutron counts.



(c) A stack plot of the total delayed neutron counts.

Figure 4.2: Delayed neutron counts and count rate contributions per fission for the top two DNP's with the largest delayed neutron count rates at each time step after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

## 4.2 Sensitivity Studies

This work includes sensitivity studies of the DNP group parameters to chemical removal, time-node configuration, and nuclear data uncertainties. For the chemical removal analysis, the MSBR chemical reprocessing scheme is used. The MSBR reprocessing scheme has large chemical removal rates, providing a conservative impact of these chemical removal rates on the DNP group parameters. The time node evaluation considers node spacing, node count, total decay time, and combinations of decay time and node number.” Finally, the nuclear data uncertainties include uncertainty in the fission yields, half-lives, and emission probabilities. To propagate these uncertainties to DNP group parameters, the nonlinear least-squares solver runs 1000 times using random normal samples of each DNP’s fission yield, half-life, and emission probability.

### 4.2.1 Chemical Removal

As shown in Appendix B, the fraction of fuel salt in the ex-core region determines the scaling of the chemical removal rate. This analysis assumes an ex-core fraction,  $f_{ex}$ , of 0.5. This value reflects the relative volumes of core zone I, core zone II, and annulus compared to the total primary-system salt volume in the MSBR [154]. The justification for using the MSBR reprocessing scheme is provided in Section 3.2. Table 3.1 shows the MSBR chemical reprocessing scheme used in this analysis, recreated here with an additional column including the ex-core fraction scaling.

Table 4.2: MSBR Online Reprocessing Removal Rates from [154].

| Reprocessing Group | Element(s)                             | Removal Rate [ $s^{-1}$ ] | Scaled Removal Rate [ $s^{-1}$ ] |
|--------------------|----------------------------------------|---------------------------|----------------------------------|
| Rare Earths        | Y, La, Ce, Pr, Nd, Pm, Sm, Gd          | $2.32 \times 10^{-7}$     | $1.16 \times 10^{-7}$            |
| Rare Earths        | Eu                                     | $2.32 \times 10^{-8}$     | $1.16 \times 10^{-8}$            |
| Noble Metals       | Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te | $5.00 \times 10^{-2}$     | $2.50 \times 10^{-2}$            |
| Seminoble Metals   | Zr, Cd, In, Sn                         | $5.79 \times 10^{-8}$     | $2.90 \times 10^{-8}$            |
| Gases              | Kr, Xe                                 | $5.00 \times 10^{-2}$     | $2.50 \times 10^{-2}$            |
| Volatile Fluorides | Br, I                                  | $1.93 \times 10^{-7}$     | $9.65 \times 10^{-8}$            |
| Discard            | Rb, Sr, Cs, Ba                         | $3.37 \times 10^{-9}$     | $1.69 \times 10^{-9}$            |
| Salt Discard       | Th, Li, Be, F                          | $3.37 \times 10^{-9}$     | $1.69 \times 10^{-9}$            |
| Protactinium       | Pa                                     | $3.86 \times 10^{-6}$     | $1.93 \times 10^{-6}$            |
| Higher Nuclides    | Np, Pu                                 | $1.98 \times 10^{-9}$     | $9.9 \times 10^{-10}$            |

The 0D scaled model underestimates chemical removal rate effects, as discussed in Section 3.2.1, . This is because it applies removal only to cumulative fission yields rather than individual nuclides and their decay chains. For example,  $^{137}\text{I}$  has a chemical removal rate of  $1.9 \times 10^{-7} [s^{-1}]$ . However, its decay parents antimony and tellurium have a chemical removal rate of  $0.05 [s^{-1}]$ , many orders of magnitude larger than the iodine removal rate. The 0D

scaled model, with cumulative fission yields and no decay chain, does not capture this impact on the concentration of iodine.

### Long Cycle Time Evaluation

Table 4.2 lists the various elements with a chemical removal rate in the MSBR. Since the sample is irradiated only for a few minutes, it is useful to assess the significance of smaller removal rates. If small removal rates have a negligible impact, then the reprocessing scheme can be greatly simplified.

The cycle time is defined as the inverse of the removal rate, and can be interpreted as the linear approximation of the time for an element to be fully removed from a volume. This analysis compares a partial MSBR reprocessing scheme consisting of 20-second cycle time elements to the full MSBR reprocessing scheme. Figure 4.3 shows the difference between the full MSBR reprocessing scheme, given in Table 4.2, and the partial scheme, which only uses noble metals and gases.

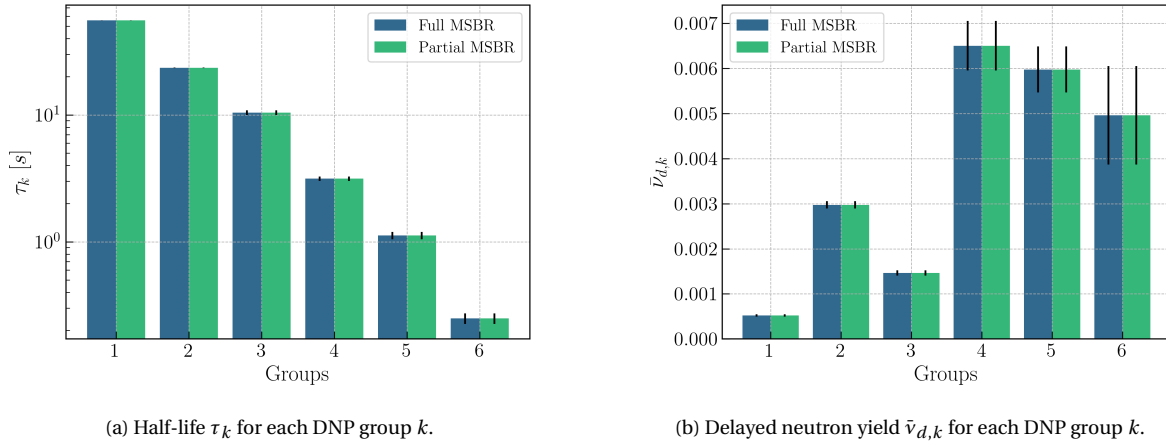


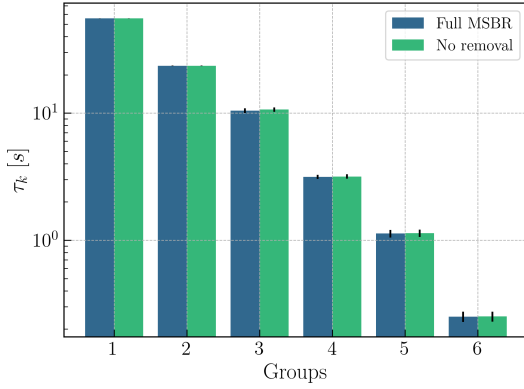
Figure 4.3: Delayed neutron group parameters calculated with (Full MSBR) and without (Partial MSBR) long cycle time chemical removal rates in the MSBR after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model. The difference between the Full MSBR and Partial MSBR simulations are minimal and well within uncertainty bounds, indicating the long cycle time chemical removal has a negligible effect on the DNP group parameters.

The differences between each DNP group parameter,  $\bar{\nu}_{d,k}$  and  $\lambda_k$ , are less than 0.001%. This analysis shows elements with cycle times longer than 20 seconds do not affect DNP group parameters for thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

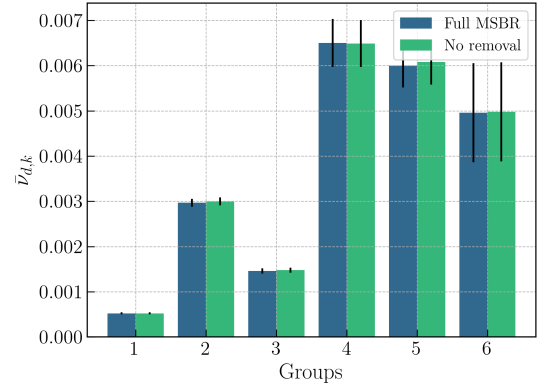
### Removal Rate Evaluation

It is also of interest to determine the sensitivity of the DNP group parameters to the chemical reprocessing scheme. The 0D scaled model underestimates chemical removal effects due to the use of cumulative fission yields without explicit nuclide tracking. This analysis still illustrates the qualitative effect of the chemical reprocessing, albeit the actual impact should be more significant. The comparison of DNP group parameters with and without the MSBR reprocessing scheme is given in Figure 4.4. The figure shows the group half-lives remain similar, while yields without removal are slightly higher.

Differences in average half-lives are negligible, while differences between the total delayed neutron yields are 0.57%, or approximately 12 pcm. This small difference likely results from the 0D scaled model underestimating



(a) Half-life  $\tau_k$  for each DNP group  $k$ .



(b) Delayed neutron yield  $\bar{v}_{d,k}$  for each DNP group  $k$ .

Figure 4.4: Delayed neutron group parameters with and without the MSBR chemical removal rates after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model. Neglecting chemical removal rates generally leads to larger yields, though all differences are within uncertainty bounds.

the effect of chemical removal. Since chemical reprocessing consistently reduces total delayed neutron yield by removing DNPs, it is important to account for this factor.

#### 4.2.2 Time Discretization

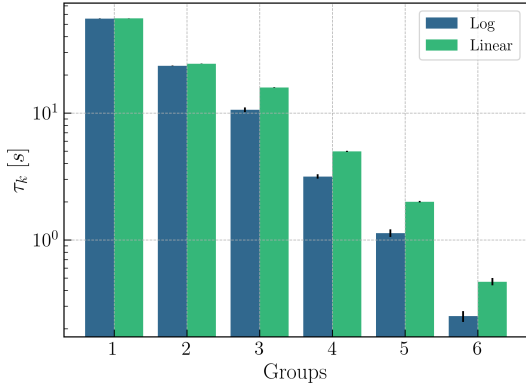
Time discretization is represented by a set of time nodes over the total decay time. The time node sensitivity is important, as many nodes may be required to fully capture the physics, but each additional node increases computational cost. Of interest are the node distribution, number of nodes, sample decay time, and combined decay time and number of nodes.

##### Distribution

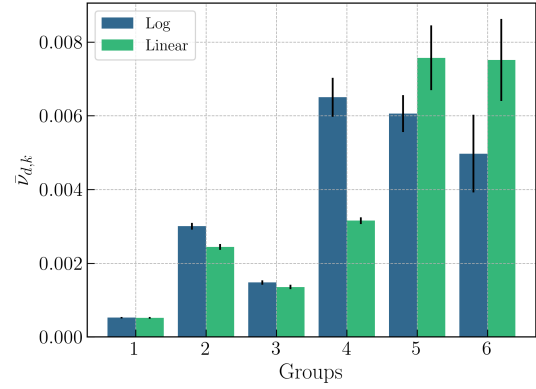
Two time node distributions are compared: logarithmic spacing from 0.01 to 600 seconds and linear spacing from 0 to 600 seconds. These distributions of nodes are compared to determine whether logarithmic spacing adequately resolves short-lived precursors while maintaining accuracy at longer times. Comparing these node distributions can inform if logarithmic spacing provides sufficient accuracy for long-lived DNP while capturing short-lived DNP, as linear spacing will struggle to capture short-lived DNP. For example, linear spacing fails to resolve behavior in the first two seconds with 300 time nodes but provides higher resolution at longer time scales. This longer time scale resolution is less useful, as many relevant DNPs quickly drop off at longer time scales, as discussed in Section 4.1.

Figure 4.5 compares DNP group parameters for logarithmic and linear spacing. Half-lives of shorter-lived groups are larger for linear spacing, consistent with expectations. Linear spacing has no data between zero and two seconds, and it places greater weight on longer-lived groups due to the data distribution. This causes shorter-lived groups to capture a larger yield, with yields shifted from longer-lived groups two, three, and four to shorter-lived groups five and six. This is because the underlying DNP data is the same, causing the shorter-lived groups to capture the short-lived DNPs as well as the longer-lived DNPs due to their longer half-lives. For example, even though group six has a longer half-life, it is still the shortest-lived group and captures the yield from the shortest-lived DNPs. Additionally, because it is longer-lived, it also captures more of the longer-lived DNPs.

The total DNP yield differs by approximately 0.1%, so changing time node spacing does not alter macroscopic



(a) Half-life  $\tau_k$  for each DNP group  $k$ .



(b) Delayed neutron yield  $\bar{v}_{d,k}$  for each DNP group  $k$ .

Figure 4.5: DNP group parameters for linear and logarithmic spacings of the time nodes after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model. The logarithmic time node spacing better captures short-lived DNPs. The differences between the DNP group parameters for the different time node spacing is outside uncertainty bounds for every group except the longest-lived group one.

DNP group behavior. Instead, it affects the distribution of delayed neutrons over time, with logarithmic spacing more accurate by capturing shorter-lived DNPs effects.

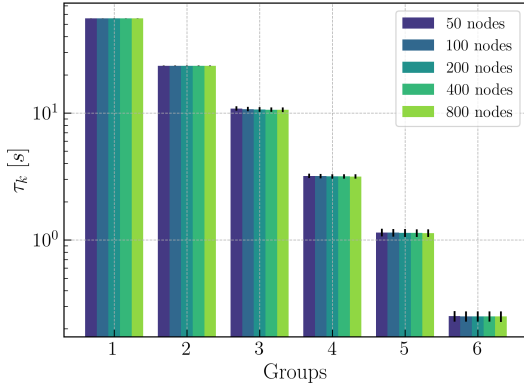
### Number of Time Nodes

Figure 4.6 shows how varying the number of nodes with a logarithmic distribution affects the results. It appears there is not a significant difference between the results when varying the number of nodes. The main difference appears in the yields of groups four and five, with increasing nodes increasing the yield of group four and decreasing the yield of group five. There are also some smaller differences, such as the yield of group six decreasing and the yields of groups two and three increasing with an increasing number of nodes.

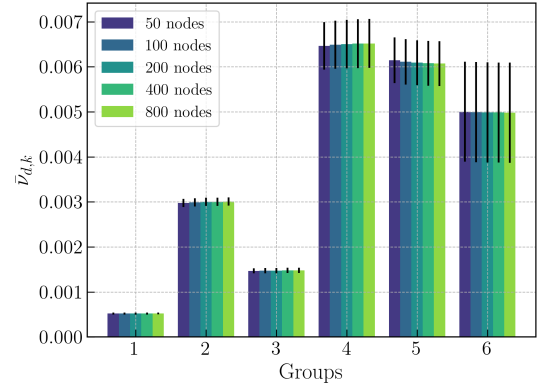
These trends arise numerically from different sampling densities rather than physical effects. The DNP group parameters with the most nodes are the most accurate, as many nodes provide the most accurate set of data for the DNP group parameters to use for least squares fitting. The logarithmic time node spacing means that the short-lived DNP groups are captured, and the long-lived DNP groups don't need as many data points to be captured. Groups four and five are slightly longer-lived, and appear to require a sufficient number of nodes for their behavior to be captured fully. The differences are within uncertainty bounds and are fairly small as the number of nodes increase, but the marginal computational cost for the increased number of nodes is negligible, as shown in Table 4.3.

The simulation with the most nodes is most accurate, which allows calculation of percent differences in DNP group yields and the associated computational cost. This is shown in Table 4.3, which shows the percent difference between each yield compared to the yield calculated with 800 nodes. The computational cost for using that number of nodes is also provided, though the cost is relatively constant. 50 nodes results in the group-five yield being off by approximately 1% compared to 800 nodes, and 200 nodes keeps differences below 0.02%. These results indicate that 300 nodes provide sufficiently accurate yields, within 0.1% of the yields using 800 nodes, while remaining in the constant computational cost regime.

Although the linear distribution of time nodes is not used in this work, the general trend of increasing nodes increasing accuracy holds. However, the shorter-lived DNP group yields vary far more significantly, as each additional node gives far more data regarding DNPs with half-lives on the order of tenths of a second. For example,



(a) Half-life  $\tau_k$  for each DNP group  $k$ .



(b) Delayed neutron yield  $\bar{v}_{d,k}$  for each DNP group  $k$ .

Figure 4.6: DNP group parameters for varying numbers of time nodes after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model. Increasing the number of nodes increases the accuracy of the DNP group parameters, with differences in the yields that are within uncertainty bounds.

Table 4.3: Comparison of how the number of nodes changes the computational cost and relative difference in yields compared to the 800 node case after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

| Nodes | Yield ( $\bar{v}_{d,k}$ ) Difference [%] |      |      |      |      |      | Time [s] |
|-------|------------------------------------------|------|------|------|------|------|----------|
|       | 1                                        | 2    | 3    | 4    | 5    | 6    |          |
| 50    | 0.30                                     | 0.90 | 0.53 | 0.75 | 0.99 | 0.59 | 297      |
| 100   | 0.17                                     | 0.49 | 0.25 | 0.39 | 0.46 | 0.77 | 297      |
| 200   | 0.01                                     | 0.02 | 0.00 | 0.01 | 0.01 | 0.09 | 300      |
| 400   | 0.00                                     | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 | 304      |
| 800   | -                                        | -    | -    | -    | -    | -    | 310      |

the group six yields for the logarithmically spaced nodes do not vary significantly, and are approximately 0.005. For the linearly spaced nodes, 50 nodes results in a group six yield of 0.015 while 800 nodes results in a group six yield of 0.0065. This is an exceedingly large difference far outside uncertainty bounds, indicating that linearly spaced nodes require many more nodes to achieve similar results as the logarithmically spaced nodes.

### Total Time Post-Irradiation

This analysis compares different total decay times for a fixed number of nodes. It is important to ensure the total length of time is accurate, as a longer decay time will introduce additional cost in the 0D flow model while a decay time that is too short will not fully capture the DNPs.

Increasing total decay time reduces node density and increases the number of nodes at later times. The decrease in time node density is because the same number of time nodes spread over a longer time frame will be less dense. The increased number of time nodes at longer times can be understood by considering an extreme example of a distribution of 100 time nodes over 100 seconds compared to 100 time nodes over 100 years; even with logarithmic spacing, most of the nodes will be placed at a longer time in the 100 year case than the 100 second case. Thus, longer decay times are expected to improve characterization of long-lived groups while logarithmic spacing should enable sufficient resolution for short-lived groups.

Because the longest-lived DNP<sup>1</sup>,  $^{87}\text{Br}$ , has a half-life of approximately one minute, it is reasonable to assume that

<sup>1</sup>The DNPs  $^{211}\text{Tl}$ ,  $^{210}\text{Tl}$ , and  $^{210}\text{Hg}$  are longer-lived and have non-zero emission probabilities, but they live only tens of seconds longer and

a final time of ten minutes offers sufficient data on all DNP decays. Figure 4.7 shows this assumption is not accurate. The DNP group parameters,  $\tau_k$  and  $\bar{\nu}_{d,k}$ , converge asymptotically as the total decay time increases, with the decay time of 1200 seconds, or 20 minutes, being the smallest decay time after which longer decay times do not change the results. The shorter total decay times inaccurately model DNPs, as they do not fully capture long-lived DNPs and have shorter group half-lives. These shorter half-lives correlate with an overestimation of some longer-lived DNP group yields.

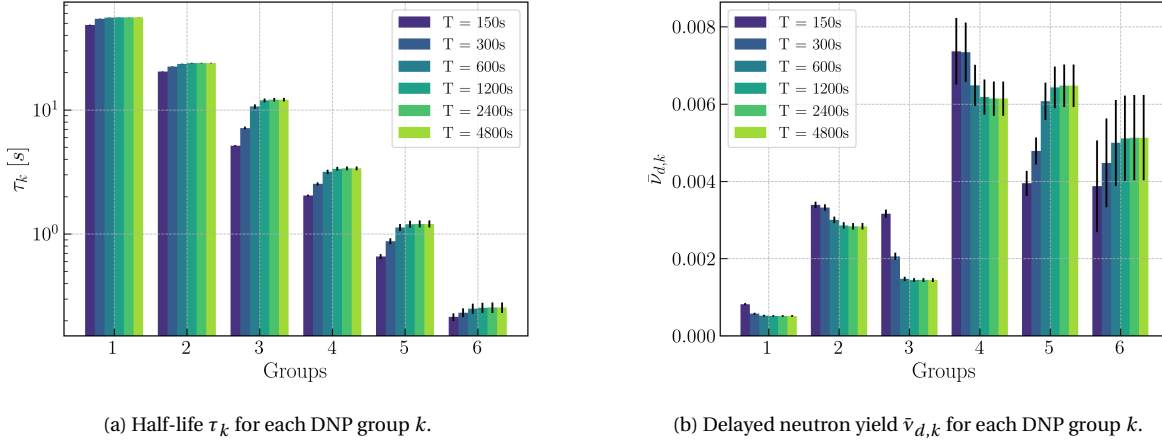


Figure 4.7: DNP group parameters for varying total decay times after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model. Collecting delayed neutron counts for longer after irradiation results in longer-lived groups with smaller yields in groups one through four and larger yields in groups five and six.

The total delayed neutron yields and average half-lives, shown in Table 4.4, provide another comparison between different decay times. The table shows differences are small and within uncertainty bounds.

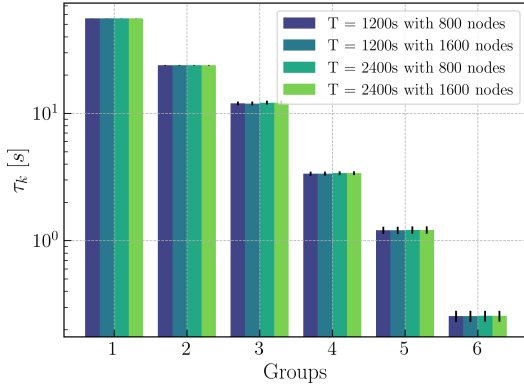
Table 4.4: Comparison of how the net decay time,  $T$  changes the total delayed neutron yield and average half-life after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

| $T$ [s] | $\bar{\nu}_d$       | $\bar{\tau}$ [s]    |
|---------|---------------------|---------------------|
| 150     | $0.0225 \pm 0.0015$ | $6.3786 \pm 0.3784$ |
| 300     | $0.0225 \pm 0.0014$ | $6.3945 \pm 0.3554$ |
| 600     | $0.0225 \pm 0.0013$ | $6.3978 \pm 0.3412$ |
| 1200    | $0.0225 \pm 0.0013$ | $6.3977 \pm 0.3415$ |
| 2400    | $0.0225 \pm 0.0013$ | $6.3977 \pm 0.3417$ |
| 4800    | $0.0225 \pm 0.0013$ | $6.3977 \pm 0.3417$ |

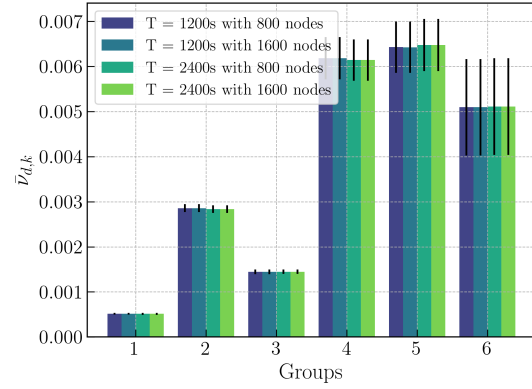
### Time Node Density

This section evaluates DNP group parameter sensitivity to the time node density, number of time nodes, and total decay time by comparing different numbers of time nodes and decay times. Figure 4.8 shows how the DNP group parameters vary with time node density. For a linearly spaced set of time nodes, this could be represented by the time step. If DNP group yields and half-lives align when the time node density values are the same but the total decay time and number of time nodes differ, then that would indicate that those parameters are less significant than the density of time nodes.

generally have negligible concentrations.



(a) Half-life  $\tau_k$  for each DNP group  $k$ .



(b) Delayed neutron yield  $\bar{\nu}_{d,k}$  for each DNP group  $k$ .

Figure 4.8: DNP group parameters,  $\tau_k$  and  $\bar{\nu}_{d,k}$ , for varying node densities after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model. Changing the number of nodes does not have a significant change on the DNP group parameters, whereas changes to the total decay time leads to significant differences.

The 1200 second case with 800 nodes has a time-node density of roughly two nodes per three seconds. The 2400 second case with 1600 nodes has the same time-node density. If the DNP group parameters primarily depend on the node density, then these two different sets of decay times and time nodes should give the same set of DNP group parameters. Instead, the results remain approximately constant with varying node numbers, with the main difference arising from the total decay time. This indicates the total decay time sensitivity analysis does not need to account for varying time node density. For a linearly distributed set of nodes, this result would likely be different, as both the number of nodes and the total decay time would be important.

### 4.2.3 Nuclear Data Uncertainties

This work uses a Monte Carlo approach to propagate the uncertainties in nuclear data and evaluate the sensitivity of the DNP group parameters to these uncertainties. For each nuclear data variable in the 0D scaled model, such as the  $^{87}\text{Br}$  half-life, a random value is sampled from a normal distribution. This process is repeated for every DNP fission yield, half-life, and emission probability. For the analyses in this section, 5000 Monte Carlo non-linear least squares solves are used with 800 decay time nodes each. The other parameters are the same as given in Table 4.1.

#### Count Rate

Figure 4.9 shows the count rate normalized to the Keepin et al. count rate. The red lines show count rates from sampled simulations, the blue dashed line shows the DNP group count rate, and the black X's indicate the mean count rate at each point. The other six-group DNP parameters plotted include Keepin (1.0, as the plot is normalized to Keepin values), Brady, and Synetos [95, 23, 187].

Beyond approximately three seconds, the current DNP group parameters agree within uncertainty bounds with most literature results, with Brady parameters showing a noticeable difference at longer times. Short time scales are inaccurate in the 0D scaled model, potentially due to the combination of issues in the DNP data used and lack of pulse irradiation data. To compare only the longer-lived group parameters, the first two groups are used while the shorter-lived groups three through six are replaced with the Keepin et al. DNP group parameters [95]. Figure 4.10 shows this modified set of DNP group parameters. A more thorough analysis of the pulse irradiation data and DNP

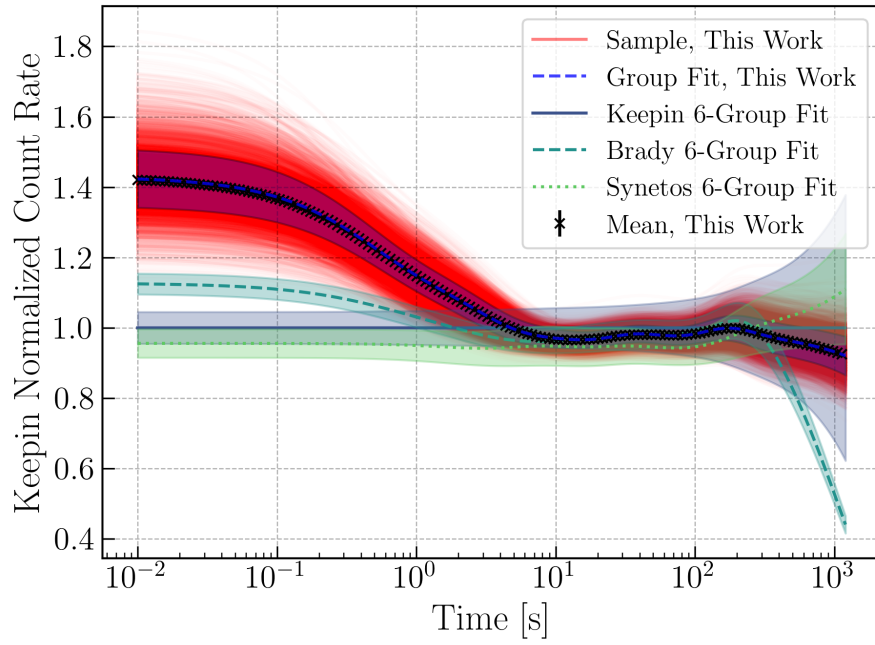


Figure 4.9: The delayed neutron count rate for thermal fission of  $^{235}\text{U}$  normalized to the Keepin et al. six-group DNP group parameters. The current work, represented by the Sampled, Group Fit, and Mean data trends shows poor agreement at times less than one second and agreement within uncertainty bounds beyond three seconds.

data will be conducted, as discussed in Chapter 5.

The modified DNP group parameters are not a perfect comparison, as the non-linear least squares solve adjusts each group's yield relative to the yields and half-lives of other groups. This indicates the potential combination of DNP data with the missing pulse irradiation data also affects longer-lived groups, though less than it affects shorter-lived groups. Adjusting the shorter-lived groups improves agreement with the literature for the 0D scaled model, particularly at times less than one second.

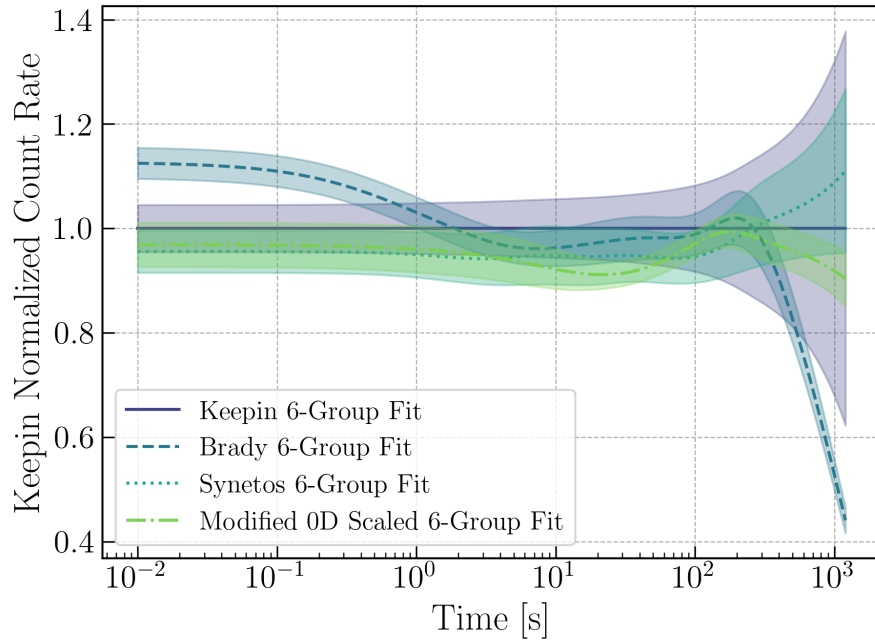


Figure 4.10: The delayed neutron count rate for thermal fission of  $^{235}\text{U}$  normalized to the Keepin et al. six-group DNP group parameters. The modified 0D scaled model has overlapping uncertainty bounds with the Keepin et al. results at all points in time.

### Individual DNPs

The Monte Carlo sampling of each DNP's nuclear data performed in this work enables sensitivity analyses on how each change affects the DNP group parameters. For example, because group one is dominated by  $^{87}\text{Br}$ , changes to its half-life, emission probability, or fission yield noticeably affect the group one DNP half-life and yield.

To find the most sensitive DNPs that experience the most difference in DNP group parameters  $\tau_k$  and  $\bar{v}_{d,k}$ , a linear regression fits the slope of the relative change in DNP group parameters to the relative change in the individual DNP nuclear data, while also calculating the Pearson correlation coefficient (PCC). The PCC quantifies the linear relationship between two variables: -1 indicates perfect negative correlation, +1 indicates perfect positive correlation, and 0 indicates no correlation [64]. Changes in the nuclear data include emission probability  $P_n$ , fission yields (affecting concentration  $N$ ), and half-lives  $\tau$ .

Correlations of interest for this analysis have absolute PCC values greater than 0.2. This threshold indicates a DNP has at least a non-negligible correlation with a DNP group parameter. Table 4.5 presents the results of this analysis.

One of the most predictable sensitivities in Table 4.5 is  $^{87}\text{Br}$ 's effect on group one half-life and yield, as it dominates this group. Specifically,  $^{87}\text{Br}$ , with a half-life of  $55.64 \pm 0.15$  seconds, has correlations with PCCs greater than 0.2 that impact the following group parameters:  $^{87}\text{Br}$ 's emission probability and concentration affect the group one yield and  $^{87}\text{Br}$ 's half-life affects the group one half-life. Figure 4.11 shows how group one parameters vary with  $^{87}\text{Br}$  data. The figure also shows an example of weakly correlated data using the  $^{87}\text{Br}$  emission probability on the group three half-life, though even this weak correlation has some non-negligible impact.

Although it may seem counterintuitive, changes to the  $^{87}\text{Br}$  emission probability affect the group three half-life, showing a non-negligible correlation. The impact of  $^{87}\text{Br}$  on group one is expected, as well as some differences

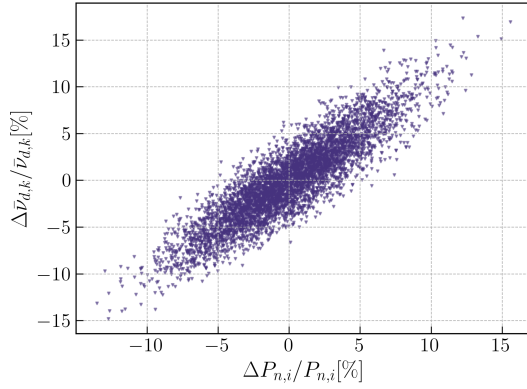
Table 4.5: PCCs with a magnitude larger than 0.2 for variation in DNP data which correspond to changes in DNP group parameters after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

| Nuclide           | Group Value         | DNP Value  | PCC       |
|-------------------|---------------------|------------|-----------|
| $^{137}\text{I}$  | $\bar{v}_{d,2} [-]$ | $P_n [-]$  | 0.483443  |
| $^{137}\text{I}$  | $\bar{v}_{d,2} [-]$ | $\tau [s]$ | -0.275226 |
| $^{137}\text{I}$  | $\bar{v}_{d,2} [-]$ | $N [-]$    | 0.736407  |
| $^{137}\text{I}$  | $\tau_2 [s]$        | $\tau [s]$ | 0.868136  |
| $^{86}\text{As}$  | $\bar{v}_{d,5} [-]$ | $N [-]$    | 0.450798  |
| $^{86}\text{As}$  | $\tau_5 [s]$        | $N [-]$    | -0.243154 |
| $^{98}\text{Y}$   | $\bar{v}_{d,4} [-]$ | $P_n [-]$  | 0.217891  |
| $^{86}\text{Ge}$  | $\bar{v}_{d,2} [-]$ | $\tau [s]$ | -0.302236 |
| $^{86}\text{Ge}$  | $\tau_2 [s]$        | $\tau [s]$ | 0.365628  |
| $^{86}\text{Ge}$  | $\bar{v}_{d,3} [-]$ | $\tau [s]$ | -0.492681 |
| $^{86}\text{Ge}$  | $\tau_3 [s]$        | $\tau [s]$ | 0.690005  |
| $^{86}\text{Ge}$  | $\bar{v}_{d,4} [-]$ | $\tau [s]$ | -0.494493 |
| $^{86}\text{Ge}$  | $\tau_4 [s]$        | $\tau [s]$ | 0.707295  |
| $^{86}\text{Ge}$  | $\tau_5 [s]$        | $\tau [s]$ | 0.644484  |
| $^{86}\text{Ge}$  | $\bar{v}_{d,6} [-]$ | $P_n [-]$  | 0.836040  |
| $^{86}\text{Ge}$  | $\bar{v}_{d,6} [-]$ | $N [-]$    | 0.485383  |
| $^{86}\text{Ge}$  | $\tau_6 [s]$        | $P_n [-]$  | -0.333414 |
| $^{86}\text{Ge}$  | $\tau_6 [s]$        | $\tau [s]$ | 0.862992  |
| $^{137}\text{Sb}$ | $\tau_3 [s]$        | $N [-]$    | -0.236469 |
| $^{137}\text{Sb}$ | $\tau_4 [s]$        | $N [-]$    | -0.234367 |
| $^{137}\text{Sb}$ | $\tau_5 [s]$        | $N [-]$    | -0.294677 |
| $^{87}\text{Br}$  | $\bar{v}_{d,1} [-]$ | $P_n [-]$  | 0.891153  |
| $^{87}\text{Br}$  | $\bar{v}_{d,1} [-]$ | $N [-]$    | 0.446840  |
| $^{87}\text{Br}$  | $\tau_1 [s]$        | $\tau [s]$ | 0.999924  |
| $^{85}\text{As}$  | $\bar{v}_{d,4} [-]$ | $N [-]$    | 0.700410  |
| $^{85}\text{As}$  | $\tau_4 [s]$        | $N [-]$    | -0.387391 |
| $^{85}\text{As}$  | $\bar{v}_{d,5} [-]$ | $N [-]$    | 0.791653  |
| $^{85}\text{As}$  | $\tau_5 [s]$        | $N [-]$    | 0.476758  |
| $^{88}\text{Br}$  | $\bar{v}_{d,3} [-]$ | $P_n [-]$  | 0.548987  |
| $^{88}\text{Br}$  | $\bar{v}_{d,3} [-]$ | $\tau [s]$ | -0.280146 |
| $^{88}\text{Br}$  | $\bar{v}_{d,3} [-]$ | $N [-]$    | 0.377025  |
| $^{88}\text{Br}$  | $\tau_3 [s]$        | $P_n [-]$  | 0.352869  |
| $^{88}\text{Br}$  | $\tau_3 [s]$        | $N [-]$    | 0.245891  |

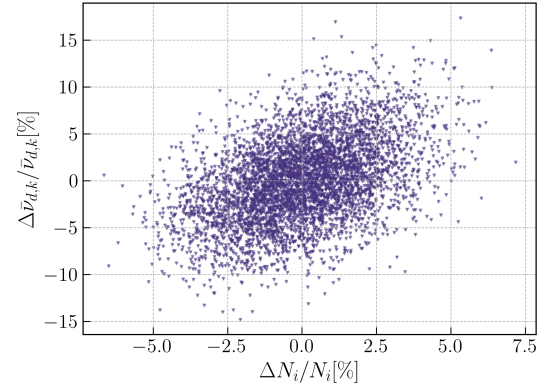
in group two, but even changes to the emission probability of  $^{87}\text{Br}$  has an effect on the half-life of group three. This is because the DNP group parameters cannot be considered in isolation; they form a collective set that best fits the delayed neutron count rate curve. Changes to that curve propagate to all group parameters, with varying magnitudes across groups and parameters.

Another DNP of interest in Table 4.5 is  $^{86}\text{Ge}$ , an important contributor to delayed neutron yield with a half-life of  $0.222 \pm 0.022$  seconds, as shown in Figure 4.1. Variation in  $^{86}\text{Ge}$  data impacts many DNP group parameters, especially due to large relative uncertainty in its half-life of 9.9%. Figures 4.12 and 4.13 show each DNP group parameter's sensitivity to variations in  $^{86}\text{Ge}$  data.

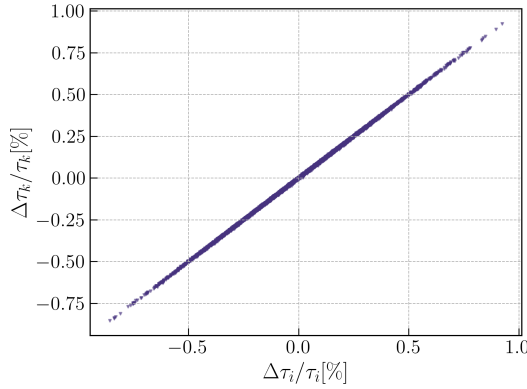
Figure 4.12 shows the largest group half-life sensitivities occur due to changes in  $^{86}\text{Ge}$  half-life. For increasingly longer-lived groups,  $^{86}\text{Ge}$  half-life has progressively less impact on group half-lives. This is intuitive, as a short-lived nuclide such as  $^{86}\text{Ge}$  should have much more impact on short-lived groups than on long-lived groups. Group six half-life sensitivity to  $^{86}\text{Ge}$  shows a lower bound with no data points, indicating asymmetry. Sensitivities typically



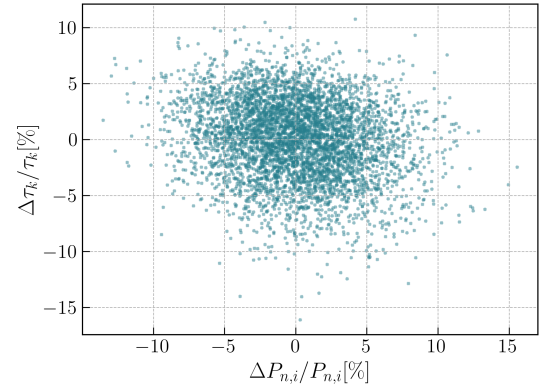
(a) DNP group 1 yield variation due to emission probability variation with a PCC of 0.90.



(b) DNP group 1 yield variation due to fission yield variation with a PCC of 0.46.



(c) DNP group 1 half-life variation due to half-life variation with a PCC of 1.00.



(d) DNP group 3 half-life variation due to emission probability variation with a PCC of -0.19.

Figure 4.11: DNP group parameter sensitivity to changes in  $^{87}\text{Br}$  nuclear data after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model. The group one PCC values are larger than that of group three because  $^{87}\text{Br}$ 's half-life is closer to that of group one than the half-life of group three.

exhibit symmetry about the line that best fits the data for a linear response. This asymmetry likely arises because many longer-lived DNPs exist, whereas  $^{86}\text{Ge}$  is one of the few very short-lived DNPs dominating this time region. Because shorter-lived DNPs are absent in this time region, group six half-life decreases less than it increases.

Figure 4.13 shows changes in the group yields due to changes in the  $^{86}\text{Ge}$  half-life. No clear trend exists from shorter to longer-lived groups, though groups three and four show negative correlations, suggesting a longer-lived  $^{86}\text{Ge}$  decreases their yields while increasing their half-lives, with half-lives shown in Figure 4.12. These groups capture fewer short-lived DNPs as the  $^{86}\text{Ge}$  half-life increases, both by increasing their group half-lives and decreasing their yields.

Another interesting sensitivity is the  $^{86}\text{Ge}$  emission probability  $P_n$  on the group six yield, showing a narrow spread for small emission probabilities that widens for larger emission probabilities. Increasing the emission probability for the short-lived  $^{86}\text{Ge}$  corresponds to an increase in the short-lived group 6 yield. This larger spread indicates a smaller magnitude of the PCC, which means that as  $^{86}\text{Ge}$  emits more delayed neutrons, its marginal impact on group six becomes smaller.

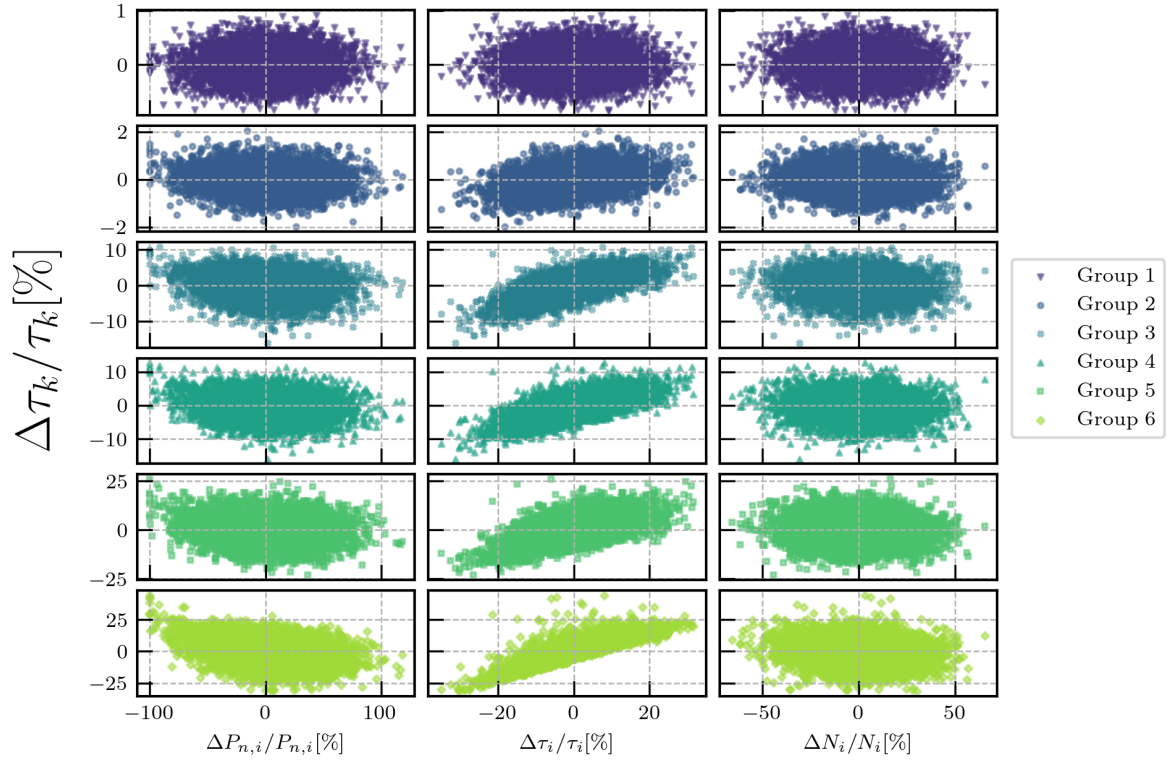


Figure 4.12: The difference in each DNP group half-life from the  $^{86}\text{Ge}$  relative changes in emission probability, half-life, and concentration after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model. The half-life of the shorter-lived groups is more sensitive to the half-life of  $^{86}\text{Ge}$ , with less correlation for longer-lived groups.

A metric to compare the overall sensitivity of the group parameters to nuclear data for each DNP is the summed magnitude of the PCCs,  $\sum |PCC|$ . This value provides insight into how changing the nuclear data for a given DNP affects all group parameters. Figure 4.14 shows these values for each DNP, where the theoretical maximum value would be a magnitude of one for each group half-life and yield, resulting in a net value of 12. A value of 12 would indicate any change to the DNP's nuclear data results in a linear response in every DNP group parameter. The largest value is for  $^{86}\text{Ge}$ , with a value of approximately 8.1, indicating that uncertainties in  $^{86}\text{Ge}$  will have a large impact on the DNP group parameters.

Table 4.6 shows the same information as Figure 4.14, but only for the top 10 DNPs. This table makes it clear that  $^{86}\text{Ge}$  is an outlier, with a summed PCC over twice as large as the next largest value. This is likely due to its large delayed neutron yield and short half-life, which leads to it affecting more groups simultaneously. Nearly every DNPs which appear in this table also appear in Table 4.5, as to have a large magnitude of PCCs, it makes sense that at least one of those magnitudes would be greater than 0.2. However,  $^{100}\text{Rb}$  does not appear in Table 4.5, indicating it has many PCCs smaller than 0.2.

These PCCs can be combined with the relative uncertainty associated with each DNP to capture the nuclides which both impact the DNP group parameters most significantly and have large uncertainties. Specifically, this is

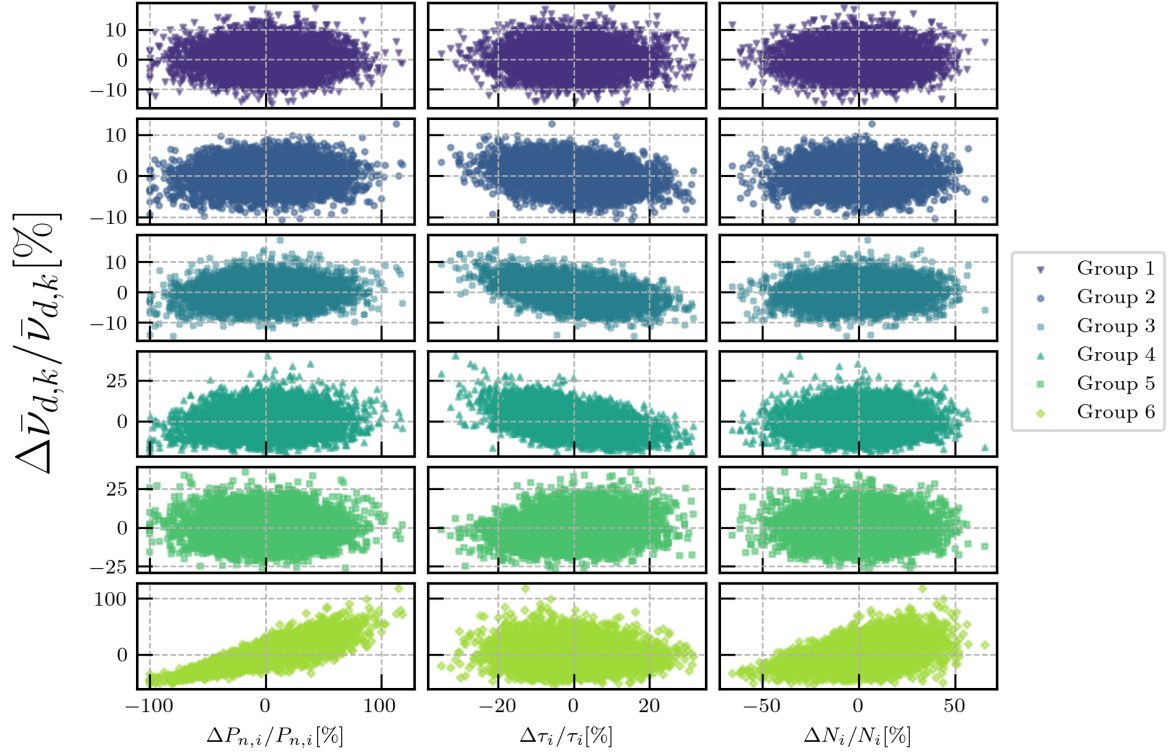


Figure 4.13: The difference in each DNP group yield from the  $^{86}\text{Ge}$  relative changes in emission probability, half-life, and concentration after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model. The group six yields are sensitive to the emission probability and concentration of  $^{86}\text{Ge}$ . The group yields are also sensitive to the half-life of  $^{86}\text{Ge}$ , as the changes in the  $^{86}\text{Ge}$  half-life affects both the group half-lives and their yields.

Table 4.6: The sum of the PCC magnitudes for the top 10 DNP on each group half-life  $\tau_k$  and yield  $\bar{v}_{d,k}$  after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

| Nuclide           | $\sum  PCC $ |
|-------------------|--------------|
| $^{86}\text{Ge}$  | 8.06         |
| $^{87}\text{Br}$  | 3.82         |
| $^{137}\text{I}$  | 3.52         |
| $^{85}\text{As}$  | 3.14         |
| $^{88}\text{Br}$  | 2.77         |
| $^{137}\text{Sb}$ | 2.08         |
| $^{86}\text{As}$  | 1.97         |
| $^{84}\text{Ga}$  | 1.62         |
| $^{98}\text{Y}$   | 1.42         |
| $^{100}\text{Rb}$ | 1.22         |

calculated as the relative uncertainties times the magnitude of the PCC for a DNP  $i$  and DNP value  $v$ , or  $U_{i,v}$ :

$$U_{i,v} = \frac{\sigma_{i,v}}{x_{i,v}} \sum_g |PCC_{i,v,g}|, \quad (4.1)$$

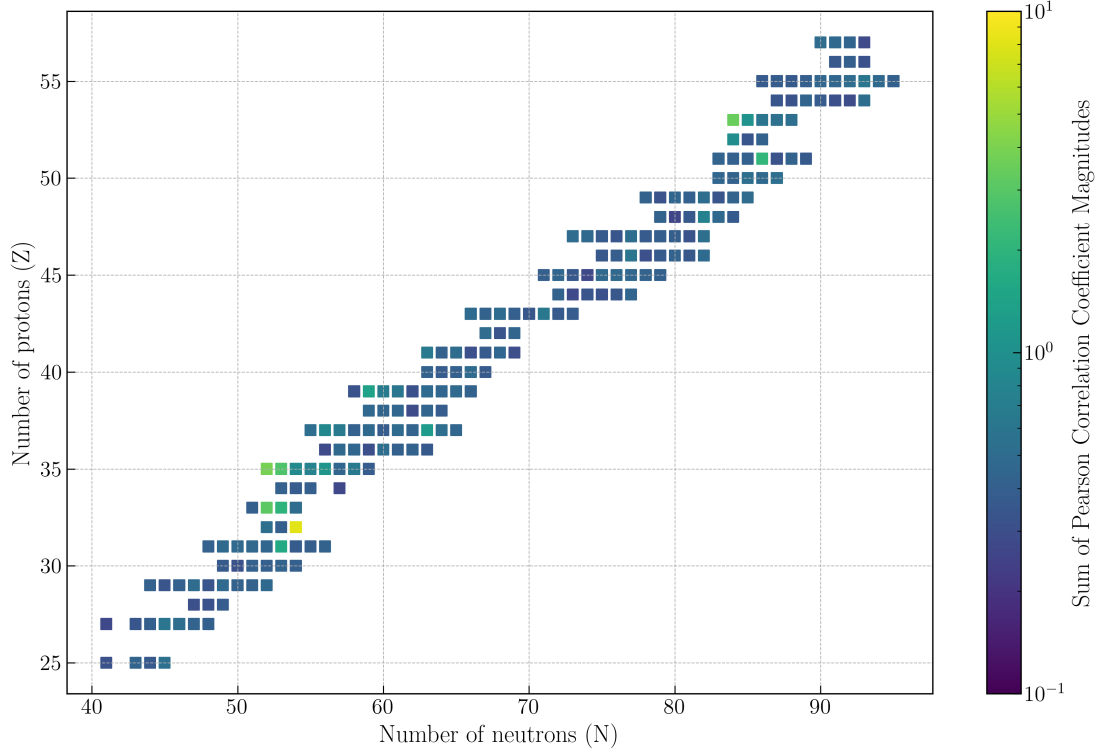


Figure 4.14: The sum of the PCC magnitudes for each DNP on each group half-life  $\tau_k$  and yield  $\bar{v}_{d,k}$  after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model. The  $^{86}\text{Ge}$  summed value is noticeably larger than the others, indicating its uncertainties have a larger impact on group parameters than other DNPs.

where

$U_{i,v}$  = the uncertainty-weighted sensitivity of DNP value  $v$  for DNP  $i$  [-]

$\sigma_{i,v}$  = the uncertainty in the DNP value  $v$  for DNP  $i$  [-]

$x_{i,v}$  = the DNP value of half-life, emission probability, or concentration ( $v$ ) of DNP  $i$  [-]

$PCC_{i,v,g}$  = the PCC for DNP  $i$  and value  $v$  with DNP group parameter  $g$  [-].

(4.2)

Figure 4.15 shows that, once again,  $^{86}\text{Ge}$  is the largest value, indicating that not only do the uncertainties in its data have a large effect on the DNP group parameters, but it also has large relative uncertainties. When compared to Figure 4.14, it is noticeable that the values are generally much smaller due to scaling by the relative uncertainties. Table 4.7 gives more insight into these values, in which  $^{86}\text{Ge}$  is approximately four times larger than the next largest value. This table shows the most DNPs that require improved data due to the combination of their importance to DNP group parameter calculation and their current uncertainties.

The only DNPs that share entries in both tables are  $^{86}\text{Ge}$  and  $^{98}\text{Y}$ , indicating that these DNPs don't have exceedingly large relative uncertainties, but rather relative uncertainties that are large relative to their impact on DNP group parameters. Figure 4.16 shows how the relative uncertainties and sensitivity combinations, represented using  $U_x$ , differ for each DNP and each set of DNP data  $x$ , such as emission probability  $P_n$  or half-life  $\tau$ . The ten

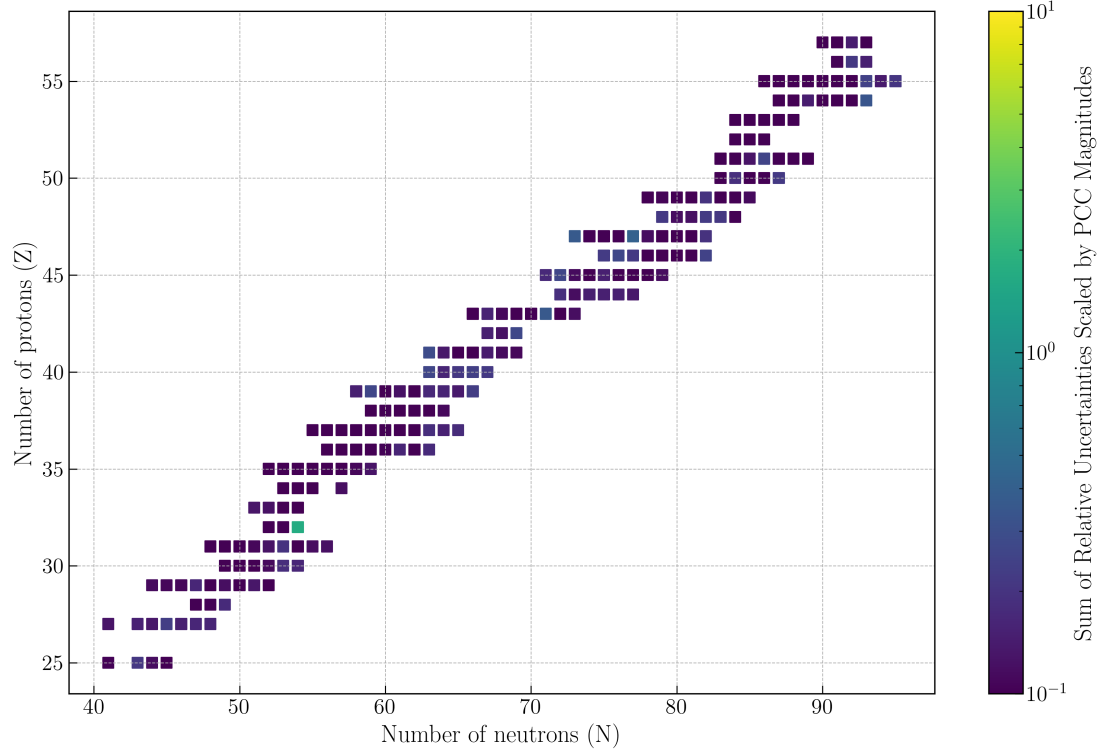


Figure 4.15: The sum of the PCC magnitudes times the relative uncertainties for each DNP on each group half-life  $\tau_k$  and yield  $\tilde{v}_{d,k}$  after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model. The  $^{86}\text{Ge}$  summed value is slightly larger than the others, though there are many DNPs which can use improved data.

Table 4.7: The sum of the relative uncertainties scaled by the PCC magnitudes for the top 10 DNP on each group half-life  $\tau_k$  and yield  $\tilde{v}_{d,k}$  after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

| Nuclide           | $\sum  PCC $ |
|-------------------|--------------|
| $^{86}\text{Ge}$  | 1.71         |
| $^{124}\text{Ag}$ | 0.41         |
| $^{120}\text{Ag}$ | 0.36         |
| $^{114}\text{Tc}$ | 0.36         |
| $^{147}\text{Xe}$ | 0.34         |
| $^{104}\text{Nb}$ | 0.29         |
| $^{111}\text{Mo}$ | 0.27         |
| $^{122}\text{Pd}$ | 0.27         |
| $^{98}\text{Y}$   | 0.26         |
| $^{103}\text{Zr}$ | 0.26         |

largest values of  $U_x$  show that half-life and emission probability DNP data has the largest effect on DNP group parameters when accounting for the relative uncertainties in the data. All of the DNPs with large values of  $U_x$  have short half-lives, with the longest half-life of the ten most significant DNPs being approximately 2 seconds from  $^{98}\text{Y}$ .

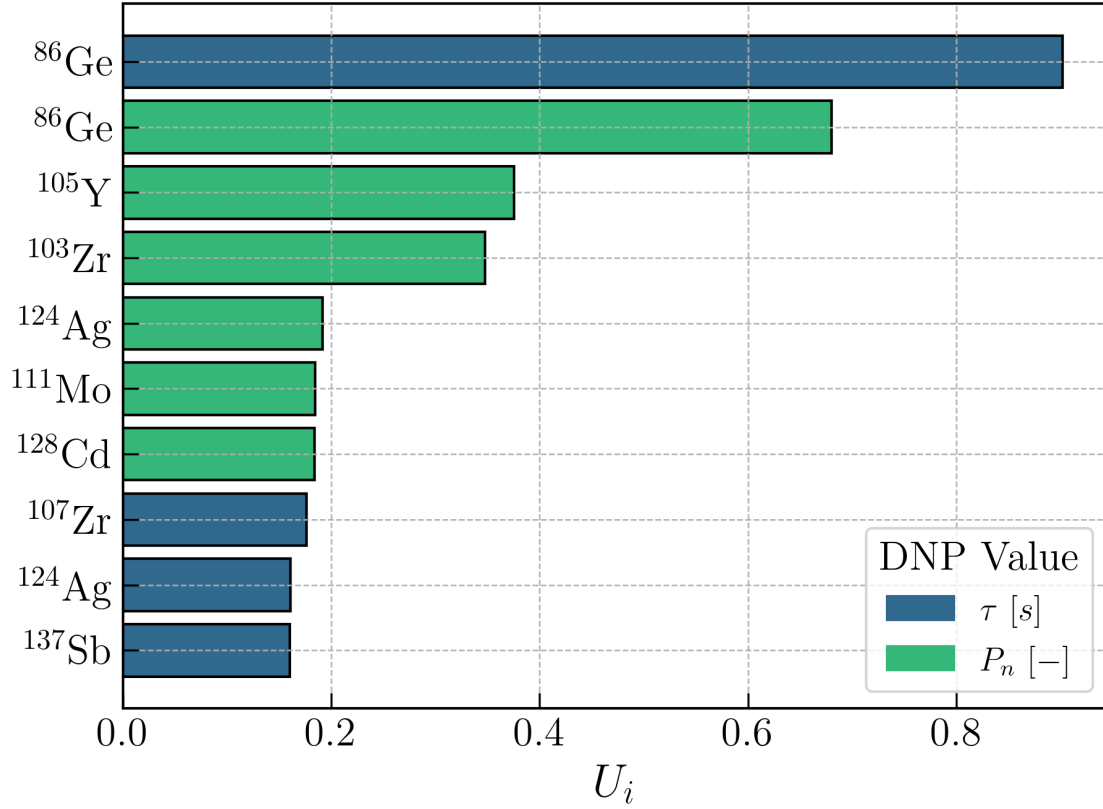


Figure 4.16: The sum of the PCC magnitudes times the relative uncertainties for each DNP  $i$ , isolated by each DNP value on each group half-life  $\tau_k$  and yield  $\bar{v}_{d,k}$  after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model. The  $^{86}\text{Ge}$  values are noticeably larger, and the most important data to improve are the half-lives and emission probabilities for short-lived DNP.

### 4.3 Verification

The 0D scaled model verification uses DNP group parameters from thermal fission of  $^{235}\text{U}$ . The first verification step calculates the delayed neutron yield  $\bar{v}_d$  and average half-life  $\bar{\tau}$ , shown in Table 4.8. These values are calculated and compared with the six group DNP group parameters formulated from those DNPs. Ideally, these values would be identical, but the six-group fit cannot fully capture all decaying DNPs behavior, producing small differences. These differences are small relative to the uncertainty bounds for each term, which verifies that the group fit is accurately capturing the general behavior of the DNPs.

Table 4.8: The delayed neutron yield  $\bar{v}_d$  and average half-life  $\bar{\tau}$  calculated by summing individual DNPs and from the 6 DNP group parameters.

| Term             | DNP                 | Group               |
|------------------|---------------------|---------------------|
| $\bar{v}_d$      | $0.0225 \pm 0.0015$ | $0.0224 \pm 0.0013$ |
| $\bar{\tau}$ [s] | $6.3852 \pm 0.3801$ | $6.4029 \pm 0.3365$ |

Verification against literature data, introduced in Section 2.1.1, is combined and recreated here with this work's results. Tables 4.9, 4.10, 4.11, and 4.12 show the DNP group parameters from literature and how they compare to the

0D scaled model DNP group parameters. As discussed, the 0D scaled model short-lived groups do not align with results in the literature, which may be from a combination of the current DNP data and from exclusion of pulse irradiation data. This appears in the yields: groups five and six differ significantly, while groups one and two match the literature more closely. The group half-lives generally agree with the literature, though groups three and five are outliers. This could be because the cumulative fission yield of short-lived DNPs, such as  $^{86}\text{Ge}$ , are too large in ENDF/B-VII.1; the emission probabilities in the IAEA database for short-lived DNP are too large; or because the lack of pulse irradiation data leads to overestimation of the contribution of the short-lived DNPs.

Table 4.9: DNP group yields from thermal fission of  $^{235}\text{U}$  (recreated from [135]).

| Source                      | Group [-] |         |         |         |         |         |
|-----------------------------|-----------|---------|---------|---------|---------|---------|
|                             | 1         | 2       | 3       | 4       | 5       | 6       |
| Keepin et al. [95]          | 0.00052   | 0.00346 | 0.00310 | 0.00624 | 0.00182 | 0.00066 |
| Charlton et al. [36]*       | 0.00054   | 0.00344 | 0.00319 | 0.00607 | 0.00186 | 0.00066 |
| ENDF/B-VI.8 [159]           | 0.00058   | 0.00302 | 0.00288 | 0.00646 | 0.00265 | 0.00111 |
| Mills et al. [125]          | 0.00056   | 0.00325 | 0.00196 | 0.00658 | 0.00287 | 0.00122 |
| Brady and England [23]      | 0.00068   | 0.00341 | 0.00292 | 0.00611 | 0.00310 | 0.00158 |
| Saleh et al. [166]          | 0.00055   | 0.00363 | 0.00297 | 0.00593 | 0.00168 | 0.00063 |
| Synetos and Williams [187]* | 0.00051   | 0.00328 | 0.00214 | 0.00583 | 0.00270 | 0.00063 |
| Tuttle [195]                | 0.00065   | 0.00362 | 0.00320 | 0.00692 | 0.00218 | 0.00044 |
| Waldo et al.[200]           | 0.00055   | 0.00366 | 0.00327 | 0.00660 | 0.00192 | 0.00070 |
| 0D scaled model             | 0.00052   | 0.00297 | 0.00146 | 0.00650 | 0.00597 | 0.00500 |

\* Modified to include a sixth group by Parish et al. [135].

Table 4.10: DNP group yield uncertainties from thermal fission of  $^{235}\text{U}$  (recreated from [135]).

| Source                      | Group Uncertainties[-] |               |               |               |               |               |
|-----------------------------|------------------------|---------------|---------------|---------------|---------------|---------------|
|                             | $\sigma_1$             | $\sigma_2$    | $\sigma_3$    | $\sigma_4$    | $\sigma_5$    | $\sigma_6$    |
| Keepin et al. [95]          | $\pm 0.00006$          | $\pm 0.00025$ | $\pm 0.00041$ | $\pm 0.00047$ | $\pm 0.00019$ | $\pm 0.00013$ |
| Charlton et al. [36]*       | $\pm 0.00003$          | $\pm 0.00005$ | $\pm 0.00024$ | $\pm 0.00013$ | $\pm 0.00014$ | $\pm 0.00014$ |
| ENDF/B-VI.8 [159]           | -                      | -             | -             | -             | -             | -             |
| Mills et al. [125]          | $\pm 0.00004$          | $\pm 0.00023$ | $\pm 0.00020$ | $\pm 0.00051$ | $\pm 0.00035$ | $\pm 0.00012$ |
| Brady and England [23]      | $\pm 0.00004$          | $\pm 0.00019$ | $\pm 0.00016$ | $\pm 0.00034$ | $\pm 0.00017$ | $\pm 0.00009$ |
| Saleh et al. [166]          | $\pm 0.00005$          | $\pm 0.00032$ | $\pm 0.00027$ | $\pm 0.00104$ | $\pm 0.00015$ | $\pm 0.00012$ |
| Synetos and Williams [187]* | $\pm 0.00004$          | $\pm 0.00020$ | $\pm 0.00036$ | $\pm 0.00037$ | $\pm 0.00029$ | $\pm 0.00012$ |
| Tuttle [195]                | $\pm 0.00007$          | $\pm 0.00013$ | $\pm 0.00041$ | $\pm 0.00019$ | $\pm 0.00021$ | $\pm 0.00007$ |
| Waldo et al.[200]           | $\pm 0.00006$          | $\pm 0.00041$ | $\pm 0.00041$ | $\pm 0.00038$ | $\pm 0.00017$ | $\pm 0.00009$ |
| 0D scaled model             | $\pm 0.00002$          | $\pm 0.00009$ | $\pm 0.00006$ | $\pm 0.00056$ | $\pm 0.00051$ | $\pm 0.00111$ |

\* Modified to include a sixth group by Parish et al. [135].

Table 4.13 compares the total delayed neutron yield for 100 fissions between the literature and the 0D scaled model. It can be seen that the data are not within uncertainty bounds of each other, and even differs between different approaches for the same dataset. Differences between microscopic and macroscopic delayed yields occur for many fissile nuclides, not only  $^{235}\text{U}$  [119]. The 0D scaled model yield exceeds other literature values, but this is a data issue and not a fitting issue. The summed DNP data give approximately the same yield as shown in Table 4.8. This indicates the DNP group parameter formulation is not responsible for the discrepancy; rather, different fission yields, half-lives, or emission probabilities may provide more consistent results with the literature. Additionally, pulse irradiation data may be needed to capture the smaller yields of shorter-lived DNP groups. This can be seen in

Table 4.11: DNP group half-lives from thermal fission of  $^{235}\text{U}$  (recreated from [135]).

| Source                      | Group [s] |          |          |         |         |         |
|-----------------------------|-----------|----------|----------|---------|---------|---------|
|                             | 1         | 2        | 3        | 4       | 5       | 6       |
| Keepin et al. [95]          | 55.89897  | 22.72614 | 6.24457  | 2.30281 | 0.60802 | 0.23028 |
| Charlton et al. [36]*       | 55.45177  | 22.43195 | 6.02737  | 2.18658 | 0.49159 | 0.23028 |
| ENDF/B-VI.8 [159]           | 51.72740  | 21.19716 | 5.73797  | 2.28913 | 0.81595 | 0.24295 |
| Mills et al. [125]          | 55.27489  | 22.80089 | 7.67860  | 2.77148 | 1.07381 | 0.28177 |
| Brady and England [23]      | 52.11633  | 21.32761 | 5.68620  | 2.18727 | 0.70114 | 0.23462 |
| Saleh et al. [166]          | 55.45177  | 22.65187 | 6.24457  | 2.31049 | 0.63013 | 0.23028 |
| Synetos and Williams [187]* | 56.35343  | 22.35959 | 7.00149  | 2.72893 | 0.77016 | 0.23028 |
| Tuttle [195]                | 54.57852  | 21.86584 | 6.02737  | 2.22877 | 0.49511 | 0.17911 |
| Waldo et al. [200]          | 54.57852  | 21.86584 | 6.02737  | 2.22877 | 0.49511 | 0.17911 |
| 0D scaled model             | 55.52977  | 23.54278 | 10.43942 | 3.14523 | 1.12586 | 0.25015 |

\* Modified to include a sixth group by Parish et al. [135].

Table 4.12: DNP group half-life uncertainties from thermal fission of  $^{235}\text{U}$  (recreated from [135]).

| Source                      | Group Uncertainties [s] |               |               |               |               |               |
|-----------------------------|-------------------------|---------------|---------------|---------------|---------------|---------------|
|                             | $\sigma_1$              | $\sigma_2$    | $\sigma_3$    | $\sigma_4$    | $\sigma_5$    | $\sigma_6$    |
| Keepin et al. [95]          | $\pm 1.35239$           | $\pm 0.67061$ | $\pm 0.22503$ | $\pm 0.09181$ | $\pm 0.08000$ | $\pm 0.02219$ |
| Charlton et al. [36]*       | $\pm 0.44361$           | $\pm 0.50817$ | $\pm 0.10482$ | $\pm 0.06208$ | $\pm 0.03138$ | $\pm 0.02219$ |
| ENDF/B-VI.8 [159]           | -                       | -             | -             | -             | -             | -             |
| Mills et al. [125]          | $\pm 0.09918$           | $\pm 0.11723$ | $\pm 0.37861$ | $\pm 0.11259$ | $\pm 0.08521$ | $\pm 0.01020$ |
| Brady and England [23]      | -                       | -             | -             | -             | -             | -             |
| Saleh et al. [166]          | $\pm 3.99253$           | $\pm 0.14805$ | $\pm 0.39380$ | $\pm 0.03851$ | $\pm 0.01719$ | $\pm 0.02219$ |
| Synetos and Williams [187]* | $\pm 0.45816$           | $\pm 0.28851$ | $\pm 0.77794$ | $\pm 0.21488$ | $\pm 0.09413$ | $\pm 0.02219$ |
| Tuttle [195]                | $\pm 1.28926$           | $\pm 0.82773$ | $\pm 0.20965$ | $\pm 0.08600$ | $\pm 0.04244$ | $\pm 0.02545$ |
| Waldo et al. [200]          | $\pm 1.28926$           | $\pm 0.82773$ | $\pm 0.20965$ | $\pm 0.08600$ | $\pm 0.04244$ | $\pm 0.02545$ |
| 0D scaled model             | $\pm 0.15676$           | $\pm 0.12560$ | $\pm 0.44990$ | $\pm 0.12771$ | $\pm 0.07629$ | $\pm 0.02325$ |

\* Modified to include a sixth group by Parish et al. [135].

Table 4.9, where the group 5 and 6 yields are overestimated relative to other DNP group yields in the literature. The modified 0D scaled model, using Keepin et al. DNP group parameters for groups three through six, produces a yield more consistent with literature values.

Table 4.14 compares average half-lives between the 0D scaled model and the literature. The average half-life from the 0D scaled model is too short because the overestimated yields of groups five and six give disproportionate weight to their short half-lives. The modified 0D scaled model, with groups three through six replaced by Keepin et al. DNP group parameters, aligns more closely with the literature. This again indicates the potential importance of the DNP data, including emission probabilities, half-lives, and fission yields; as well as the potential importance of the pulse irradiation data to capture short-lived DNP behavior.

Table 4.13: Total delayed neutron yields of 100 fissions from thermal fission of  $^{235}\text{U}$  (based on [84]).

| Approach    | Data                       | Delayed Yield [-] |
|-------------|----------------------------|-------------------|
| Microscopic | JEF-2.2 [129]              | $1.708 \pm 0.12$  |
| Microscopic | JEF-3.1.1 [167]            | $1.477 \pm 0.08$  |
| Microscopic | ENDF/B-VII.1 [34]          | $1.905 \pm 0.10$  |
| Microscopic | ENDF/B-VI.8 [159]          | 1.670             |
| Microscopic | Mills et al. [125]         | $1.645 \pm 0.07$  |
| Microscopic | Brady and England [23]     | $1.780 \pm 0.05$  |
| Microscopic | 0D scaled model            | $2.256 \pm 0.13$  |
| Hybrid      | Modified 0D scaled model   | $1.531 \pm 0.09$  |
| Macroscopic | JEF-2.2 [129]              | 1.655             |
| Macroscopic | JEF-3.1.1 [167]            | 1.620             |
| Macroscopic | ENDF/B-VII.0 [33]          | 1.585             |
| Macroscopic | Keepin et al. [95]         | $1.580 \pm 0.07$  |
| Macroscopic | Charlton et al. [36]       | $1.577 \pm 0.03$  |
| Macroscopic | Saleh et al. [166]         | $1.645 \pm 0.11$  |
| Macroscopic | Synetos and Williams [187] | $1.510 \pm 0.06$  |
| Macroscopic | Tuttle [195]               | $1.700 \pm 0.05$  |
| Macroscopic | Waldo et al.[200]          | $1.670 \pm 0.07$  |

Table 4.14: Average half-life from thermal fission of  $^{235}\text{U}$ .

| Approach    | Data                       | Average half-life [s] |
|-------------|----------------------------|-----------------------|
| Microscopic | ENDF/B-VI.8 [159]          | 7.66                  |
| Microscopic | Mills et al. [125]         | $8.63 \pm 0.52$       |
| Microscopic | Brady and England [23]     | $7.90 \pm 0.34$       |
| Microscopic | 0D scaled model            | $6.39 \pm 0.34$       |
| Hybrid      | Modified 0D scaled model   | $8.74 \pm 0.46$       |
| Macroscopic | Keepin et al. [95]         | $9.03 \pm 0.63$       |
| Macroscopic | Charlton et al. [36]       | $8.92 \pm 0.28$       |
| Macroscopic | Saleh et al. [166]         | $9.49 \pm 0.90$       |
| Macroscopic | Synetos and Williams [187] | $8.96 \pm 0.56$       |
| Macroscopic | Tuttle [195]               | $8.84 \pm 0.45$       |
| Macroscopic | Waldo et al.[200]          | $8.72 \pm 0.73$       |

## 4.4 Summary

This chapter presented results of the 0D scaled model on thermal fission of  $^{235}\text{U}$ , which demonstrated generation of improved DNP group parameters that account for chemical removal in the MSBR. These results consist of analysis of individual DNPs' yields, concentrations, and count rates; various sensitivities; and verification of DNP group parameters with results in the literature. The 0D scaled model results show good agreement with the literature for the longer-lived groups, but have poor agreement for the shorter-lived groups. This indicates the need for improvements to the model and analysis of DNP data, as discussed in Chapter 5. The sensitivities include analysis of chemical removal rates, time nodes, and nuclear data uncertainties. These sensitivities are useful for understanding how the chemical removal affects the DNP group parameters and analyzing how each group responds to changes in nuclear data based on its uncertainty.

Of the MSR physical phenomena considered, only chemical removal is accounted for in the 0D scaled model. The 0D scaled model does not account for decay chains, removing residence time dependence and causing the fission rate scaling to be identical to the DNP generation rate scaling. This causes the in-core and ex-core residence

times to not have any effect in the 0D scaled model. The proposed 0D flow model will be able to model the effects of residence time, as it accounts for time-dependent decays and parasitic absorptions. Although the 0D scaled model is rapid and enables more simulations, it does not incorporate fuel salt circulation into improved DNP group parameter generation and fails to properly model short-lived DNP groups.

The failure to model short-lived DNP groups may be due to a combination of issues in the fission yields, half-lives, or emission probabilities, as well as the lack of pulse irradiation data. The proposed work will investigate both of these possibilities, as described in Chapter 5. Whether combined via simultaneous solution of a larger matrix or solved separately and then combined, the macroscopic approach indicates pulse irradiation results improve DNP group parameters for shorter-lived groups [95]. The proposed work will indicate if this holds true for the microscopic approach as well. A simple investigation into the effects of pulse and saturation data is given in Appendix D. The investigation shows that the saturation irradiation generally captures more DNPs than the pulse irradiation, though as fewer DNP groups are used, the saturation irradiation has a larger average half-life while the pulse irradiation's is shorter. This is reflected in the 0D scaled model results, which show greater yields in the short-lived DNP groups, leading to a shorter average half-life. This may be from the lack of pulse irradiation data as well as the DNP nuclear data. The 0D scaled model cannot model pulse irradiations because the cumulative fission yields do not provide accurate concentrations for non-equilibrium concentrations. Section 5.1.1 discusses this further.

These preliminary results showed the importance of nuclear data and uncertainties, highlighting  $^{86}\text{Ge}$  as the DNP which has both the greatest impact on DNP group parameters and the largest uncertainty-weighted sensitivity. The sensitivity analysis showed the importance of the short-lived DNPs on the DNP group parameters, which also provides further motivation to use a model better able to capture these short-lived DNPs. The importance of the short-lived DNP also motivates a more detailed investigation into the different sensitivities and uncertainty-weighted sensitivities in other datasets. The 0D flow model will capture these short-lived DNPs more effectively and calculate improved DNP group parameters that account for both the recirculation and chemical removal in MSRs.

## Chapter 5

# Proposed Work

Completion of the proposed work will result in a comprehensive analysis of improved DNP group parameters for thermal fission of  $^{235}\text{U}$  in the MSRE and the effects of these improved parameters on transient simulations. MoSDeN will serve as a tool to investigate the effects of DNP group parameters with MSR physical phenomena included and will have an initial release as part of this work. The proposed work will focus on thermal fission of  $^{235}\text{U}$  specifically to mitigate scope creep. It will provide a foundation for broader analyses, such as those described in Section 5.4.

The preliminary results have shown that the current 0D scaled model can capture long-lived group behavior but fails to represent residence time, chemical removal, and short-lived DNP groups. Additionally, the preliminary work has analyzed time node distributions, the number of nodes, the length of the decay time, and the node density; all of which will be useful in the 0D flow model. Sensitivities of the DNP group parameters due to the nuclear data were also presented, which can be replicated using the 0D flow model to provide more accurate results. Because the 0D scaled model runs much faster than the 0D flow model is expected to run, it will be useful to compare their results and determine for what use cases the 0D scaled model can provide acceptable accuracy.

The proposed work includes developing the 0D flow model, validating and verifying it, generating improved group spectra and DNP group parameters, and applying the parameters in transient simulations. The development of the 0D flow model will provide more complete time-dependent data, enabling non-equilibrium irradiations and residence time analyses. Validation and verification of the 0D flow model is critical to ensure the results align with experiments and the literature. The improved DNP group spectra are used for calculation of the effective delayed neutron fraction, so developing functionality for MoSDeN to generate the improved group spectra is useful. The transient simulations will provide insight into the impact of the improved DNP groups and clarify the conditions under which the modified groups significantly influence reactor behavior.

Potential future work includes development of the spatially resolved model, uncertainty tracking of MSR physical phenomena in the DNP group parameters, and investigation into the effects of varying the incident neutron energy spectrum on the DNP group parameters. The spatially resolved model could be used for modeling chemical removal or diversion within the reactor or for more accurately modeling vortices and other thermal-hydraulic effects. Uncertainty tracking could extend beyond nuclear data to include chemical removal and thermal-hydraulic uncertainties. Investigating the incident neutron energy's effect on the DNP groups would be useful because the literature presents the groups for broad classes of energy spectra, such as thermal or fast. Sensitivity studies could be conducted to see how much the DNP group parameters can vary while changing the energy spectrum, such as emulating the spectral changes due to fuel depletion over the reactor lifetime.

Figure 5.1 shows the project timeline, divided into development, simulation, writing, and review categories.

Most tasks take one month, but four are assumed to take two months: verifying the 0D flow model, setting up the transient simulations, running the various MSRE transients, and writing the results and conclusions chapters.

Verifying the 0D flow model is expected to take two months because the simulation time will be slower than the 0D scaled model, and will include analysis of multiple datasets as well as different combinations of pulse and saturation irradiation data. Setting up the transient simulations is expected to take two months because it will require depletion simulations for the safeguards diversion scenarios as well as multiple OpenMC solves of the MSRE for cross-section data to pass into Moltres. Development of the Moltres model is expected to take longer because there may be unexpected issues with the meshing of the upper and lower plena, data collection, or model development itself. The MSRE transient simulations are given two months of simulation time because there are two transient simulations for validation (MSRE pump start-up and coast-down), four for safety (ULOF, UTOP, ULOHS, and FSOC), three for security (UTOP combined with each of the others), and two for safeguards (one SQ diversion of plutonium and a diversion of  $^{235}\text{U}$ ), as discussed in Section 3.5. As will be discussed, these transient simulations also require nine transport solves: three for fresh fuel salt in the MSRE prior to a transient with higher and lower temperatures, three with one SQ removal of  $^{235}\text{U}$  over one year, and three with one SQ removal of plutonium over one year. The diversion scenarios both require a depletion simulation, which also slightly increases the cost of the simulations. The results chapter is given two months because there will be many new results to discuss, whereas the other chapters will not have significant changes.

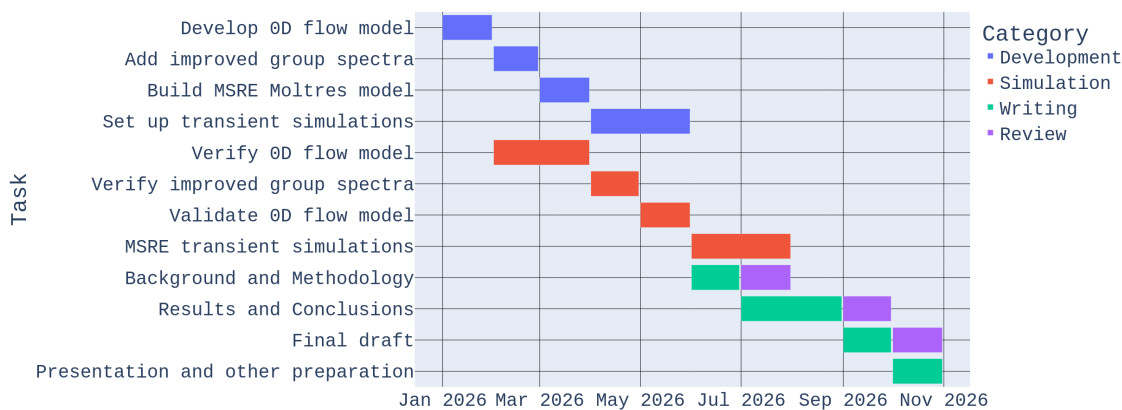


Figure 5.1: Gantt chart of proposed work.

## 5.1 0D Flow Model

The next model developed beyond the 0D scaled model is the 0D flow model. The 0D scaled model generates saturation irradiation results and scales the flux and chemical removal based on the fraction of salt in-core and ex-core. It also does not track nuclide decays into DNPs during post-irradiation decay. Because the 0D scaled model uses cumulative fission yields to calculate concentrations, it cannot capture non-equilibrium behavior and is unable to accurately capture residence time effects. The chemical removal results are smaller than expected in the 0D scaled model since intermediate nuclides cannot be removed. Despite these limitations, the 0D scaled model runs quickly and enables rapid testing of many parameter combinations.

The 0D flow model will improve on the fidelity offered by the 0D scaled model by providing both saturation and pulse irradiation results, handling in-core and ex-core regions in time rather than simply scaling the flux, and

modeling decay chains. Modeling the decay chains improves the accuracy of the formulated concentrations when they are not at equilibrium, which is especially important for pulse irradiations. Additionally, having separate regions rather than scaling the flux with the time dependence of the decay chains means that nuclides with larger cross-sections will have the correct amount of time to decay between irradiations, rather than experiencing a constant scaled irradiation.

This section outlines the proposed work for the 0D flow model, the combined use of pulse and saturation data in the least-squares solve, the use of multiple data sources, and the model's verification and validation.

### 5.1.1 Least Squares Treatment

The 0D scaled model did not account for pulse irradiation data, and instead only used saturation irradiation data. The work will compare different methods for combining pulse and saturation irradiation data. This comparison will provide insight into if and when the pulse irradiation data is necessary, as well as insight into the differences in results based on the  $l^2$ -norm of the residual.

The results of this analysis will be used to evaluate which method can provide a solution with the least computational cost while also generating accurate group parameters. The group parameters will be evaluated by calculating the relative residual for the least squares problem using the improved DNP group parameters over all times for both pulse and saturation irradiations.

There are different methodologies for combining pulse and saturation irradiation DNP group parameters that will be compared. One approach, used by Brady and Keepin et al., is combining two different fits after they have been generated. Brady used the group 5 and 6 parameters from the pulse irradiation with the remaining parameters from the saturation irradiation to form the six DNP group parameters [24]. Keepin et al. fit four groups with each irradiation, and then renormalized at the second group [95]. Another approach to combining the pulse and saturation irradiation data is combining the data to formulate the relative residual to minimize, as shown in Equation (5.1) as:

$$r = \min_x \left\| \frac{\vec{b}_p - A_p \vec{x}}{\vec{b}_p} \right\|_2^2 + \left\| \frac{\vec{b}_s - A_s \vec{x}}{\vec{b}_s} \right\|_2^2, \quad (5.1)$$

where

- $r$  = the minimization function of the problem [-]
- $A_p$  = the pulse irradiation coefficient matrix [ $s^{-1}$ ]
- $b_p$  = the pulse irradiation delayed neutron count rate [ $\# \cdot s^{-1}$ ]
- $A_s$  = the saturation irradiation coefficient matrix [ $s^{-1}$ ]
- $b_s$  = the saturation irradiation delayed neutron count rate [ $\# \cdot s^{-1}$ ]
- $x$  = the solution vector containing group yields [-].

This formulation fits both pulse and saturation data simultaneously to generate improved DNP group parameters.

### 5.1.2 Data Source Analysis

As shown in Section 4.3, experimental results for delayed neutron yields vary non-negligibly. This proposed analysis will use different data sources to generate the improved DNP group parameters. The purpose of this analysis is to provide a more robust understanding of the uncertainties and what differences can be expected due to the

data source used. Huynh et al. compare the effects of cumulative fission yields and emission probabilities from Joint Evaluated Fission and Fusion (JEFF), ENDF, and Japanese Evaluated Nuclear Data Library (JENDL) on the total delayed neutron yield [84]. A similar comparison can also be found in the JEFF report 17 [129]. Mathieu et al. investigated the National Nuclear Data Center (NNDC) website, which sources data from Evaluated Nuclear Structure Data File (ENSDF), and various data compilations [119, 143, 6, 161].

These comparisons can be expanded to include the IAEA database, NNDC website, JEFF, ENDF, JENDL, ENSDF, and other data compilations for emission probabilities (delayed neutron branching ratios), half-lives, fission yields, and cross sections. Comparing these sources will clarify the differences that arise from nuclear data discrepancies across datasets. Table 5.1 shows the data sources used in the comparison. The "Concentration" column refers to data including fission yields and cross-sections that affect the DNP concentration term. Although half-lives also influence concentration, data sources without a concentration contribution provide only half-life and emission probability data. Audi et al., Pfeiffer and Kratz, and Rudstam et al. presented data from the literature rather than official databases. Audi et al. used ENSDF, Atomic Mass Evaluation (AME), experimental data from the literature, and trends in neighboring nuclides. Pfeiffer and Kratz presented experimental data, an update to the Kratz Herrmann formula (KHF), and data from a unified macroscopic-microscopic model within the quasi-particle random-phase approximation (QRPA). Rudstam et al. presented experimental data.

Table 5.1: Various data sources as well as the data which they contain.

| Data name          | Year | Source | Half-life | Emission probability | Concentration |
|--------------------|------|--------|-----------|----------------------|---------------|
| IAEA database      | 2018 | [51]   | Yes       | Yes                  | No            |
| JEFF 3.3           | 2017 | [146]  | Yes       | Yes                  | Yes           |
| ENDF/B-VII.1       | 2011 | [34]   | Yes       | Yes                  | Yes           |
| ENDF/B-VIII.0      | 2018 | [29]   | Yes       | Yes                  | Yes           |
| JENDL-5            | 2021 | [90]   | Yes       | Yes                  | Yes           |
| ENSDF              | 2025 | [57]   | Yes       | Yes                  | No            |
| Audi et al.        | 2003 | [6]    | Yes       | Yes                  | No            |
| Pfeiffer and Kratz | 2002 | [143]  | Yes       | Yes                  | No            |
| Rudstam et al.     | 1993 | [161]  | Yes       | Yes                  | No            |

Recently updated datasets and official databases are expected to be more accurate due to improved measurement techniques and broader validation. Official databases are expected to be more reliable, as they have many users reporting errors and have undergone extensive development over time. Investigating these data sources and comparing the DNP group parameters generated from each will be useful when analyzing uncertainty. If the differences are large, understanding the uncertainty propagation will be important. If the differences are negligible, this will simplify further uncertainty analysis.

### 5.1.3 Verification and Validation

Verifying and validating the DNP group parameters is essential to ensure their practical utility. Verification against the literature ensures that the fundamental approach behaves correctly. Since the literature does not contain DNP group parameters that include MSR physical phenomena, this thesis will include a verification with the literature only for the stationary solve.

After this, although the group parameters will be verified with the literature, that does not necessarily mean they capture all physical phenomena of interest. Validation identifies potential weaknesses in the approach by

highlighting discrepancies between predicted and experimental results.

Verification of the DNP group parameters will be of the same form as in Section 4.3. It has already been shown that these results should have a percent difference of 9% for thermal fission in  $^{235}\text{U}$  for JEFF-3.1.1 data, so a similar percent difference can be expected for other datasets [119]. Results will vary based on the source of the data, but the DNP group parameters should closely match the results from the literature.

Validation of the DNP group parameters will be conducted through the MSRE pump start-up and coast-down experimental data. This data assesses the accuracy of the group parameters in modeling reactivity changes from DNPs [148]. The validation requires modeling the MSRE core and upper and lower plena in Moltres [63]. The reactivity effects of the DNPs will be calculated for both transients using static and improved DNP group parameters. The improved group parameters are expected to predict reactivity more accurately as they account for more physical phenomena in the MSRE. If neither set of reactivity responses aligns with the experimental results, then that will indicate an issue in the kinetics simulation, as Moltres has previously demonstrated the ability to model these transients [150, 137]. These transients have also been evaluated using other tools, providing additional comparison [59, 48]. If the static group parameters predict reactivity more accurately than the improved group parameters, this will suggest an issue with the improved group parameter formulation. The issue must be resolved before proceeding, as the improved DNP group parameters should provide higher accuracy in MSR modeling.

## 5.2 Improved Group Spectra

The DNP group spectra will be generated using using the half-life grouping and fractional grouping methods. The delayed neutron energy spectrum,  $\chi_d$  is important for accurate prediction of reactor kinetics [46]. This can be seen in the equation for the effective delayed neutron fraction in Equation (2.36), recreated here as:

$$\beta_{eff} = \frac{\int \Phi^\dagger \chi_d \beta \bar{\nu} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}{\int \Phi^\dagger \chi \bar{\nu} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}.$$

Because the DNP group yields and half-lives are affected by the MSR physical phenomena, it follows that the DNP group spectra are also affected. The methodology to evaluate the delayed neutron spectra is discussed in Section 2.1.2. The DNP group spectra will be compared with the spectra generated from a stationary problem with no chemical removal to determine the effect of MSR phenomena on the spectra.

## 5.3 Transient Simulations

Transient simulations will apply the improved DNP group parameters in the MSRE to demonstrate how they affect power and temperature responses. They are critical for validating the DNP group parameters using MSRE transients and for investigating transients affecting safety, security, and safeguards design decisions. Section 3.5 discusses the proposed simulations in detail. These simulations will use OpenMC to generate multigroup cross sections at three temperatures: a higher than nominal, a lower than nominal, and the nominal MSRE temperature.

For safety design, the ULOF, UTOP, ULOHS, and FSOC will be investigated for various reactor parameters to assess the impact of the improved DNP group parameters. For security design, sabotage-initiated transients will build on those studied for safety design. For safeguards design, comparisons are made between diversion scenarios. The diversion scenarios alter the fuel composition, affecting the effective delayed neutron fraction and reactor behavior during transients [202]. The improved DNP group parameters will be used to investigate the extent of SNM removal which could remain undetected while using previous DNP group parameters.

The kinetics simulations will use Moltres models of the MSRE to evaluate how the group parameters affect reactivity and temperature responses. These results can be compared with other analyses from the literature [78, 181]. The results will identify which transients are more sensitive to the improved data and highlight potential concerns.

## 5.4 Future Work

This section contains various work that would be useful and interesting extensions of the proposed work. The three main categories of potential future work include development of a spatially resolved model for improved DNP group parameters, tracking of additional contributors to uncertainty, and sensitivity of DNP group parameters to varying incident neutron energy spectra. Additional miscellaneous work of interest could also include:

- Use a sinusoidal flux profile instead of a Heaviside function for the derivation given in Appendix A.
- Compare exponential stripping, least squares, and set half-life DNP group parameter fitting methods.
- Add intermediate models between the 0D scaled and 0D flow models
  - 0D scaled with explicit calculation of concentration rather than scaled flux (based on Appendix A.
  - 0D scaled with cross-sections included.
  - 0D scaled with simplified decay chains included.
- Determine if an intermediate length irradiation improves the fitting compared to combined pulse and saturation irradiations.
- Investigate if there may be a way to structure DNP group parameters such that fuel salt movement is accounted for in the delayed neutron count rate (capturing circulation in the parameters themselves).
- Include the effects of photofission.
- Include the effects of photoneutrons.
- Investigate the sensitivity of the DNP group parameters to the evolution of the neutron energy spectrum over the lifetime of a reactor.
- Determine if weighted least squares can improve the fit or the conditioning.

### 5.4.1 Spatially Resolved Model

Although the 0D flow model provides useful insights, it does not provide a high level of fidelity for any specific geometry. A spatially resolved model can be developed to evaluate the accuracy of the 0D flow model. This model would verify whether the 0D flow model produces sufficiently accurate DNP group parameters. Appendix E discusses the spatially resolved model in detail. This spatially resolved model can compare multiple levels of fidelity for generating DNP group parameters: 0D scaled model using cumulative fission yields, 0D flow model with depletion, spatially resolved flow model using cumulative fission yields, and spatially resolved flow model with depletion. Each step increases computational cost, so if a simpler method generates DNP group parameters with negligible differences, it will be used preferentially.

The comparison between the 0D flow model and the spatially resolved model can be conducted as follows: the spatially resolved model is first executed to generate DNP group parameters and residence times; the residence times are supplied to the 0D flow model; and the resulting DNP group parameters are compared between the two models. If the results are within the uncertainty bounds of each other, then this indicates that the 0D flow model only needs residence time data for a given MSR to generate the DNP group parameters. If they do not match, then a spatially resolved approach may be necessary to properly generate the MSR DNP group parameters.

### Sensitivity to Number of DNPs

Because each nuclide tracked in the spatially resolved model increases computational cost, it would be useful to determine the sensitivity of the simulation to the number of tracked DNPs. The 0D flow model's computational cost increases negligibly with more DNPs, while the spatially resolved flow model requires significantly more memory and operations to track each DNP over space and time. To improve computational efficiency, minimizing the number of DNPs is desirable. If the 0D flow model generates DNP group parameters with negligible differences compared to the spatially resolved flow model, minimizing the number of DNPs becomes less critical. However, it is of interest to optimize the spatially resolved flow model for potential future analyses which could include spatially resolved reprocessing and other effects.

The impact of each DNP is given in Equation (2.31), replicated here:

$$\tilde{v}_{d,i} = P_{n,i} N_i(0).$$

Dividing  $\tilde{v}_{d,i}$  by the total delayed neutron yield,  $\tilde{v}_d$ , provides the relative contribution of each DNP to the total delayed neutron yield. The DNPs with the most impact can be modeled preferentially to determine the minimum number of DNPs to model with negligible changes to the DNP group parameters. A full sensitivity study can be conducted to determine a reasonable number of DNPs to include in the spatially resolved flow model.

### Reactor Generalization

If the DNP group parameters can be applied generally to different MSRs with similar residence times, then the amount of computational work required is drastically reduced. However, it is necessary that sufficient evidence is provided demonstrating that the improved parameters are identical for different MSRs.

Specifically, the MSRE and the MSFR can be simulated using the spatially resolved model [155, 28]. The models can be adjusted such that the in-core and ex-core residence times are identical for each. The results would be a set of group parameters for two distinct reactors but with the same residence times. Instead of calculating the neutron flux, both reactors will be given an incident neutron energy spectrum for irradiating the sample. The impact of the incident neutron energy spectra is discussed in Section 5.4.3, so differences caused by the spectra will not be studied in this section. Additionally, the fuel salt and fuel temperature in the MSFR will be replaced with those from the MSRE.

The two different reactor simulations would have all of the same input parameters aside from the geometry and thermal-hydraulic parameters. If the DNP group parameters have negligible differences, then the geometry and thermal-hydraulic parameters are not necessary for these two exceedingly different MSR designs when generating the group parameters. This would indicate that any MSR of a similar design would also not require geometry and thermal-hydraulic parameters to generate the improved DNP group parameters. Instead, all that is needed are the other input parameters, meaning the work does not need to be replicated for every different reactor geometry and thermal-hydraulic configuration. If the DNP group parameters have non-negligible differences, then each

MSR needs to have DNP group parameters generated for its own specific geometry and flow conditions. Further investigation may also be warranted to determine if there is potential for a reduced order model in which one or two parameters which could be used to capture these differences, thus reducing the total amount of work necessary to model all MSR designs.

### 5.4.2 MSR Uncertainties

This analysis would add additional uncertainty tracking relevant to MSR recirculation and chemical removal. Although several sources of uncertainty are currently considered, including half-lives, emission probabilities, cross-sections, and fission yields, uncertainties also arise from MSR physical phenomena, including flow rate and reprocessing rates, where the reprocessing rate is correlated with the flow rate. These uncertainties are important because they propagate to the DNP group parameters, which affect the reactor's response to different transients [149]. It would be useful to understand this for safety, security, and safeguards design, as the design should account for these uncertainties. In particular, uncertainties of interest are in the residence times and chemical removal rates.

The residence time correlates linearly with the flow rate, which is dependent on the temperature. Although the flow rate would also have uncertainty based on variance in the pump, this uncertainty may be beyond the scope of future work in MoSDeN. The temperature affects fuel salt density, buoyant force, and viscosity, all influencing the flow rate and residence time. The temperature is influenced by the fission rate, which has uncertainties from the flux and fission cross section. For the 0D flow model, the residence time uncertainties can be conservatively approximated by treating the difference between the inlet and outlet temperatures as the temperature uncertainty. Assuming a constant mass flow rate, the density change due to the large temperature uncertainty provides a velocity uncertainty, which determines the variance in the residence time. If this conservative approach to temperature uncertainty results in negligible residence time uncertainty, it can be neglected for the spatially resolved flow model. Otherwise, the spatially resolved flow model can account for this uncertainty using the Stochastic Tools module to directly determine residence time uncertainties. The Stochastic Tools module is a toolbox designed for performing stochastic analysis for MOOSE-based applications [182]. The Stochastic Tools module enables setting a range of temperatures and simulating each temperature, which can be used to generate a histogram of residence times as a function of core temperature.

The chemical removal uncertainty would be challenging to treat, as there are different chemical processes for removing elements, such as sparging the fuel salt in the core, filtering elements with a nickel filter, and removing elements via liquid-liquid reductive extraction [164]. The removal efficiency of xenon in a gas separation system is given by [139, 164] as:

$$\epsilon = \frac{1 - e^{-k_L a A_C L \left( \frac{1}{Q_{salt}} + \frac{RT}{H} \frac{1}{Q_{He}} \right)}}{1 + \frac{RT}{H} \frac{Q_{salt}}{Q_{He}}}, \quad (5.2)$$

where

$R$  = the universal gas constant [ $L \cdot Pa \cdot mol^{-1} \cdot K^{-1}$ ]

$T$  = the salt temperature [ $K$ ]

$H$  = Henry's law constant for solute gas [ $Pa \cdot mol^{-1} \cdot L$ ]

$Q_{salt}$  = the volumetric salt flow rate [ $m^3 \cdot s^{-1}$ ]

$Q_{He}$  = the volumetric helium flow rate [ $m^3 \cdot s^{-1}$ ]

$k_L$  = the liquid phase mass transfer coefficient [ $m \cdot s^{-1}$ ]

$a$  = the gas-liquid interfacial area per unit volume [ $m^{-1}$ ]

$A_C$  = the contactor cross-section [ $m^2$ ]

$L$  = the contactor length [ $m$ ].

The liquid-liquid reductive extraction, such as with bismuth, has distribution coefficients and equilibrium constants dependent on the element, temperature, and concentration [126]. In general, an increase in the temperature leads to an increased distribution coefficient and a decreased equilibrium constant. The distribution coefficient represents the ratio of the bismuth phase concentration to the salt phase concentration, and can be thought of as an efficiency term that is multiplied with the mass flow rate to determine the removal rate. However, the distribution coefficient changes with temperature and flow rate differently for each element and fuel salt used.

Propagating the uncertainty in the chemical removal of elements from the fuel salt is challenging, as there are many variables to consider. These include the fuel salt composition, flow rate, redox potential, hydrostatic pressure, interfacial tension, and temperature [175, 192]. To stay within scope, propagation of uncertainty would not account for every variable in every elemental chemical removal. Instead, only the chemical removal of noble gases and noble metals from the MSRE would be investigated, as in Lo et al. [113]. The noble gases are actively pumped out via the off-gas system while noble metals, including molybdenum, technetium, ruthenium, rhodium, palladium, silver, and antimony plate out via reduction by  $UF_6$ . Some elements, specifically selenium, niobium, and tellurium, vary in their behavior. These elements only sometimes plate out based on the redox potential of the salt; a well reduced salt causes them to plate out, otherwise they act as salt seekers [93, 113].

The removal rate,  $\lambda_r$ , of noble gases from the pump bowl via the off-gas system is [91, 113]:

$$\lambda_r = \frac{\dot{V}}{V} \epsilon, \quad (5.3)$$

where

$\dot{V}$  = the salt flow rate through the gas-liquid separator loop [ $m^3 \cdot s^{-1}$ ]

$V$  = the total volume of salt in the primary loop [ $m^3$ ]

$\epsilon$  = the removal efficiency [-].

For the MSRE, the values are a salt flow rate of  $0.003153 m^3 \cdot s^{-1}$ , a total salt volume of  $1.996 m^3$ , and a removal efficiency of 0.026 [91, 155]. For the 0D flow model, this removal occurs while the sample is in the ex-core region and stops while the sample is in the in-core region. This is because the removal occurs in the pump bowl and not inside the core. For the spatially resolved flow model, this removal would also occur in the ex-core region.

The removal rate,  $\lambda_{i,l,r}$ , of noble metal  $i$  to waste stream  $l$  via plate-out is [113]:

$$\lambda_{i,l,r} = \frac{h_{i,l}A_l}{V}, \quad (5.4)$$

where

$h_{i,l}$  = the mass transfer rate of element  $i$  to waste stream  $l$  [ $m \cdot s^{-1}$ ]

$A_l$  = the surface area of waste stream  $l$  [ $m^2$ ].

The removal of noble metals does not occur in a single region, unlike the noble gas removal. Instead, there is a distribution of noble metals: 50% are in the hastalloy-N surfaces in the loop, 49% are in the ex-core region, and 1% are in the graphite surfaces in the in-core region [93]. Because 99% of the plate-out occurs in the ex-core region, the removal can be treated as occurring only in the ex-core region. The surface areas and mass transfer rates for the noble metals are given in Table 5.2, recreated from [113].

Table 5.2: Noble metal mass transfer rates, surface areas, and resulting removal rates (recreated from [113] with data from [93]).

| Component                              | $h_{i,l}$ [ $m \cdot s^{-1}$ ] | $A_l$ [ $m^2$ ]     | $\lambda_{i,l,r}$ [ $s^{-1}$ ] |
|----------------------------------------|--------------------------------|---------------------|--------------------------------|
| Heat Exchanger                         | $4.657 \times 10^{-5}$         | $2.93 \times 10^1$  | $6.827 \times 10^{-4}$         |
| Fuel loop piping, core and pump volute | $1.041 \times 10^{-4}$         | $6.60 \times 10^0$  | $3.441 \times 10^{-4}$         |
| Core wall cooling annulus              | $4.318 \times 10^{-5}$         | $1.43 \times 10^1$  | $3.095 \times 10^{-4}$         |
| Core graphite (exposed to salt)        | $5.334 \times 10^{-6}$         | $1.36 \times 10^2$  | $3.637 \times 10^{-4}$         |
| Miscellaneous                          | -                              | -                   | $1.695 \times 10^{-4}$         |
| Bubbles                                | $4.233 \times 10^{-4}$         | $3.205 \times 10^1$ | $6.7978 \times 10^{-3}$        |
| Total                                  | -                              | -                   | $8.6674 \times 10^{-3}$        |

### 5.4.3 Incident Neutron Energy Spectra

The incident neutron energy spectrum directly affects the DNP group parameters [23]. However, it would be computationally expensive to generate new DNP group parameters for every unique incident neutron energy spectrum. Some coarser resolution is required, so it is of interest to determine a reasonable categorization of the DNP group parameters for different energy spectra.

This sensitivity study could investigate different levels of fidelity for organizing the energy spectra: fine, intermediate, coarse, and single-value spectra. The approach can use the 0D flow model, with changes only to the neutron source energy spectra. The fine, middling, and coarse spectra include the full incident neutron energy spectra but with different bin sizes. The single-value spectra include several categories: neutron energy classification, mean energy, and median energy. The neutron energy classification is provided in Table 5.3, recreated from [116]. The study will also examine hand-crafted spectra where the mean and median remain unchanged, but the spectrum itself varies.

Table 5.3: Classification of neutron energy categories by energy range from [116].

| Category     | Energy Range [ $MeV$ ]                |
|--------------|---------------------------------------|
| Cold         | $< 3 \times 10^{-9}$                  |
| Thermal      | $3 \times 10^{-9} - 4 \times 10^{-7}$ |
| Epithermal   | $4 \times 10^{-7} - 1 \times 10^{-4}$ |
| Intermediate | $1 \times 10^{-4} - 2 \times 10^{-1}$ |
| Fast         | $2 \times 10^{-1} - 1 \times 10^1$    |
| Relativistic | $> 1 \times 10^1$                     |

This sensitivity study can determine the necessary level of detail in the incident neutron energy spectrum for generating DNP group parameters. Organizing by category or mean energy is expected to suffice for most cases. For cases where this approach is inadequate, a more robust method can be selectively applied.

## Appendix A

# Derivation of the Delayed Neutron Count Rate in the 0D Flow Model

The delayed neutron count rate will be different in an MSR than a stationary sample, as the sample is irradiated for the in-core residence time and not irradiated for the ex-core residence time. The derivation presented in this appendix captures this effect. This derivation begins with Equations (AA.1) and (AA.2) provided by Wilson and England [204], modified to a single DNP group  $k$  as:

$$R_k(t) = \bar{v}_{d,k} \lambda_k e^{-\lambda_k t}, \quad (\text{AA.1})$$

and

$$\dot{n}_{d,k}(T, t) = \bar{v}_{d,k} \int_0^T F(\tau) R_k(T - \tau + t) d\tau, \quad (\text{AA.2})$$

where

$K$  = the number of DNP groups [#]

$\bar{v}_{d,k}$  = the group  $k$  delayed neutron yield [-]

$\lambda_k$  = the group  $k$  decay constant [ $s^{-1}$ ]

$T$  = the duration of the fission history [s]

$t$  = the decay time from the end of the fission history [s]

$\dot{n}_d(T, t)$  = the delayed neutron count rate [ $\# \cdot s^{-1}$ ]

$R_k(t)$  = the probability of delayed neutron emission from group  $k$  at time  $t$  [ $\# \cdot s^{-1}$ ]

$F(\tau)$  = the fission rate history [ $\frac{\#}{s}$ ].

Because the results are normalized to a single delayed neutron, the  $\bar{v}_d$  term is divided out.

Although the saturation, or infinite, irradiation derivation makes use of assuming the time is infinite, due to the spatial dependence of this problem, the in-core and ex-core timing must be handled properly. If the sample is initially in the core inlet, then the final portion of the fission rate history for each circulation must consist purely of decay. If the sample is initially in the center of the core, then the final portion of the fission rate history for each circulation must consist of irradiation.

For this derivation, the sample is assumed to exist as a point at the core inlet. It will circulate through the

in-core region, then the ex-core region, and repeat until the net irradiation time  $T$  has elapsed. At that point, the delayed neutron count rate will be collected with  $t$  starting at zero. This means the irradiation consists of an in-core residence time of fission followed by an ex-core residence time of decay, after which the cycle repeats until the net irradiation time is met. The normalized fission rate history is therefore given as a square wave composed of steps up and down:

$$\frac{F(\tau)}{F_0} = \sum_{j=0}^{J_{in}} \overbrace{H(\tau - j \cdot \tau_{net})}^{\text{Step up}} - \overbrace{H(\tau - j \cdot \tau_{net} - \tau_{in})}^{\text{Step down}}, \quad (\text{AA.3})$$

where

$$\begin{aligned} H &= \text{the Heaviside function [-]} \\ \tau_{net} &= \tau_{in} + \tau_{ex} \\ &= \text{the total time per circulation [s]} \\ J_{in} &= \left\lfloor \frac{T}{\tau_{net}} \right\rfloor \\ &= \text{the number of times the sample enters the core minus one [#]} \\ F_0 &= \text{the magnitude of the constant fission rate } \left[ \frac{\#}{s} \right]. \end{aligned}$$

This function could be made to sum over infinity instead of  $J_{in}$ , since any Heaviside in which the value of  $j\tau_{net}$  is larger than  $T$  would become zero. However, to make the equation cleaner, this clearer limit of  $J_{in}$  is used.

Plugging this into Equation (AA.2) for group  $k$  yields:

$$\begin{aligned} \dot{n}_{d,k}(T, t) &= \int_0^T \sum_{j=0}^{J_{in}} H(\tau - j \cdot \tau_{net}) \tilde{v}_{d,k} \lambda_k e^{-\lambda_k(T-\tau+t)} d\tau \\ &\quad - \int_0^T \sum_{j=0}^{J_{in}} H(\tau - j \cdot \tau_{net} - \tau_{in}) \tilde{v}_{d,k} \lambda_k e^{-\lambda_k(T-\tau+t)} d\tau, \end{aligned} \quad (\text{AA.4})$$

from which the common terms of  $F_0$ ,  $\tilde{v}_{d,k}$ , and  $\lambda_k$  can be moved to the left-hand side:

$$\begin{aligned} \frac{\dot{n}_{d,k}(T, t)}{F_0 \tilde{v}_{d,k} \lambda_k} &= \int_0^T \sum_{j=0}^{J_{in}} H(\tau - j \cdot \tau_{net}) e^{-\lambda_k(T-\tau+t)} d\tau \\ &\quad - \int_0^T \sum_{j=0}^{J_{in}} H(\tau - j \cdot \tau_{net} - \tau_{in}) e^{-\lambda_k(T-\tau+t)} d\tau. \end{aligned} \quad (\text{AA.5})$$

Solving the integrals yields:

$$\begin{aligned} \frac{\dot{n}_{d,k}(T, t)}{F_0 \tilde{v}_{d,k} \lambda_k} &= \sum_{j=0}^{J_{in}} \frac{H(T - j \cdot \tau_{net})}{\lambda_k} \left( e^{-\lambda_k t} - e^{-\lambda_k(t+T-j \cdot \tau_{net})} \right) \\ &\quad - \sum_{j=0}^{J_{in}} \frac{H(T - j \cdot \tau_{net} - \tau_{in})}{\lambda_k} \left( e^{-\lambda_k t} - e^{-\lambda_k(t+T-j \cdot \tau_{net}-\tau_{in})} \right). \end{aligned} \quad (\text{AA.6})$$

The Heaviside function in the first summation resolves to unity for every value of  $j$  up to  $J_{in}$ , and is zero for any values of  $j$  larger than  $J_{in}$ . The second Heaviside is modified by  $\tau_{in}$ , so the limits of summation must be adjusted.

The problem becomes:

$$\frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{v}_{d,k}} = \sum_{j=0}^{J_{in}} \left( e^{-\lambda_k t} - e^{-\lambda_k(t+T-j\tau_{net})} \right) - \sum_{j=0}^{J_{ex}} \left( e^{-\lambda_k t} - e^{-\lambda_k(t+T-j\tau_{net}-\tau_{in})} \right), \quad (\text{AA.7})$$

where

$$J_{ex} = \left\lfloor \frac{T - \tau_{in}}{\tau_{net}} \right\rfloor$$

= the number of times the sample exits the core minus one [#]

which, because  $J_{ex}$  is always less than  $J_{in}$ , can be further simplified to:

$$\frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{v}_{d,k}} = \sum_{j=0}^{J_{ex}} \left( e^{-\lambda_k t} - e^{-\lambda_k(t+T-j\tau_{net})} - e^{-\lambda_k t} + e^{-\lambda_k(t+T-j\tau_{net}-\tau_{in})} \right) + \sum_{j=J_{ex}+1}^{J_{in}} \left( e^{-\lambda_k t} - e^{-\lambda_k(t+T-j\tau_{net})} \right). \quad (\text{AA.8})$$

The exponential terms can be moved outside of the summations:

$$\frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{v}_{d,k}} = e^{-\lambda_k t} \sum_{j=0}^{J_{ex}} \left( e^{-\lambda_k(T-\tau_{in}-j\tau_{net})} - e^{-\lambda_k(T-j\tau_{net})} \right) + e^{-\lambda_k t} \sum_{j=J_{ex}+1}^{J_{in}} \left( 1 - e^{-\lambda_k(T-j\tau_{net})} \right). \quad (\text{AA.9})$$

The closed form of the geometric series can be calculated, which enables the summation terms to collapse and be written as:

$$\frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{v}_{d,k}} = e^{-\lambda_k t} \left( (J_{in} - J_{ex}) + \frac{e^{-\lambda_k T}}{d - 1} \left( e^{\lambda_k \tau_{in}} d^{J_{ex}+1} - e^{\lambda_k \tau_{in}} + 1 - d^{J_{in}+1} \right) \right), \quad (\text{AA.10})$$

where

$$d = e^{\lambda_k \tau_{net}} \quad (\text{AA.11})$$

If the net irradiation time  $T$  is infinite and the ex-core residence time is zero, then the delayed neutron count rate simplifies to:

$$\dot{n}_{d,k}(T, t) = F_0 \bar{v}_{d,k} e^{-\lambda_k t}, \quad (\text{AA.12})$$

which matches the result from Wilson and England [204]. However, for cases with non-zero ex-core residence times, the behavior is noticeably different. For example, if an in-core residence time  $\tau_{in}$  is selected which is relatively small compared to the ex-core residence time  $\tau_{ex}$ , Equation (AA.12) would not vary at all, even though the fission rate history would be significantly different. By isolating the delayed neutron count rate  $\dot{n}_{d,k}$  and summing over all  $K$

groups, the finalized equation can be formulated as:

$$\dot{n}_d(T, t) = F_0 \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t} \left( (J_{in} - J_{ex}) - \frac{e^{-\lambda_k T}}{1-d} \left( e^{\lambda_k \tau_{in}} d^{J_{ex}+1} - e^{\lambda_k \tau_{in}} + 1 - d^{J_{in}+1} \right) \right). \quad (\text{AA.13})$$

This equation is not stable for all in-core and ex-core residence times, and is unstable if the ex-core time goes to zero. The summation form given in Equation (AA.9) is stable for general problems. The summation form is similar to the form presented by Wilson and England for an arbitrary irradiation time:

$$\dot{n}_d(T, t) = F_0 \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t} \left( 1 - e^{-\lambda_k T} \right) \quad (\text{AA.14})$$

If the in-core residence time is zero, then the delayed neutron count rate goes to zero, which is the expected behavior.

## Appendix B

# Derivation of the Equilibrium Concentration from the Cumulative Fission Yield

This derivation shows how the equilibrium concentration for a given nuclide can be calculated, particularly for DNPs. This is used for the 0D scaled model which is intended to study only the long-lived DNP. This derivation begins with the Bateman equation, given in Equation (2.59) and recreated here as:

$$\frac{dN_j}{dt} = \sum_{i \neq j} \left[ \overbrace{(\gamma_{i \rightarrow j} \sigma_{f,i} \phi + \lambda_{i \rightarrow j} + \sigma_{i \rightarrow j} \phi) N_i}^{\text{Source}} \right] - \underbrace{(\lambda_j + \sigma_j \phi + \lambda_{r,j})}_{\text{Loss}} N_j + \dot{N}_{f,j},$$

where

- $N_j$  = the atomic density of isotope  $j$   $\left[ \frac{\#}{cm^3} \right]$
- $\gamma_{i \rightarrow j}$  = the fractional fission product yield of  $j$  from the fission of isotope  $i$  [-]
- $\sigma_{f,i}$  = the microscopic fission cross-section of isotope  $i$   $[cm^2]$
- $\phi$  = the spectrum-averaged scalar flux in the fuel region  $\left[ \frac{\#}{cm^2 \cdot s} \right]$
- $\lambda_{i \rightarrow j}$  = the decay constant of decay  $i \rightarrow j$   $[s^{-1}]$
- $\sigma_{i \rightarrow j}$  = the microscopic transmutation cross-section of reaction  $i \rightarrow j$   $[cm^2]$
- $N_i$  = the atomic density of isotope  $i$   $\left[ \frac{\#}{cm^3} \right]$
- $\lambda_j$  = the decay constant of isotope  $j$   $[s^{-1}]$
- $\lambda_{r,j}$  = the removal constant for removal of isotope  $j$   $[s^{-1}]$
- $\sigma_j$  = the microscopic total transmutation cross-section of isotope  $j$   $[cm^2]$
- $\dot{N}_{f,j}$  = the feed rate for isotope  $j$   $\left[ \frac{\#}{cm^3 \cdot s} \right]$ .

This can be rewritten to the simplified form given in Equation (2.63) by collapsing the source and loss terms and

assuming a feed rate of zero, recreated here as:

$$\frac{dN_j}{dt} = S_j(t) - \lambda_{net,j} N_j(t),$$

where

$$\begin{aligned} S_j &= \text{the source term for nuclide } j, \\ &\quad \text{containing fission, decay, and transmutation sources } [\frac{\#}{cm^3 s}] \\ \lambda_{net,j} &= \text{the loss term for nuclide } j, \\ &\quad \text{containing decay and transmutation losses } [s^{-1}]. \end{aligned}$$

This can be rewritten using the scaled flux method, which is useful for flowing fuel reactors as it scales the flux term by the fraction of fuel salt in the core. Additionally, the transmutation term is small for DNPs, and is assumed to be negligible. This changes Equation (2.63) to Equation (BB.1), assuming chemical removal only in the ex-core region, given as:

$$\frac{dN_j}{dt} = f_{in} S_j(t) - \lambda_j N_j(t) - f_{ex} \lambda_{r,j} N_j(t), \quad (BB.1)$$

where

$$\begin{aligned} f_{in} &= \text{the fraction of fuel salt in the core } [-] \\ f_{ex} &= \text{the fraction of fuel salt not in the core } [-] \\ \lambda_{r,j} &= \text{the chemical removal rate of } j [s^{-1}]. \end{aligned}$$

The source term,  $S_j$ , can then be written using the cumulative fission yield as shown in Equation (BB.2). The cumulative fission yield is the total number of some nuclide formed per fission event, including decay chains that lead to that nuclide. Because the cumulative fission yield captures decays into the nuclide  $j$ , the  $\lambda_{i \rightarrow j}$  term is not included. Additionally, parasitic absorption is neglected in this derivation, which leads to:

$$S_j(t) = CFY_j \phi \Sigma_f, \quad (BB.2)$$

where

$$\begin{aligned} \phi &= \text{the neutron flux } [\# \cdot cm^{-2} \cdot s^{-1}] \\ \Sigma_f &= \text{the macroscopic fission cross section } [cm^{-1}] \\ CFY_i &= \text{the cumulative fission yield of nuclide } i [-]. \end{aligned}$$

Finally, the fission reaction rate term,  $\phi \Sigma_f$ , is assumed to be equal to 1. Equation (BB.3) can be used to calculate the concentration as a function of time using cumulative fission yields, given as:

$$\frac{dN_j}{dt} = f_{in} CFY_i - \lambda_j N_j(t) - f_{ex} \lambda_{r,j} N_j(t) \quad (BB.3)$$

Assuming a stationary fuel with no recirculation or chemical removal, the equation simplifies greatly. The

solution to that equation, assuming no initial concentration of nuclide  $j$ , is given as:

$$N_j(t) = \frac{CFY_j}{\lambda_j} \left(1 - e^{-\lambda_j t}\right) \quad (\text{BB.4})$$

As previously mentioned, the cumulative fission yield gives the total amount of the nuclide generated including from decay chains, so the time dependent result is non-physical. For a saturation irradiation however, the sample is irradiated for a sufficiently long time such that all DNPs reach equilibrium. This means that the equation should be taken as time approaches infinity, which leads to the form [84]:

$$N_j(\infty) = \frac{CFY_j}{\lambda_j}. \quad (\text{BB.5})$$

For the case of a nuclide in a circulating fuel reactor with a non-zero chemical removal rate, the concentration as a function of time comes out as:

$$N_j(t) = \frac{f_{in}CFY_j}{f_{ex}\lambda_{r,j} + \lambda_j} \left(1 - e^{-f_{ex}\lambda_{r,j}t - \lambda_j t}\right) \quad (\text{BB.6})$$

Equation (BB.6) can also be taken as time goes to infinity, resulting in:

$$N_j(\infty) = \frac{f_{in}CFY_j}{f_{ex}\lambda_{r,j} + \lambda_j} \quad (\text{BB.7})$$

## Appendix C

# Discussion of Turbulent Modeling in Moltres

Section 3.1.3 discusses the proposed use of Moltres to simulate transients in the MSRE, including how the DNPs can be modeled. The recirculation of DNPs could include the effects of turbulence, though such an effect is not necessary for the comparison between stationary and improved DNP group parameters. However, including turbulence would be useful for verification of transient responses to experimental data.

The proposed transient simulations, discussed in Section 3.5, using Moltres, discussed in Section 3.1.3, could include turbulent effects. Turbulence would not affect the DNP group parameters, but it would affect the concentration distribution for each group. The recirculation of DNPs could include the effects of turbulence, though such an effect is not necessary for the comparison between stationary and improved DNP group parameters. However, including turbulence would be useful for verification of transient responses to experimental data. The in-core region of the MSRE consists of flow through channels, which would not have large residence time differences from turbulence. Turbulence would cause mixing in the upper and lower plena, so the Moltres simulation without turbulence will assume perfect mixing in those regions.

In the proposed transient simulations, neglecting turbulence may affect residence time in the upper and lower plena. Specifically, turbulent effects would increase the ex-core residence time relative to the in-core residence time. Because Moltres can model turbulence using the Spalart-Allmaras turbulence model, it is straightforward to include in the transient simulations, though it does introduce more variables [183, 137]. The purpose of the transient simulations is to demonstrate the difference between simulations using improved DNP group parameters compared to existing group parameters. Modeling turbulence would not significantly change this difference, but would improve the simulation accuracy.

For these reasons, modeling turbulence in the Moltres transient simulations will be treated as an optional extension. The models without turbulence will assume perfect mixing in the upper and lower plena. If sufficient time is available, turbulent modeling will be included. This will also include an analysis of the difference between transients with and without turbulence will be conducted for the MSRE pump start-up and shut-down transients. This analysis will compare the reactivity difference and any changes in the in-core and ex-core residence times with and without turbulence modeled. If insufficient time is available, this analysis will be moved to the future work section.

## Appendix D

# An Analysis of Pulse and Saturation Data

The effects of using pulse or saturation irradiation data can be demonstrated through an illustrative toy problem. Instead of using several hundred DNPs, only four will be considered:  $A$ ,  $B$ ,  $C$ , and  $D$ . Each of these DNPs will have an emission probability  $P_n$  of unity and will only be formed via fission, having an equivalent cumulative fission yield  $CFY$  and independent fission yield  $IFY$  of 0.25. The only difference between these DNPs will be in their half-lives, given as:  $\lambda_A = 0.001$ ,  $\lambda_B = 0.01$ ,  $\lambda_C = 0.1$ , and  $\lambda_D = 1 \text{ s}^{-1}$ .

The concentration of each nuclide can be calculated using Equation (3.6), recreated here:

$$N_i = \frac{CFY_i}{\lambda_i}.$$

This gives concentrations of  $N_A = 250$ ,  $N_B = 25$ ,  $N_C = 2.5$ , and  $N_D = 0.25$  atoms.

Because all of the DNPs come directly from fission, the independent fission yield can be used to calculate the concentration after a single normalized fission event. This gives concentrations of  $N_A = N_B = N_C = N_D = 0.25$  atoms.

Next, to complete the comparison, these four DNPs can be fit into two DNP groups to determine what parameters best fit the delayed neutron count rates produced. The total delayed neutron yield should be one because one delayed neutron is released per fission event, and the average half-life is calculated to be approximately 192.5 seconds. Discussion on how these values are calculated is given in Section 2.1.2. The results comparing the data of these four DNPs to the different group fits are given in Table D.1.

Table D.1: A comparison of the raw DNP data to the different irradiation methods and the corresponding yields and average half-lives.

| Method             | Delayed Yield [-] | Average Half-life [s] |
|--------------------|-------------------|-----------------------|
| Data               | 1.00              | 192.5                 |
| 4 Group Saturation | 1.00              | 192.5                 |
| 3 Group Saturation | 0.98              | 188.9                 |
| 2 Group Saturation | 0.92              | 182.3                 |
| 1 Group Saturation | 0.62              | 223.8                 |
| 4 Group Pulse      | 1.00              | 192.5                 |
| 3 Group Pulse      | 0.91              | 91.65                 |
| 2 Group Pulse      | 0.80              | 61.0                  |
| 1 Group Pulse      | 0.42              | 1.3                   |

Because each DNP is equally important, the 3 group structure captures 75% of the total delayed neutrons,

while the 2 group structure is only able to capture 50% of the total delayed neutrons. The delayed neutron results show that both the pulse and saturation irradiations generally capture the delayed neutrons, provided there are sufficiently many groups available. As the number of groups decreases, the delayed yield decreases.

The average half-life shows that the saturation irradiation tends to over-estimate the average half-life as the number of groups becomes too few, though until that point it has more accurate values for the yield and average half-life. The pulse irradiation tends to under-estimate the average half-life noticeably, and the yield decreases faster than the saturation irradiation results.

Although this example does not exactly represent reality, it provides a useful understanding of the effects of both the pulse and saturation irradiations on the DNP group parameters. The results here provide further motivation to investigate combining the pulse and saturation irradiation data in the DNP group parameter formulation, as discussed in Section 5.1.

## Appendix E

# Spatially Resolved Model

The spatially resolved model is potential future work that could extend from the 0D flow model and analyses in this work. The 0D flow model will rapidly generate improved DNP group parameters, but it simplifies the treatment of flow. Instead of modeling the actual fuel salt flow, it will estimate fuel circulation using residence times for each region. A spatially resolved model would be useful for capturing the effects that the 0D flow model only capture with residence times for specific MSR geometries. Figure E.1 shows the flow diagram for the spatially resolved model, which could model irradiation of a target fissile sample using Moltres.

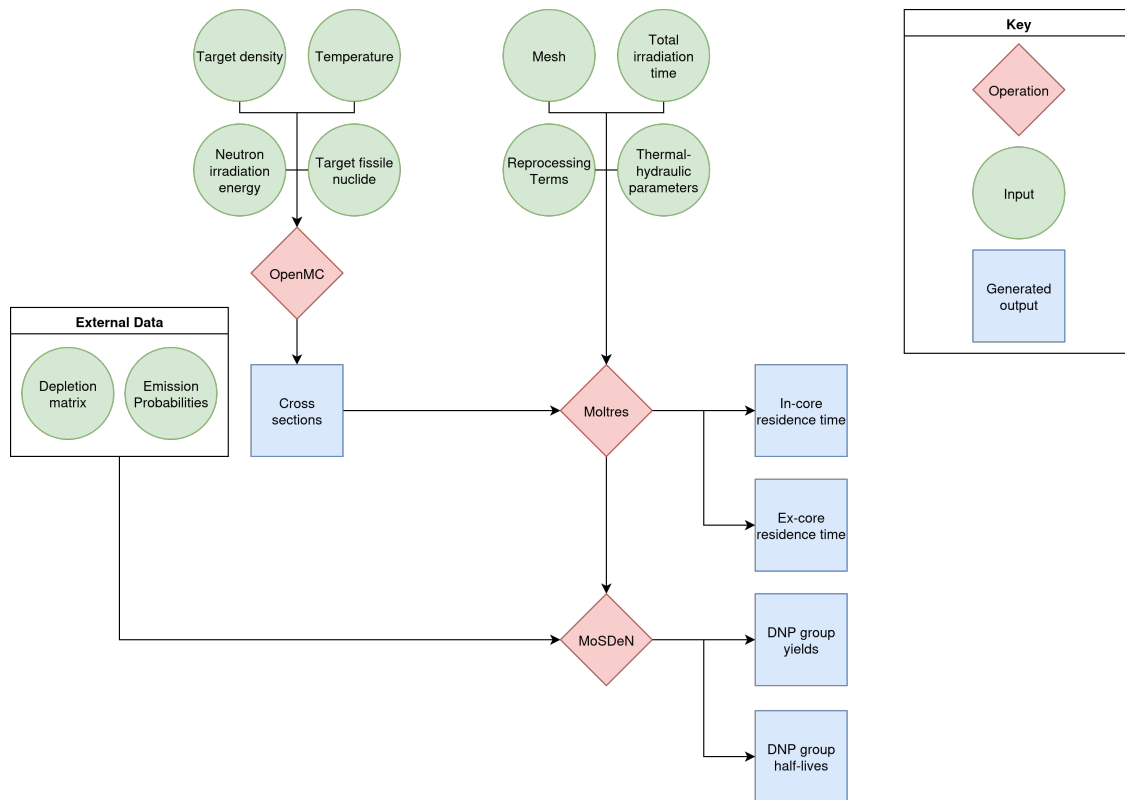


Figure E.1: Flow diagram of spatially resolved flow model used to generate improved DNP group parameters.

The spatially resolved flow model introduces two new parameters absent in the 0D flow model. The first is the

mesh, which defines the flow loop geometry. MOOSE provides built-in meshing tools, though complex geometries may require external tools such as Gmsh [68]. The second is the thermal-hydraulics parameters, which capture physical phenomena enabled by spatial resolution, as discussed in Section 3.1.3. Additionally, since the depletion would happen in Moltres instead of OpenMC, OpenMC does not need as much data.

In the spatially resolved model, the in-core and ex-core residence times would be outputs rather than inputs. These times derive from the movement of the irradiated sample over time in the fuel salt. The in-core and ex-core residence times vary as the irradiated sample diffuses and mixes further into the surrounding salt. The net residence time,  $\tau_{net}$ , remains constant unless the model changes, such as an increased temperature difference or induced flow rate. Equation (EE.1) defines the in-core residence time as:

$$\tau_{in}(t) = \frac{\int_{V_{in}} \int_0^t N(\vec{x}, \tau) d\tau d\vec{x}}{\int_V \int_0^t N(\vec{x}, \tau) d\tau d\vec{x}} \tau_{net}, \quad (\text{EE.1})$$

where

$\tau_{in}$  = the in-core residence time [s]

$V_{in}$  = the in-core volume [ $cm^3$ ]

$V$  = the total primary volume [ $cm^3$ ]

$N(\vec{x}, t)$  = the time and space resolved sample concentration [ $\# \cdot cm^{-3}$ ]

$\tau_{net}$  = the net residence, or circulation, time of the fuel salt [s].

The ex-core residence time can be calculated in the same way, or alternatively as shown in Equation (EE.2) as:

$$\tau_{ex} = \tau_{net} - \tau_{in}. \quad (\text{EE.2})$$

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